

- Hence the function $F'(q)$ is the remainder when we approximate $f'(q)$ with an order $k - 1$ Taylor polynomial. The remainder formula, equation (3.4.33), then tells us that there is a number c between a and q so that

$$\begin{aligned} F'(q) &= g(q) - \left(g(a) + g'(a)(q-a) + \frac{1}{2}g''(a)(q-a)^2 + \cdots + \frac{1}{(k-1)!}g^{(k-1)}(a)(q-a)^{k-1} \right) \\ &= \frac{1}{k!}g^{(k)}(c)(q-a)^k = \frac{1}{k!}f^{(k+1)}(c)(q-a)^k \end{aligned}$$

Notice that here we have assumed that $f^{(k+1)}(x)$ exists.

- Now substitute this back into our equation above

$$\begin{aligned} F(x) &= \frac{(x-a)^{k+1}}{(k+1)(q-a)^k} \cdot F'(q) \\ &= \frac{(x-a)^{k+1}}{(k+1)(q-a)^k} \cdot \frac{1}{k!}f^{(k+1)}(c)(q-a)^k \\ &= \frac{1}{(k+1)k!} \cdot f^{(k+1)}(c) \cdot \frac{(x-a)^{k+1}(q-a)^k}{(q-a)^k} \\ &= \frac{1}{(k+1)!} \cdot f^{(k+1)}(c) \cdot (x-a)^{k+1} \end{aligned}$$

as required.

So we now know that

- if, for some k , the remainder formula (with $n = k$) is true for all k times differentiable functions,
- then the remainder formula is true (with $n = k + 1$) for all $k + 1$ times differentiable functions.

Repeatedly applying this for $k = 1, 2, 3, 4, \dots$ (and recalling that we have shown the remainder formula is true when $n = 0, 1$) gives equation (3.4.33) for all $n = 0, 1, 2, \dots$ $\square$

3.5 ▲ Optimisation

One important application of differential calculus is to find the maximum (or minimum) value of a function. This often finds real world applications in problems such as the following.

Example 3.5.1

A farmer has 400m of fencing materials. What is the largest rectangular paddock that can be enclosed?

Solution. We will describe a general approach to these sorts of problems in Sections 3.5.2 and 3.5.3 below, but here we can take a stab at starting the problem.

- Begin by defining variables and their units (more generally we might draw a picture too); let the dimensions of the paddock be x by y metres.
- The area enclosed is then Am^2 where

$$A = x \cdot y$$

At this stage we cannot apply the calculus we have developed since the area is a function of two variables and we only know how to work with functions of a single variable. We need to eliminate one variable.

- We know that the perimeter of the rectangle (and hence the dimensions x and y) are constrained by the amount of fencing materials the farmer has to hand:

$$2x + 2y \leq 400$$

and so we have

$$y \leq 200 - x$$

Clearly the area of the paddock is maximised when we use all the fencing possible, so

$$y = 200 - x$$

- Now substitute this back into our expression for the area

$$A = x \cdot (200 - x)$$

Since the area cannot be negative (and our lengths x, y cannot be negative either), we must also have

$$0 \leq x \leq 200$$

- Thus the question of the largest paddock enclosed becomes the problem of finding the maximum value of

$$A = x \cdot (200 - x) \quad \text{subject to the constraint } 0 \leq x \leq 200.$$

Example 3.5.1

The above example is sufficiently simple that we can likely determine the answer by several different methods. In general, we will need more systematic methods for solving problems of the form

Find the maximum value of $y = f(x)$ subject to $a \leq x \leq b$

To do this we need to examine what a function looks like near its maximum and minimum values.

3.5.1 ► Local and Global Maxima and Minima

We start by asking:

Suppose that the maximum (or minimum) value of $f(x)$ is $f(c)$ then what does that tell us about c ?

Notice that we have not yet made the ideas of maximum and minimum very precise. For the moment think of maximum as “the biggest value” and minimum as “the smallest value”.

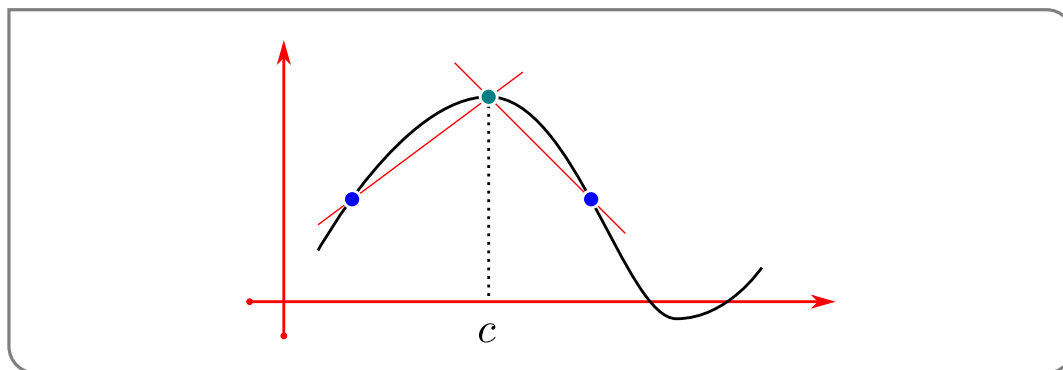
Warning 3.5.2.

It is important to distinguish between “the smallest value” and “the smallest magnitude”. For example, because

$$-5 < -1$$

the number -5 is smaller than -1 . But the magnitude of -1 , which is $|-1| = 1$, is smaller than the magnitude of -5 , which is $|-5| = 5$. Thus the smallest number in the set $\{-1, -5\}$ is -5 , while the number in the set $\{-1, -5\}$ that has the smallest magnitude is -1 .

Now back to thinking about what happens around a maximum. Suppose that the maximum value of $f(x)$ is $f(c)$, then for all “nearby” points, the function should be smaller.



Consider the derivative of $f'(c)$:

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}.$$

Split the above limit into the left and right limits:

- Consider points to the right of $x = c$, For all $h > 0$,

$$\begin{aligned} f(c+h) &\leq f(c) \\ f(c+h) - f(c) &\leq 0 \\ \frac{f(c+h) - f(c)}{h} &\leq 0 \end{aligned}$$

which implies that
which also implies
since $\frac{\text{negative}}{\text{positive}} = \text{negative}$.

But now if we squeeze $h \rightarrow 0$ we get

$$\lim_{h \rightarrow 0^+} \frac{f(c+h) - f(c)}{h} \leq 0$$

(provided the limit exists).

- Consider points to the left of $x = c$. For all $h < 0$,

$$\begin{aligned} f(c+h) &\leq f(c) && \text{which implies that} \\ f(c+h) - f(c) &\leq 0 && \text{which also implies} \\ \frac{f(c+h) - f(c)}{h} &\geq 0 && \text{since } \frac{\text{negative}}{\text{negative}} = \text{positive.} \end{aligned}$$

But now if we squeeze $h \rightarrow 0$ we get

$$\lim_{h \rightarrow 0^-} \frac{f(c+h) - f(c)}{h} \geq 0$$

(provided the limit exists).

- So if the derivative $f'(c)$ exists, then the above right- and left-hand limits must agree, which forces $f'(c) = 0$.

Thus we can conclude that

If the maximum value of $f(x)$ is $f(c)$ and $f'(c)$ exists, then $f'(c) = 0$.

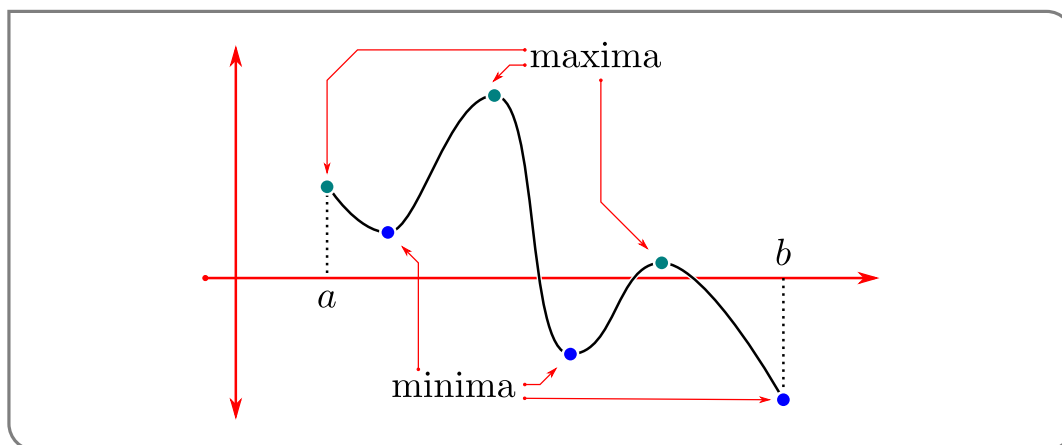
Using similar reasoning one can also see that

If the minimum value of $f(x)$ is $f(c)$ and $f'(c)$ exists, then $f'(c) = 0$.

Notice two things about the above reasoning:

- Firstly, in order for the argument to work we only need that $f(x) < f(c)$ for x close to c — it does not matter what happens for x values far from c .
- Secondly, in the above argument we needed to consider $f(x)$ for x both to the left of and to the right of c . If the function $f(x)$ is defined on a closed interval $[a, b]$, then the above argument only applies when $a < c < b$ — not when c is either of the endpoints a and b .

Consider the function below



This function has only 1 maximum value (the middle green point in the graph) and 1 minimum value (the rightmost blue point), however it has 4 points at which the derivative is zero. In the small intervals around those points where the derivative is zero, we can see that function is *locally* a maximum or minimum, even if it is not the *global* maximum or minimum. We clearly need to be more careful distinguishing between these cases.

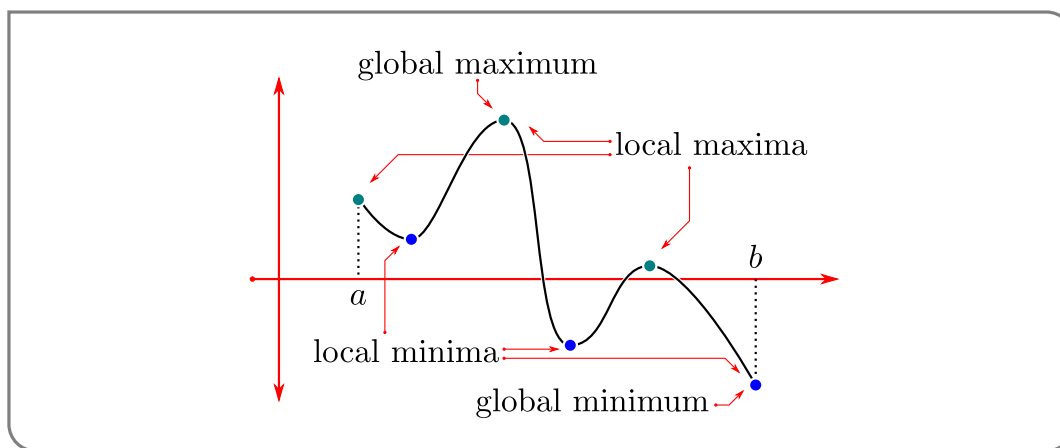
Definition 3.5.3.

Let I be an interval, like (a, b) or $[a, b]$ for example, and let the function $f(x)$ be defined for all $x \in I$. Now let $c \in I$. Then

- we say that $f(x)$ has a *global (or absolute) minimum on the interval I* at the point $x = c$ if $f(x) \geq f(c)$ for all $x \in I$.
- Similarly, we say that $f(x)$ has a *global (or absolute) maximum on I* at $x = c$ if $f(x) \leq f(c)$ for all $x \in I$.
- We say that $f(x)$ has a *local⁵¹ minimum on I* at $x = c$ if $f(x) \geq f(c)$ for all $x \in I$ that are near c . Precisely, if there is a $\delta > 0$ such that $f(x) \geq f(c)$ for all $x \in I$ that are within a distance δ of c .
- Similarly, we say that $f(x)$ has a *local maximum on I* at $x = c$ if $f(x) \leq f(c)$ for all $x \in I$ that are near c . Precisely, if there is a $\delta > 0$ such that $f(x) \leq f(c)$ for all $x \in I$ that are within a distance δ of c .

The global maxima and minima of a function are called the global extrema of the function, while the local maxima and minima are called the local extrema.

Consider again the function we showed in the figure above



It has 3 local maxima and 3 local minima on the interval $[a, b]$. The global maximum occurs at the middle green point (which is also a local maximum), and the global minimum occurs at the rightmost blue point (which is also a local minimum).

51 Beware that, while many textbooks use these definitions of local minimum and maximum, some textbooks exclude the endpoints a, b of the interval $[a, b]$ from their definitions. Our definitions allow the endpoints a and b to be local minima and maxima. Note that, under our definitions, every global minimum (maximum) is also a local minimum (maximum).

Using the above definition we can summarise what we have learned above as the following theorem⁵²:

Theorem 3.5.4.

Let the function $f(x)$ be defined on the interval I and let a, b, c be points in I with $a < c < b$. If $f(x)$ has a local maximum or local minimum at $x = c$ and if $f'(c)$ exists, then $f'(c) = 0$.

- It is often (but not always) the case that, when $f(x)$ has a local maximum at $x = c$, the function $f(x)$ increases strictly as x approaches c from the left and decreases strictly as x leaves c to the right. That is, $f'(x) > 0$ for x just to the left of c and $f'(x) < 0$ for x just to the right of c . Then, it is often the case, because $f'(x)$ is decreasing as x increases through c , that $f''(c) < 0$.
- Conversely, if $f'(c) = 0$ and $f''(c) < 0$, then, just to the right of c , $f'(x)$ must be negative, so that $f(x)$ is decreasing, and just to the left of c , $f'(x)$ must be positive, so that $f(x)$ is increasing. So $f(x)$ has a local maximum at c .
- Similarly, it is often the case that, when $f(x)$ has a local minimum at $x = c$, $f'(x) < 0$ for x just to the left of c and $f'(x) > 0$ for x just to the right of c and $f''(x) > 0$.
- Conversely, if $f'(c) = 0$ and $f''(c) > 0$, then, just to the right of c , $f'(x)$ must be positive, so that $f(x)$ is increasing, and, just to the left of c , $f'(x)$ must be negative, so that $f(x)$ is decreasing. So $f(x)$ has a local minimum at c .

Theorem 3.5.5.

Let $f(x)$ be defined on the interval I and let $a, b, c \in I$ with $a < c < b$.

If $f'(c) = 0$ and $f''(c) < 0$, then $f(x)$ has a local maximum at c .

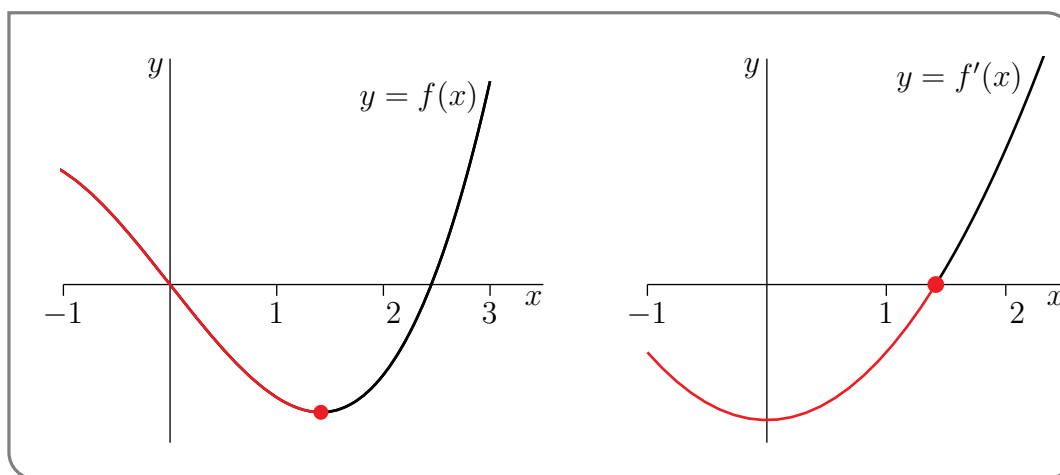
If $f'(c) = 0$ and $f''(c) > 0$, then $f(x)$ has a local minimum at c .

Note the strict inequalities.

Theorem ^{3.5.4} says that, when $f(x)$ has a local maximum or minimum on an interval I at the point $x = c$, there are three possibilities.

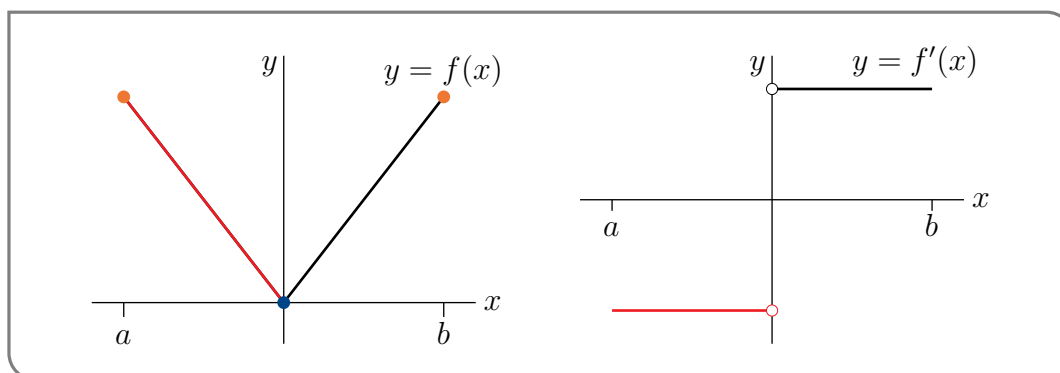
- The derivative $f'(c) = 0$. This case is illustrated in the following figure.

52 This is one of several important mathematical contributions made by Pierre de Fermat, a French government lawyer and amateur mathematician, who lived in the first half of the seventeenth century.



Observe that, in this example, $f'(x)$ changes continuously from negative to positive at the local minimum, taking the value zero at the local minimum (the red dot).

- The derivative $f'(c)$ does not exist. This case is illustrated in the following figure.



Observe that, in this example, $f'(x)$ changes discontinuously from negative to positive at the local minimum ($x = 0$) and $f'(0)$ does not exist.

- The point c is an endpoint of the interval $I = [a, b]$. This case is also illustrated in the above figure. The endpoints a and b are both local maxima. But $f'(a)$ and $f'(b)$ are not zero.

This theorem demonstrates that the points at which the derivative is zero or does not exist are very important. It simplifies the discussion that follows if we give these points names.

Definition 3.5.6.

Let $f(x)$ be a function that is defined on the interval $a < x < b$ and let $a < c < b$. Then

- if $f'(c)$ exists and is zero we call $x = c$ a critical point of the function, and
- if $f'(c)$ does not exist then we call $x = c$ a singular point⁵³ of the function.

Warning 3.5.7.

Note that some people (and texts) will combine both of these cases and call $x = c$ a critical point when either the derivative is zero or does not exist. The reader should be aware of the lack of convention on this point⁵⁴ and should be careful to understand whether the more inclusive definition of critical point is being used, or if the text is using the more precise definition that distinguishes critical and singular points.

We'll now look at a few simple examples involving local maxima and minima, critical points and singular points. Then we will move on to global maxima and minima.

Example 3.5.8

In this example, we'll look for local maxima and minima of the function $f(x) = x^3 - 6x$ on the interval $-2 \leq x \leq 3$.

- First compute the derivative

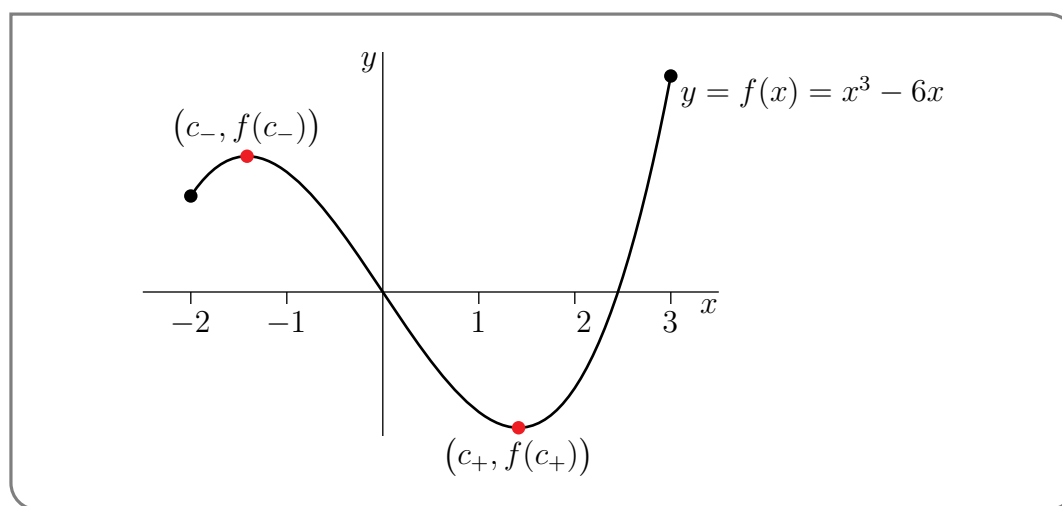
$$f'(x) = 3x^2 - 6.$$

Since this is a polynomial it is defined everywhere on the domain and so there will not be any singular points. So we now look for critical points.

- To do so we look for zeroes of the derivative

$$f'(x) = 3x^2 - 6 = 3(x^2 - 2) = 3(x - \sqrt{2})(x + \sqrt{2}).$$

This derivative takes the value 0 at two different values of x . Namely $x = c_- = -\sqrt{2}$ and $x = c_+ = \sqrt{2}$. Here is a sketch of the graph of $f(x)$.



53 For c to be a local maximum or minimum of f , the function f must obviously be defined at c . So here we are considering only points c in the domain of f . We will later, in Section 3.6.2, extend the definition of singular points of f to points that are not in the domain of f .

54 No pun intended.

From the figure we see that

- $f(x)$ has a local minimum at the endpoint $x = -2$ (i.e. we have $f(x) \geq f(-2)$ whenever $x \geq -2$ is close to -2) and
 - $f(x)$ has a local minimum at $x = c_+$ (i.e. we have $f(x) \geq f(c_+)$ whenever x is close to c_+) and
 - $f(x)$ has a local maximum at $x = c_-$ (i.e. we have $f(x) \leq f(c_-)$ whenever x is close to c_-) and
 - $f(x)$ has a local maximum at the endpoint $x = 3$ (i.e. we have $f(x) \leq f(3)$ whenever $x \leq 3$ is close to 3) and
 - the global minimum of $f(x)$, for x in the interval $-2 \leq x \leq 3$, is at $x = c_+$ (i.e. we have $f(x) \geq f(c_+)$ whenever $-2 \leq x \leq 3$) and
 - the global maximum of $f(x)$, for x in the interval $-2 \leq x \leq 3$, is at $x = 3$ (i.e. we have $f(x) \leq f(3)$ whenever $-2 \leq x \leq 3$).
- Note that we have carefully constructed this example to illustrate that the global maximum (or minimum) of a function on an interval may or may not also be a critical point of the function.

Example 3.5.8

Example 3.5.9

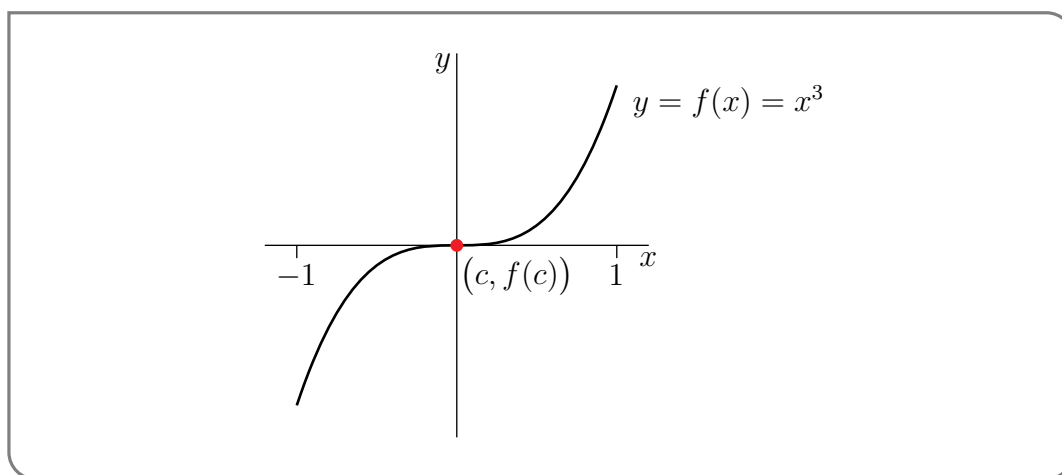
In this example, we'll look for local maxima and minima of the function $f(x) = x^3$ on the interval $-1 < x < 1$.

- First compute the derivative:

$$f'(x) = 3x^2.$$

Again, this is a polynomial and so defined on all of the domain. The function will not have singular points, but may have critical points.

- The derivative is zero only when $x = 0$, so $x = c = 0$ is the only critical point of the function.
- The graph of $f(x)$ is sketched below. From that sketch we see that $f(x)$ has *neither* a local maximum *nor* a local minimum at $x = c$ despite the fact that $f'(c) = 0$ — we have $f(x) < f(c) = 0$ for all $x < c = 0$ and $f(x) > f(c) = 0$ for all $x > c = 0$.



- Note that this example has been constructed to illustrate that a critical point (or singular point) of a function *need not be a local maximum or minimum* for the function.
- Reread Theorem [3.5.4](#). It says ⁵⁵“Let $\dots$. If $f(x)$ has a local maximum/minimum at $x = c$ and if $f'(c)$ exists, then $f'(c) = 0$ ”. It *does not say* that “if $f'(c) = 0$ then f has a local maximum/minimum at $x = c$ ”.

Example 3.5.9

Example 3.5.10

In this example, we'll look for local maxima and minima of the function

$$f(x) = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

on the interval $-1 < x < 1$ and we'll also look for local maxima and minima of the function

$$g(x) = x^{2/3}$$

on the interval $-1 < x < 1$.

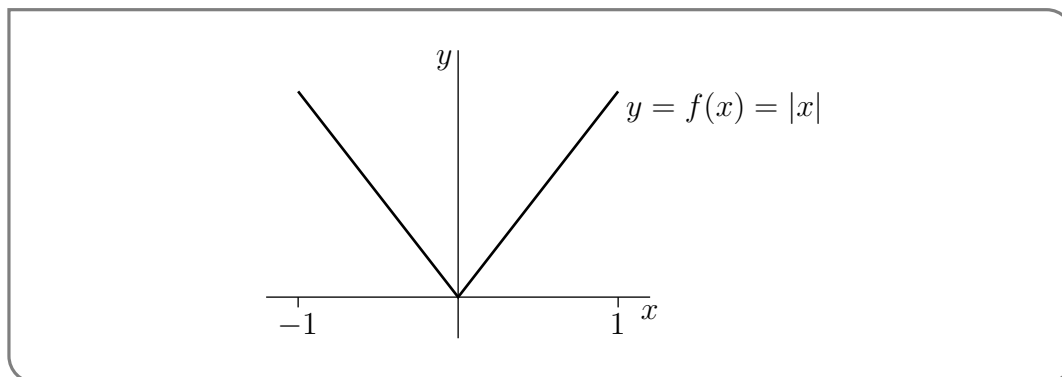
- Again, start by computing the derivatives (reread Example [2.2.10](#)):

$$f'(x) = \begin{cases} 1 & \text{if } x > 0 \\ \text{undefined} & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$$

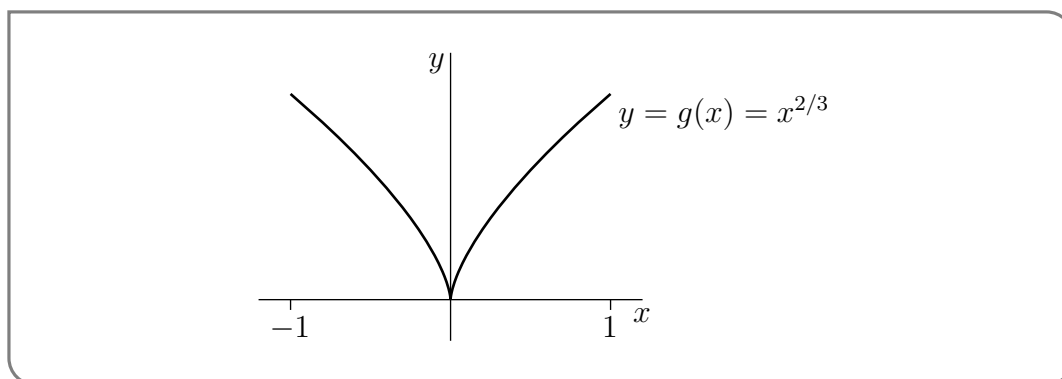
$$g'(x) = \begin{cases} \frac{2}{3}x^{-1/3} & \text{if } x \neq 0 \\ \text{undefined} & \text{if } x = 0 \end{cases}$$

⁵⁵ A very common error of logic that people make is “Affirming the consequent”. When the statement “if P then Q” is true, observing Q does *not* imply P. (“Affirming the consequent” eliminates “not” from the previous sentence.) For example, “If he is Shakespeare, then he is dead,” and “That man is dead.” does not imply “He must be Shakespeare.”. Or you may have also seen someone use this reasoning: “If a person is a genius before their time then they are misunderstood.” “I am misunderstood.” “So I must be a genius before my time.”.

- These derivatives *never* take the value 0, so the functions $f(x)$ and $g(x)$ do not have any critical points. However both derivatives do not exist at the point $x = 0$, so that point is a singular point for both $f(x)$ and $g(x)$.
- Here is a sketch of the graph of $f(x)$



and a sketch of the graph of $g(x)$.



From the figures we see that both $f(x)$ and $g(x)$ have a local (and in fact global) minimum at $x = 0$ despite the fact that $x = 0$ is not a critical point.

- Reread Theorem [3.5.4](#) yet again. It says “Let $\dots$. If $f(x)$ has a local maximum or local minimum at $x = c$ and if f is differentiable at $x = c$, then $f'(c) = 0$ ”. It says nothing about what happens at points where the derivative does not exist. Indeed that is why we have to consider both critical points and singular points when we look for maxima and minima.

Example 3.5.10

3.5.2 ► Finding Global Maxima and Minima

We now have a technique for finding local maxima and minima — just look at endpoints of the interval of interest and for values of x for which either $f'(x) = 0$ or $f'(x)$ does not exist. What about finding global maxima and minima? We'll start by stating explicitly that, under appropriate hypotheses, global maxima and minima are guaranteed to exist.

Theorem 3.5.11.

Let the function $f(x)$ be defined and continuous on the closed, finite interval $-\infty < a \leq x \leq b < \infty$. Then $f(x)$ attains a maximum and a minimum at least once. That is, there exist numbers $a \leq x_m, x_M \leq b$ such that

$$f(x_m) \leq f(x) \leq f(x_M) \quad \text{for all } a \leq x \leq b$$

So let's again consider the question

Suppose that the maximum (or minimum) value of $f(x)$, for $a \leq x \leq b$, is $f(c)$. What does that tell us about c ?

If c obeys $a < c < b$ (note the strict inequalities), then f has a local maximum (or minimum) at $x = c$ and Theorem 3.5.4 tells us that either $f'(c) = 0$ or $f'(c)$ does not exist. The only other place that a maximum or minimum can occur are at the ends of the interval. We can summarise this as:

Theorem 3.5.12.

If $f(x)$ has a global maximum or global minimum, for $a \leq x \leq b$, at $x = c$ then there are 3 possibilities. Either

- $f'(c) = 0$, or
- $f'(c)$ does not exist, or
- $c = a$ or $c = b$.

That is, a global maximum or minimum must occur either at a critical point, a singular point or at the endpoints of the interval.

This theorem provides the basis for a method to find the maximum and minimum values of $f(x)$ for $a \leq x \leq b$:

56 The hypotheses that $f(x)$ be continuous and that the interval be finite and closed are all essential. We suggest that you find three functions $f_1(x)$, $f_2(x)$ and $f_3(x)$ with f_1 defined but not continuous on $0 \leq x \leq 1$, f_2 defined and continuous on $-\infty < x < \infty$, and f_3 defined and continuous on $0 < x < 1$, and with none of f_1 , f_2 and f_3 attaining either a global maximum or a global minimum.

Corollary 3.5.13.

Let $f(x)$ be a function on the interval $a \leq x \leq b$. Then to find the global maximum and minimum of the function:

- Make a list of all values of c , with $a \leq c \leq b$, for which
 - $f'(c) = 0$, or
 - $f'(c)$ does not exist, or
 - $c = a$ or $c = b$.

That is — compute the function at all the critical points, singular points, and endpoints.

- Evaluate $f(c)$ for each c in that list. The largest (or smallest) of those values is the largest (or smallest) value of $f(x)$ for $a \leq x \leq b$.

Let's now demonstrate how to use this strategy. The function in this first example is not too simple — but it is a good example of a function that contains both a singular point and a critical point.

Example 3.5.14

Find the largest and smallest values of the function $f(x) = 2x^{5/3} + 3x^{2/3}$ for $-1 \leq x \leq 1$.

Solution. We will apply the method in Corollary 3.5.13. It is perhaps easiest to find the values at the endpoints of the intervals and then move on to the values at any critical or singular points.

- Before we get into things, notice that we can rewrite the function by factoring it:

$$f(x) = 2x^{5/3} + 3x^{2/3} = x^{2/3} \cdot (2x + 3)$$

- Let's compute the function at the endpoints of the interval:

$$\begin{aligned} f(1) &= 2 + 3 = 5 \\ f(-1) &= 2 \cdot (-1)^{5/3} + 3 \cdot (-1)^{2/3} = -2 + 3 = 1 \end{aligned}$$

- To compute the function at the critical and singular points we first need to find the derivative:

$$\begin{aligned} f'(x) &= 2 \cdot \frac{5}{3} x^{2/3} + 3 \cdot \frac{2}{3} x^{-1/3} \\ &= \frac{10}{3} x^{2/3} + 2x^{-1/3} \\ &= \frac{10x + 6}{3x^{1/3}} \end{aligned}$$

- Notice that the numerator and denominator are defined for all x . The only place the derivative is undefined is when the denominator is zero. Hence the only singular point is at $x = 0$. The corresponding function value is

$$f(0) = 0$$

- To find the critical points we need to solve $f'(x) = 0$:

$$0 = \frac{10x + 6}{3x^{1/3}}$$

Hence we must have $10x = -6$ or $x = -3/5$. The corresponding function value is

$$\begin{aligned} f(x) &= x^{2/3} \cdot (2x + 3) && \text{recall this from above, then} \\ f(-3/5) &= (-3/5)^{2/3} \cdot \left(2 \cdot \frac{-3}{5} + 3\right) \\ &= \left(\frac{9}{25}\right)^{1/3} \cdot \frac{-6 + 15}{5} \\ &= \left(\frac{9}{25}\right)^{1/3} \cdot \frac{9}{5} \approx 1.28 \end{aligned}$$

Note that if we do not want to approximate the root (if, for example, we do not have a calculator handy), then we can also write

$$\begin{aligned} f(-3/5) &= \left(\frac{9}{25}\right)^{1/3} \cdot \frac{9}{5} \\ &= \left(\frac{9}{25}\right)^{1/3} \cdot \frac{9}{25} \cdot 5 \\ &= 5 \cdot \left(\frac{9}{25}\right)^{4/3} \end{aligned}$$

Since $0 < 9/25 < 1$, we know that $0 < \left(\frac{9}{25}\right)^{4/3} < 1$, and hence

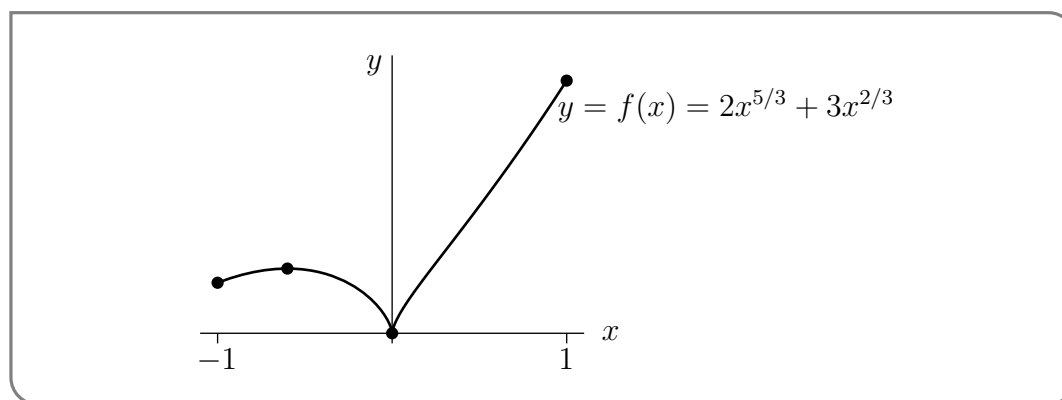
$$0 < f(-3/5) = 5 \cdot \left(\frac{9}{25}\right)^{4/3} < 5.$$

- We summarise our work in this table

c	$-\frac{3}{5}$	0	-1	1
type	critical point	singular point	endpoint	endpoint
$f(c)$	$\frac{9}{5} \sqrt[3]{\frac{9}{25}} \approx 1.28$	0	1	5

- The largest value of f in the table is 5 and the smallest value of f in the table is 0.
- Thus on the interval $-1 \leq x \leq 1$ the global maximum of f is 5, and is taken at $x = 1$, while the global minimum value of $f(x)$ is 0, and is taken at $x = 0$.

- For completeness we also sketch the graph of this function on the same interval.



Later (in Section 3.6) we will see how to construct such a sketch without using a calculator or computer.

Example 3.5.14

3.5.3 ► Max/Min Examples

As noted at the beginning of this section, the problem of finding maxima and minima is a very important application of differential calculus in the real world. We now turn to a number of examples of this process. But to guide the reader we will describe a general procedure to follow for these problems.

- (1) Read — read the problem carefully. Work out what information is given in the statement of the problem and what we are being asked to compute.
- (2) Diagram — draw a diagram. This will typically help you to identify what you know about the problem and what quantities you need to work out.
- (3) Variables — assign variables to the quantities in the problem along with their units. It is typically a good idea to make sensible choices of variable names: A for area, h for height, t for time etc.
- (4) Relations — find relations between the variables. By now you should know the quantity we are interested in (the one we want to maximise or minimise) and we need to establish a relation between it and the other variables.
- (5) Reduce — the relation down to a function of one variable. In order to apply the calculus we know, we must have a function of a single variable. To do this we need to use all the information we have to eliminate variables. We should also work out the domain of the resulting function.
- (6) Maximise or minimise — we can now apply the methods of Corollary 3.5.13 to find the maximum or minimum of the quantity we need (as the problem dictates).

- (7) Be careful — make sure your answer makes sense. Make sure quantities are physical. For example, lengths and areas cannot be negative.
- (8) Answer the question — be sure your answer really answers the question asked in the problem.

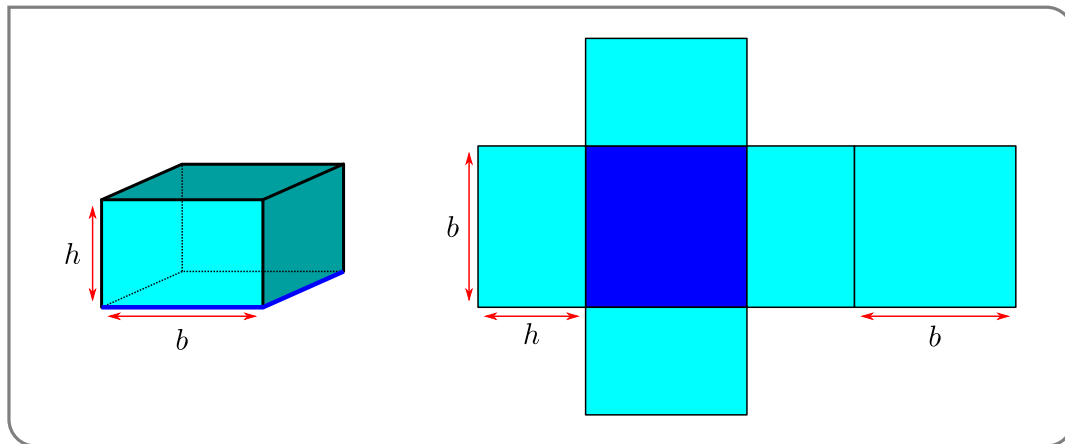
Let us start with a relatively simple problem:

Example 3.5.15

A closed rectangular container with a square base is to be made from two different materials. The material for the base costs \$5 per square meter, while the material for the other five sides costs \$1 per square meter. Find the dimensions of the container which has the largest possible volume if the total cost of materials is \$72.

Solution. We can follow the steps we outlined above to find the solution.

- We need to determine the area of the two types of materials used and the corresponding total cost.
- Draw a picture of the box.



The more useful picture is the unfolded box on the right.

- In the picture we have already introduced two variables. The square base has side-length b metres and it has height h metres. Let the area of the base be A_b and the area of the other five sides be A_s (both in m^2), and the total cost be C (in dollars). Finally let the volume enclosed be Vm^3 .
- Some simple geometry tells us that

$$A_b = b^2$$

$$A_s = 4bh + b^2$$

$$V = b^2h$$

$$C = 5 \cdot A_b + 1 \cdot A_s = 5b^2 + 4bh + b^2 = 6b^2 + 4bh.$$

- To eliminate one of the variables we use the fact that the total cost is \$72.

$$\begin{aligned}
 C &= 6b^2 + 4bh = 72 && \text{rearrange} \\
 4bh &= 72 - 6b^2 && \text{isolate } h \\
 h &= \frac{72 - 6b^2}{4b} = \frac{3}{2} \cdot \frac{12 - b^2}{b}
 \end{aligned}$$

Substituting this into the volume gives

$$V = b^2h = \frac{3b}{2}(12 - b^2) = 18b - \frac{3}{2}b^3$$

Now note that since b is a length it cannot be negative, so $b \geq 0$. Further since volume cannot be negative, we must also have

$$12 - b^2 \geq 0$$

and so $b \leq \sqrt{12}$.

- Now we can apply Corollary 3.5.13 on the above expression for the volume with $0 \leq b \leq \sqrt{12}$. The endpoints give:

$$\begin{aligned}
 V(0) &= 0 \\
 V(\sqrt{12}) &= 0
 \end{aligned}$$

The derivative is

$$V'(b) = 18 - \frac{9b^2}{2}$$

Since this is a polynomial there are no singular points. However we can solve $V'(b) = 0$ to find critical points:

$$\begin{aligned}
 18 - \frac{9b^2}{2} &= 0 && \text{divide by 9 and multiply by 2} \\
 4 - b^2 &= 0
 \end{aligned}$$

Hence $b = \pm 2$. Thus the only critical point in the domain is $b = 2$. The corresponding volume is

$$\begin{aligned}
 V(2) &= 18 \times 2 - \frac{3}{2} \times 2^3 \\
 &= 36 - 12 = 24.
 \end{aligned}$$

So by Corollary 3.5.13, the maximum volume is 24 when $b = 2$ and

$$h = \frac{3}{2} \cdot \frac{12 - b^2}{b} = \frac{3}{2} \cdot \frac{12 - 4}{2} = 6.$$

- All our quantities make sense; lengths, areas and volumes are all non-negative.

- Checking the question again, we see that we are asked for the dimensions of the container (rather than its volume) so we can answer with

The container with dimensions $2 \times 2 \times 6m$ will be the largest possible.

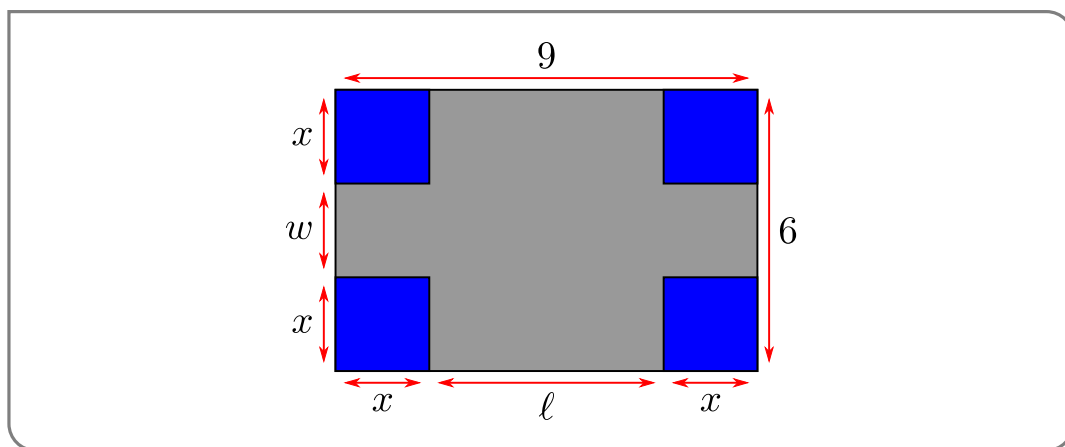
Example 3.5.15

Example 3.5.16

A rectangular sheet of cardboard is 6 inches by 9 inches. Four identical squares are cut from the corners of the cardboard, as shown in the figure below, and the remaining piece is folded into an open rectangular box. What should the size of the cut out squares be in order to maximize the volume of the box?

Solution. This one is quite similar to the previous one, so we perhaps don't need to go into so much detail.

- After reading carefully we produce the following picture:



- Let the height of the box be x inches, and the base be $\ell \times w$ inches. The volume of the box is then V cubic inches.
- Some simple geometry tells us that $\ell = 9 - 2x$, $w = 6 - 2x$ and so

$$\begin{aligned} V &= x(9 - 2x)(6 - 2x) \text{ cubic inches} \\ &= 54x - 30x^2 + 4x^3. \end{aligned}$$

Notice that since all lengths must be non-negative, we must have

$$x, \ell, w \geq 0$$

and so $0 \leq x \leq 3$ (if $x > 3$ then $w < 0$).

- We can now apply Corollary [3.5.13](#). First the endpoints of the interval give

$$V(0) = 0$$

$$V(3) = 0$$

The derivative is

$$\begin{aligned} V'(x) &= 54 - 60x + 12x^2 \\ &= 6(9 - 10x + 2x^2) \end{aligned}$$

Since this is a polynomial there are no singular points. To find critical points we solve $V'(x) = 0$ to get

$$\begin{aligned} x_{\pm} &= \frac{10 \pm \sqrt{100 - 4 \times 2 \times 9}}{4} \\ &= \frac{10 \pm \sqrt{28}}{4} = \frac{10 \pm 2\sqrt{7}}{4} = \frac{5 \pm \sqrt{7}}{2} \end{aligned}$$

We can then use a calculator to approximate

$$x_+ \approx 3.82 \qquad x_- \approx 1.18.$$

So x_- is inside the domain, while x_+ lies outside.

Alternatively⁵⁷, we can bound $x_{\pm}$ by first noting that $2 \leq \sqrt{7} \leq 3$. From this we know that

$$\begin{aligned} 1 &= \frac{5-3}{2} \leq x_- = \frac{5-\sqrt{7}}{2} \leq \frac{5-2}{2} = 1.5 \\ 3.5 &= \frac{5+2}{2} \leq x_+ = \frac{5+\sqrt{7}}{2} \leq \frac{5+3}{2} = 4 \end{aligned}$$

- Since the volume is zero when $x = 0, 3$, it must be the case that the volume is maximised when $x = x_- = \frac{5-\sqrt{7}}{2}$.
- Notice that since $0 < x_- < 3$ we know that the other lengths are positive, so our answer makes sense. Further, the question only asks for the length x and not the resulting volume so we have answered the question.

Example 3.5.16

There is a new wrinkle in the next two examples. Each involves finding the minimum value of a function $f(x)$ with x running over all real numbers, rather than just over a finite interval as in Corollary 3.5.13. Both in Example 3.5.18 and in Example 3.5.19 the function $f(x)$ tends to $+\infty$ as x tends to either $+\infty$ or $-\infty$. So the minimum value of $f(x)$ will be achieved for some finite value of x , which will be a local minimum as well as a global minimum.

⁵⁷ Say if we do not have a calculator to hand, or your instructor insists that the problem be done without one.

Theorem 3.5.17.

Let $f(x)$ be defined and continuous for all $-\infty < x < \infty$. Let c be a finite real number.

- (a) If $\lim_{x \rightarrow +\infty} f(x) = +\infty$ and $\lim_{x \rightarrow -\infty} f(x) = +\infty$ and if $f(x)$ has a global minimum at $x = c$, then there are 2 possibilities. Either
- $f'(c) = 0$, or
 - $f'(c)$ does not exist

That is, a global minimum must occur either at a critical point or at a singular point.

- (b) If $\lim_{x \rightarrow +\infty} f(x) = -\infty$ and $\lim_{x \rightarrow -\infty} f(x) = -\infty$ and if $f(x)$ has a global maximum at $x = c$, then there are 2 possibilities. Either
- $f'(c) = 0$, or
 - $f'(c)$ does not exist

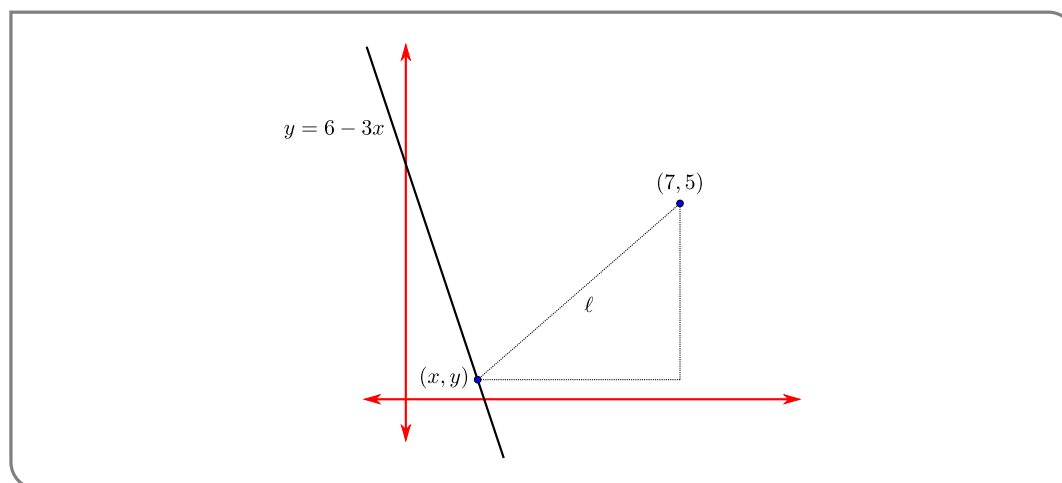
That is, a global maximum must occur either at a critical point or at a singular point.

Example 3.5.18

Find the point on the line $y = 6 - 3x$ that is closest to the point $(7, 5)$.

Solution. In this problem

- A simple picture



- Some notation is already given to us. Let a point on the line have coordinates (x, y) , and we do not need units. And let ℓ be the distance from the point (x, y) to the point $(7, 5)$.

- Since the points are on the line the coordinates (x, y) must obey

$$y = 6 - 3x$$

Notice that x and y have no further constraints. The distance ℓ is given by

$$\ell^2 = (x - 7)^2 + (y - 5)^2$$

- We can now eliminate the variable y :

$$\begin{aligned}\ell^2 &= (x - 7)^2 + (y - 5)^2 \\ &= (x - 7)^2 + (6 - 3x - 5)^2 = (x - 7)^2 + (1 - 3x)^2 \\ &= x^2 - 14x + 49 + 1 - 6x + 9x^2 = 10x^2 - 20x + 50 \\ &= 10(x^2 - 2x + 5) \\ \ell &= \sqrt{10} \cdot \sqrt{x^2 - 2x + 5}\end{aligned}$$

Notice that as $x \rightarrow \pm\infty$ the distance $\ell \rightarrow +\infty$.

- We can now apply Theorem [3.5.17](#)
 - Since the distance is defined for all real x , we do not have to check the endpoints of the domain — there are none.
 - Form the derivative:

$$\frac{d\ell}{dx} = \sqrt{10} \frac{2x - 2}{2\sqrt{x^2 - 2x + 5}}$$

It is zero when $x = 1$, and undefined if $x^2 - 2x + 5 < 0$. However, since

$$x^2 - 2x + 5 = (x^2 - 2x + 1) + 4 = \underbrace{(x - 1)^2}_{\geq 0} + 4$$

we know that $x^2 - 2x + 5 \geq 4$. Thus the function has no singular points and the only critical point occurs at $x = 1$. The corresponding function value is then

$$\ell(1) = \sqrt{10}\sqrt{1 - 2 + 5} = 2\sqrt{10}.$$

- Thus the minimum value of the distance is $\ell = 2\sqrt{10}$ and occurs at $x = 1$.
- This answer makes sense — the distance is not negative.
- The question asks for the point that minimises the distance, not that minimum distance. Hence the answer is $x = 1, y = 6 - 3 = 3$. I.e.

The point that minimises the distance is $(1, 3)$.

Notice that we can make the analysis easier by observing that the point that minimises the distance also minimises the squared-distance. So that instead of minimising the function ℓ , we can just minimise ℓ^2 :

$$\ell^2 = 10(x^2 - 2x + 5)$$

The resulting algebra is a bit easier and we don't have to hunt for singular points.

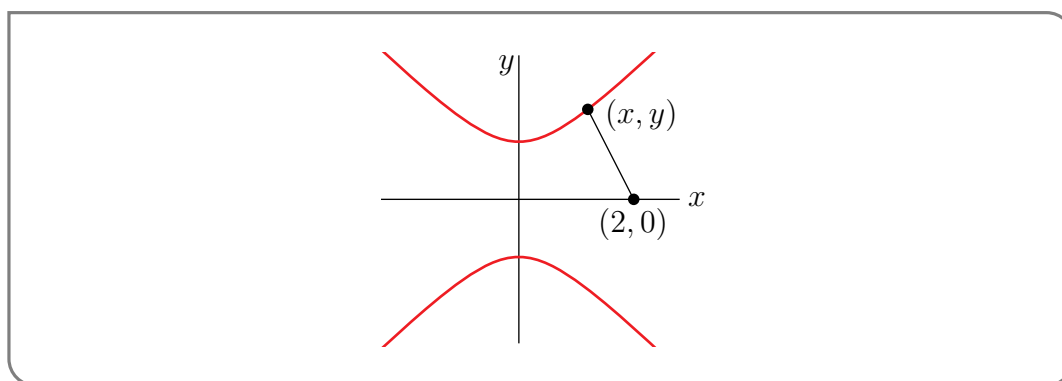
Example 3.5.18

Example 3.5.19

Find the minimum distance from $(2, 0)$ to the curve $y^2 = x^2 + 1$.

Solution. This is very much like the previous question.

- After reading the problem carefully we can draw a picture



- In this problem we do not need units and the variables x, y are supplied. We define the distance to be ℓ and it is given by

$$\ell^2 = (x - 2)^2 + y^2.$$

As noted in the previous problem, we will minimise the squared-distance since that also minimises the distance.

- Since x, y satisfy $y^2 = x^2 + 1$, we can write the distance as a function of x :

$$\ell^2 = (x - 2)^2 + y^2 = (x - 2)^2 + (x^2 + 1)$$

Notice that as $x \rightarrow \pm\infty$ the squared-distance $\ell^2 \rightarrow +\infty$.

- Since the squared-distance is a polynomial it will not have any singular points, only critical points. The derivative is

$$\frac{d}{dx}\ell^2 = 2(x - 2) + 2x = 4x - 4$$

so the only critical point occurs at $x = 1$.

- When $x = 1, y = \pm\sqrt{2}$ and the distance is

$$\ell^2 = (1 - 2)^2 + (1 + 1) = 3 \qquad \ell = \sqrt{3}$$

and thus the minimum distance from the curve to $(2, 0)$ is $\sqrt{3}$.

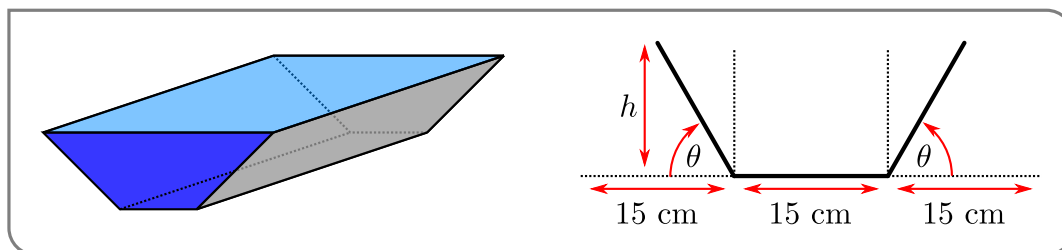
Example 3.5.19

Example 3.5.20

A water trough is to be constructed from a metal sheet of width 45 cm by bending up one third of the sheet on each side through an angle θ . Which θ will allow the trough to carry the maximum amount of water?

Solution. Clearly $0 \leq \theta \leq \pi$, so we are back in the domain⁵⁸ of Corollary 3.5.13.

- After reading the problem carefully we should realise that it is really asking us to maximise the cross-sectional area. A figure really helps.



- From this we are led to define the height h cm and cross-sectional area A cm². Both are functions of θ .

$$h = 15 \sin \theta$$

while the area can be computed as the sum of the central $15 \times h$ rectangle, plus two triangles. Each triangle has height h and base $15 \cos \theta$. Hence

$$\begin{aligned} A &= 15h + 2 \cdot \frac{1}{2} \cdot h \cdot 15 \cos \theta \\ &= 15h (1 + \cos \theta) \end{aligned}$$

- Since $h = 15 \sin \theta$ we can rewrite the area as a function of just θ :

$$A(\theta) = 225 \sin \theta (1 + \cos \theta)$$

where $0 \leq \theta \leq \pi$.

⁵⁸ Again, no pun intended.

- Now we use Corollary 3.5.13. The ends of the interval give

$$\begin{aligned} A(0) &= 225 \sin 0(1 + \cos 0) = 0 \\ A(\pi) &= 225 \sin \pi(1 + \cos \pi) = 0 \end{aligned}$$

The derivative is

$$\begin{aligned} A'(\theta) &= 225 \cos \theta \cdot (1 + \cos \theta) + 225 \sin \theta \cdot (-\sin \theta) \\ &= 225 [\cos \theta + \cos^2 \theta - \sin^2 \theta] && \text{recall } \sin^2 \theta = 1 - \cos^2 \theta \\ &= 225 [\cos \theta + 2 \cos^2 \theta - 1] \end{aligned}$$

This is a continuous function, so there are no singular points. However we can still hunt for critical points by solving $A'(\theta) = 0$. That is

$$\begin{aligned} 2 \cos^2 \theta + \cos \theta - 1 &= 0 && \text{factor carefully} \\ (2 \cos \theta - 1)(\cos \theta + 1) &= 0 \end{aligned}$$

Hence we must have $\cos \theta = -1$ or $\cos \theta = \frac{1}{2}$. On the domain $0 \leq \theta \leq \pi$, this means $\theta = \pi/3$ or $\theta = \pi$.

$$\begin{aligned} A(\pi) &= 0 \\ A(\pi/3) &= 225 \sin(\pi/3)(1 + \cos(\pi/3)) \\ &= 225 \cdot \frac{\sqrt{3}}{2} \cdot \left(1 + \frac{1}{2}\right) \\ &= 225 \cdot \frac{3\sqrt{3}}{4} \approx 292.28 \end{aligned}$$

- Thus the cross-sectional area is maximised when $\theta = \frac{\pi}{3}$.

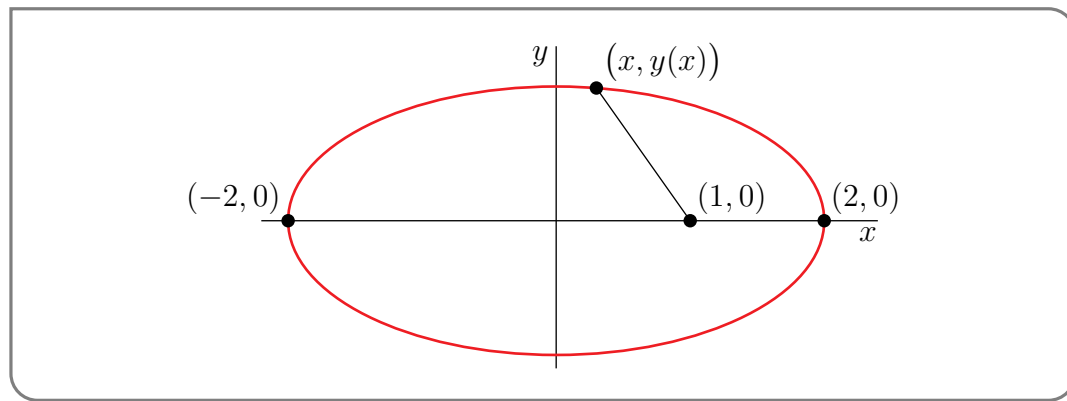
Example 3.5.20

Example 3.5.21

Find the points on the ellipse $\frac{x^2}{4} + y^2 = 1$ that are nearest to and farthest from the point $(1, 0)$.

Solution. While this is another distance problem, the possible values of x, y are bounded, so we need Corollary 3.5.13 rather than Theorem 3.5.17.

- We start by drawing a picture:



- Let ℓ be the distance from the point (x, y) on the ellipse to the point $(1, 0)$. As was the case above, we will maximise the squared-distance.

$$\ell^2 = (x - 1)^2 + y^2.$$

- Since (x, y) lies on the ellipse we have

$$\frac{x^2}{4} + y^2 = 1$$

Note that this also shows that $-2 \leq x \leq 2$ and $-1 \leq y \leq 1$.

Isolating y^2 and substituting this into our expression for ℓ^2 gives

$$\ell^2 = (x - 1)^2 + \underbrace{1 - x^2/4}_{=y^2}.$$

- Now we can apply Corollary [3.5.13](#). The endpoints of the domain give

$$\ell^2(-2) = (-2 - 1)^2 + 1 - (-2)^2/4 = 3^2 + 1 - 1 = 9$$

$$\ell^2(2) = (2 - 1)^2 + 1 - 2^2/4 = 1 + 1 - 1 = 1$$

The derivative is

$$\frac{d}{dx}\ell^2 = 2(x - 1) - x/2 = \frac{3x}{2} - 2$$

Thus there are no singular points, but there is a critical point at $x = 4/3$. The corresponding squared-distance is

$$\begin{aligned}\ell^2(4/3) &= \left(\frac{4}{3} - 1\right)^2 + 1 - \frac{(4/3)^2}{4} \\ &= (1/3)^2 + 1 - (4/9) = 6/9 = 2/3.\end{aligned}$$

- To summarise (and giving distances and coordinates of points):

x	(x, y)	ℓ
-2	$(-2, 0)$	3
$4/3$	$(4/3, \pm\sqrt{5}/3)$	$\sqrt{2/3}$
2	$(2, 0)$	1

The point of maximum distance is $(-2, 0)$, and the point of minimum distance is $(4/3, \pm\sqrt{5}/3)$.

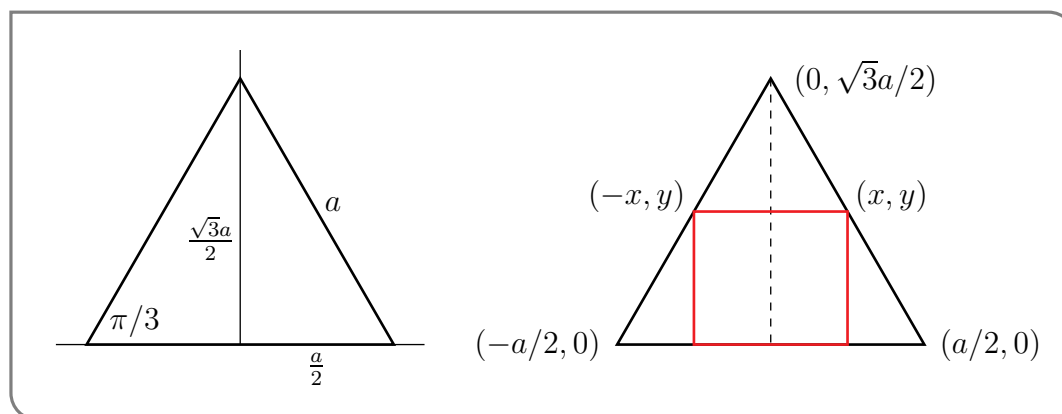
Example 3.5.21

Example 3.5.22

Find the dimensions of the rectangle of largest area that can be inscribed in an equilateral triangle of side a if one side of the rectangle lies on the base of the triangle.

Solution. Since the rectangle must sit inside the triangle, its dimensions are bounded and we will end up using Corollary 3.5.13.

- Carefully draw a picture:



We have drawn (on the left) the triangle in the xy -plane with its base on the x -axis. The base has been drawn running from $(-a/2, 0)$ to $(a/2, 0)$ so its centre lies at the origin. A little Pythagoras (or a little trigonometry) tells us that the height of the triangle is

$$\sqrt{a^2 - (a/2)^2} = \frac{\sqrt{3}}{2} \cdot a = a \cdot \sin \frac{\pi}{3}$$

Thus the vertex at the top of the triangle lies at $(0, \frac{\sqrt{3}}{2} \cdot a)$.

- If we construct a rectangle that does not touch the sides of the triangle, then we can increase the dimensions of the rectangle until it touches the triangle and so make its area larger. Thus we can assume that the two top corners of the rectangle touch the triangle as drawn in the right-hand figure above.

- Now let the rectangle be $2x$ wide and y high. And let A denote its area. Clearly

$$A = 2xy.$$

where $0 \leq x \leq a/2$ and $0 \leq y \leq \frac{\sqrt{3}}{2}a$.

- Our construction means that the top-right corner of the rectangle will have coordinates (x, y) and lie on the line joining the top vertex of the triangle at $(0, \sqrt{3}a/2)$ to the bottom-right vertex at $(a/2, 0)$. In order to write the area as a function of x alone, we need the equation for this line since it will tell us how to write y as a function of x . The line has slope

$$\text{slope} = \frac{\sqrt{3}a/2 - 0}{0 - a/2} = -\sqrt{3}.$$

and passes through the point $(0, \sqrt{3}a/2)$, so any point (x, y) on that line satisfies:

$$y = -\sqrt{3}x + \frac{\sqrt{3}}{2}a.$$

- We can now write the area as a function of x alone

$$\begin{aligned} A(x) &= 2x \left(-\sqrt{3}x + \frac{\sqrt{3}}{2}a \right) \\ &= \sqrt{3}x(a - 2x). \end{aligned}$$

with $0 \leq x \leq a/2$.

- The ends of the domain give:

$$A(0) = 0$$

$$A(a/2) = 0.$$

The derivative is

$$A'(x) = \sqrt{3}(x \cdot (-2) + 1 \cdot (a - 2x)) = \sqrt{3}(a - 4x).$$

Since this is a polynomial there are no singular points, but there is a critical point at $x = a/4$. There

$$\begin{aligned} A(a/4) &= \sqrt{3} \cdot \frac{a}{4} \cdot (a - a/2) = \sqrt{3} \cdot \frac{a^2}{8}. \\ y &= -\sqrt{3} \cdot (a/4) + \frac{\sqrt{3}}{2}a = \sqrt{3} \cdot \frac{a}{4}. \end{aligned}$$

- Checking the question again, we see that we are asked for the dimensions rather than the area, so the answer is $2x \times y$:

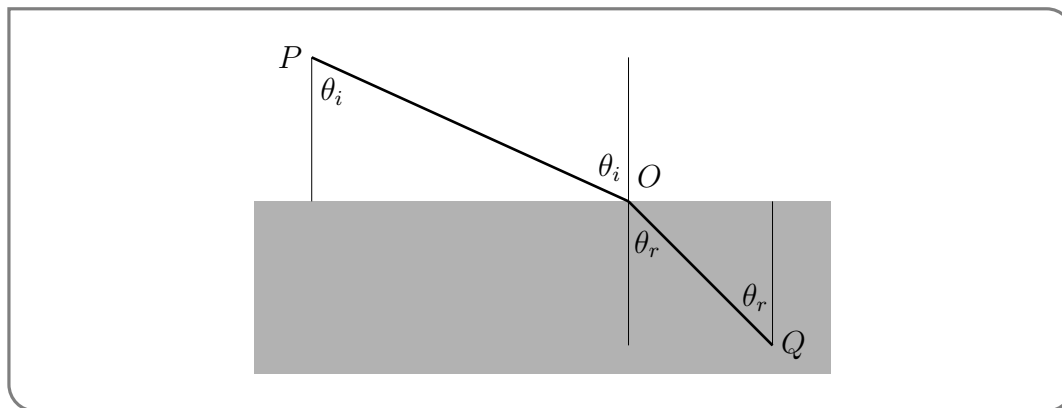
The largest such rectangle has dimensions $\frac{a}{2} \times \frac{\sqrt{3}a}{4}$.

Example 3.5.22

This next one is a good physics example. In it we will derive Snell's Law⁵⁹ from Fermat's principle⁶⁰

Example 3.5.23

Consider the figure below which shows the trajectory of a ray of light as it passes through two different mediums (say air and water).



Let c_a be the speed of light in air and c_w be the speed of light in water. Fermat's principle states that a ray of light will always travel along a path that minimises the time taken. So if a ray of light travels from P (in air) to Q (in water) then it will "choose" the point O (on the interface) so as to minimise the total time taken. Use this idea to show Snell's law,

$$\frac{\sin \theta_i}{\sin \theta_r} = \frac{c_a}{c_w}$$

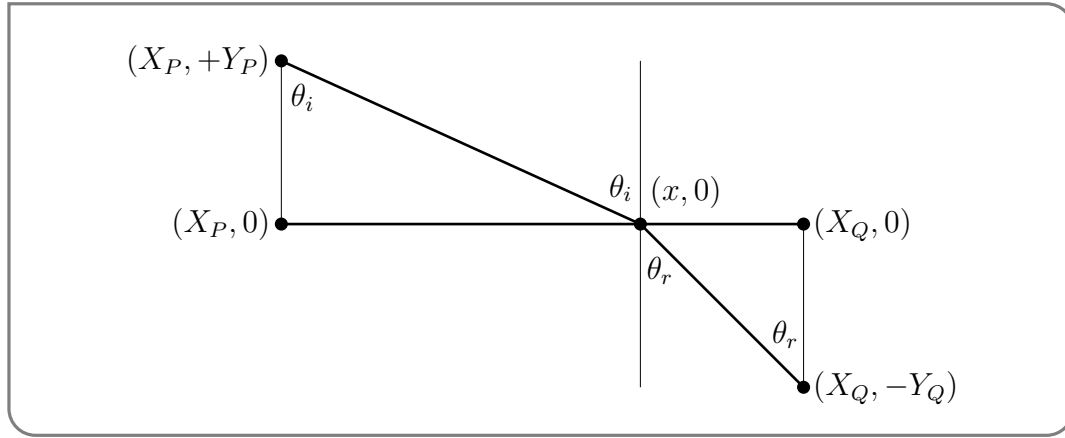
where θ_i is the angle of incidence and θ_r is the angle of refraction (as illustrated in the figure above).

Solution. This problem is a little more abstract than the others we have examined, but we can still apply Theorem 3.5.17

- We are given a figure in the statement of the problem and it contains all the relevant points and angles. However it will simplify things if we decide on a coordinate system. Let's assume that the point O lies on the x -axis, at coordinates $(x, 0)$. The point P then lies above the axis at $(X_P, +Y_P)$, while Q lies below the axis at $(X_Q, -Y_Q)$. This is drawn below.

59 Snell's law is named after the Dutch astronomer Willebrord Snellius who derived it in around 1621, though it was first stated accurately in 984 by Ibn Sahl.

60 Named after Pierre de Fermat who described it in a letter in 1662. The beginnings of the idea, however, go back as far as Hero of Alexandria in around 60CE. Hero is credited with many inventions including the first vending machine, and a precursor of the steam engine called an aeolipile.



- The statement of Snell's law contains terms $\sin \theta_i$ and $\sin \theta_r$, so it is a good idea for us to see how to express these in terms of the coordinates we have just introduced:

$$\sin \theta_i = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{(x - X_P)}{\sqrt{(X_P - x)^2 + Y_P^2}}$$

$$\sin \theta_r = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{(X_Q - x)}{\sqrt{(X_Q - x)^2 + Y_Q^2}}$$

- Let ℓ_P denote the distance PO , and ℓ_Q denote the distance OQ . Then we have

$$\ell_P = \sqrt{(X_P - x)^2 + Y_P^2}$$

$$\ell_Q = \sqrt{(X_Q - x)^2 + Y_Q^2}$$

If we then denote the total time taken by T , then

$$T = \frac{\ell_P}{c_a} + \frac{\ell_Q}{c_w} = \frac{1}{c_a} \sqrt{(X_P - x)^2 + Y_P^2} + \frac{1}{c_w} \sqrt{(X_Q - x)^2 + Y_Q^2}$$

which is written as a function of x since all the other terms are constants.

- Notice that as $x \rightarrow +\infty$ or $x \rightarrow -\infty$ the total time $T \rightarrow \infty$ and so we can apply Theorem [3.5.17](#). The derivative is

$$\frac{dT}{dx} = \frac{1}{c_a} \frac{-2(X_P - x)}{2\sqrt{(X_P - x)^2 + Y_P^2}} + \frac{1}{c_w} \frac{-2(X_Q - x)}{2\sqrt{(X_Q - x)^2 + Y_Q^2}}$$

Notice that the terms inside the square-roots cannot be zero or negative since they are both sums of squares and $Y_P, Y_Q > 0$. So there are no singular points, but there is a critical point when $T'(x) = 0$, namely when

$$\begin{aligned} 0 &= \frac{1}{c_a} \frac{X_P - x}{\sqrt{(X_P - x)^2 + Y_P^2}} + \frac{1}{c_w} \frac{X_Q - x}{\sqrt{(X_Q - x)^2 + Y_Q^2}} \\ &= \frac{-\sin \theta_i}{c_a} + \frac{\sin \theta_r}{c_w} \end{aligned}$$

Rearrange this to get

$$\frac{\sin \theta_i}{c_a} = \frac{\sin \theta_r}{c_w} \quad \text{move sines to one side}$$

$$\frac{\sin \theta_i}{\sin \theta_r} = \frac{c_a}{c_w}$$

which is exactly Snell's law.

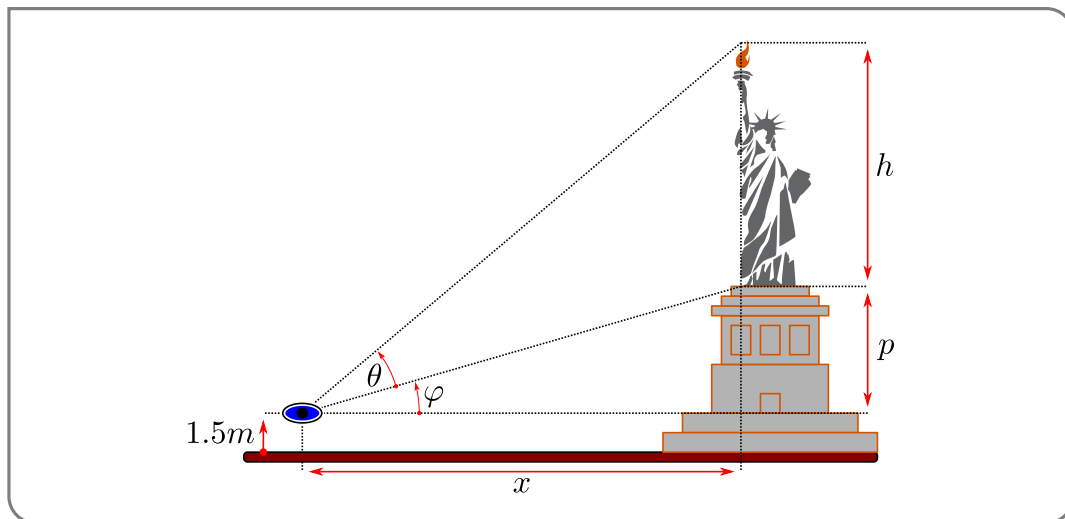
Example 3.5.23

Example 3.5.24

The Statue of Liberty has height 46m and stands on a 47m tall pedestal. How far from the statue should an observer stand to maximize the angle subtended by the statue at the observer's eye, which is 1.5m above the base of the pedestal?

Solution. Obviously if we stand too close then all the observer sees is the pedestal, while if they stand too far then everything is tiny. The best spot for taking a photograph is somewhere in between.

- Draw a careful picture⁶¹



and we can put in the relevant lengths and angles.

- The height of the statue is $h = 46\text{m}$, and the height of the pedestal (above the eye) is $p = 47 - 1.5 = 45.5\text{m}$. The horizontal distance from the statue to the eye is x . There are two relevant angles. First θ is the angle subtended by the statue, while φ is the angle subtended by the portion of the pedestal above the eye.

61 And make some healthy use of public domain clip art.

- Some trigonometry gives us

$$\begin{aligned}\tan \varphi &= \frac{p}{x} \\ \tan(\varphi + \theta) &= \frac{p+h}{x}\end{aligned}$$

Thus

$$\begin{aligned}\varphi &= \arctan \frac{p}{x} \\ \varphi + \theta &= \arctan \frac{p+h}{x}\end{aligned}$$

and so

$$\theta = \arctan \frac{p+h}{x} - \arctan \frac{p}{x}.$$

- If we allow the viewer to stand at any point in front of the statue, then $0 \leq x < \infty$. Further observe that as $x \rightarrow \infty$ or $x \rightarrow 0$ the angle $\theta \rightarrow 0$, since

$$\lim_{x \rightarrow \infty} \arctan \frac{p+h}{x} = \lim_{x \rightarrow \infty} \arctan \frac{p}{x} = 0$$

and

$$\lim_{x \rightarrow 0^+} \arctan \frac{p+h}{x} = \lim_{x \rightarrow 0^+} \arctan \frac{p}{x} = \frac{\pi}{2}$$

Clearly the largest value of θ will be strictly positive and so has to be taken for some $0 < x < \infty$. (Note the strict inequalities.) This x will be a local maximum as well as a global maximum. As θ is not singular at any $0 < x < \infty$, we need only search for critical points. A careful application of the chain rule shows that the derivative is

$$\begin{aligned}\frac{d\theta}{dx} &= \frac{1}{1 + (\frac{p+h}{x})^2} \cdot \left(\frac{-(p+h)}{x^2} \right) - \frac{1}{1 + (\frac{p}{x})^2} \cdot \left(\frac{-p}{x^2} \right) \\ &= \frac{-(p+h)}{x^2 + (p+h)^2} + \frac{p}{x^2 + p^2}\end{aligned}$$

So a critical point occurs when

$$\begin{aligned}\frac{(p+h)}{x^2 + (p+h)^2} &= \frac{p}{x^2 + p^2} && \text{cross multiply} \\ (p+h)(x^2 + p^2) &= p(x^2 + (p+h)^2) && \text{collect } x \text{ terms} \\ x^2(p+h-p) &= p(p+h)^2 - p^2(p+h) && \text{clean up} \\ hx^2 &= p(p+h)(p+h-p) = ph(p+h) && \text{cancel common factors} \\ x^2 &= p(p+h) \\ x &= \pm \sqrt{p(p+h)} \approx \pm 64.9m\end{aligned}$$

- Thus the best place to stand approximately 64.9m in front or behind the statue. At that point $\theta \approx 0.348$ radians or 19.9° .

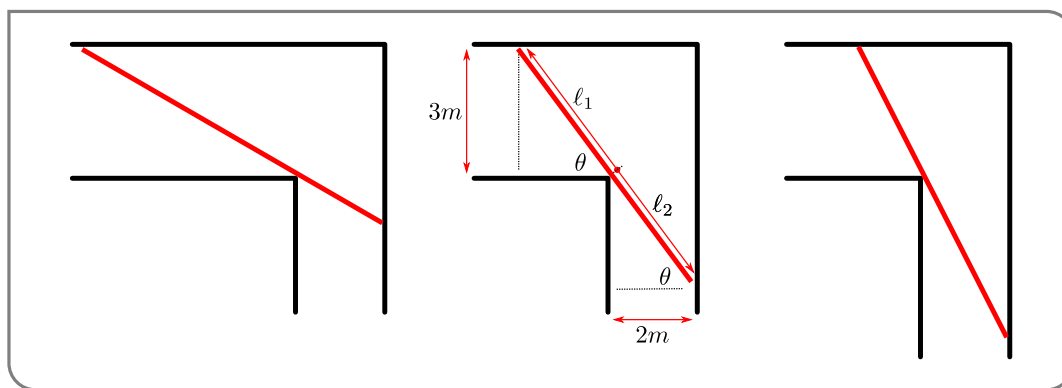
Example 3.5.24

Example 3.5.25

Find the length of the longest rod that can be carried horizontally (no tilting allowed) from a corridor 3m wide into a corridor 2m wide. The two corridors are perpendicular to each other.

Solution.

- Suppose that we are carrying the rod around the corner, then if the rod is as long as possible it must touch the corner and the outside walls of both corridors. A picture of this is shown below.



You can see that this gives rise to two similar triangles, one inside each corridor. Also the maximum length of the rod changes with the angle it makes with the walls of the corridor.

- Suppose that the angle between the rod and the inner wall of the 3m corridor is θ , as illustrated in the figure above. At the same time it will make an angle of $\frac{\pi}{2} - \theta$ with the outer wall of the 3m corridor. Denote by $\ell_1(\theta)$ the length of the part of the rod forming the hypotenuse of the upper triangle in the figure above. Similarly, denote by $\ell_2(\theta)$ the length of the part of the rod forming the hypotenuse of the lower triangle in the figure above. Then

$$\ell_1(\theta) = \frac{3}{\sin \theta} \quad \ell_2(\theta) = \frac{2}{\cos \theta}$$

and the total length is

$$\ell(\theta) = \ell_1(\theta) + \ell_2(\theta) = \frac{3}{\sin \theta} + \frac{2}{\cos \theta}$$

where $0 \leq \theta \leq \frac{\pi}{2}$.

- The length of the longest rod we can move through the corridor in this way is the minimum of $\ell(\theta)$. Notice that $\ell(\theta)$ is not defined at $\theta = 0, \frac{\pi}{2}$. Indeed we find that as $\theta \rightarrow 0^+$ or $\theta \rightarrow \frac{\pi}{2}^-$, the length $\ell \rightarrow +\infty$. (You should be able to picture what happens to our rod in those two limits). Clearly the minimum allowed $\ell(\theta)$ is going to be finite and will be achieved for some $0 < \theta < \frac{\pi}{2}$ (note the strict inequalities) and so will be a local minimum as well as a global minimum. So we only need to find zeroes of $\ell'(\theta)$. Differentiating ℓ gives

$$\frac{d\ell}{d\theta} = -\frac{3\cos\theta}{\sin^2\theta} + \frac{2\sin\theta}{\cos^2\theta} = \frac{-3\cos^3\theta + 2\sin^3\theta}{\sin^2\theta\cos^2\theta}.$$

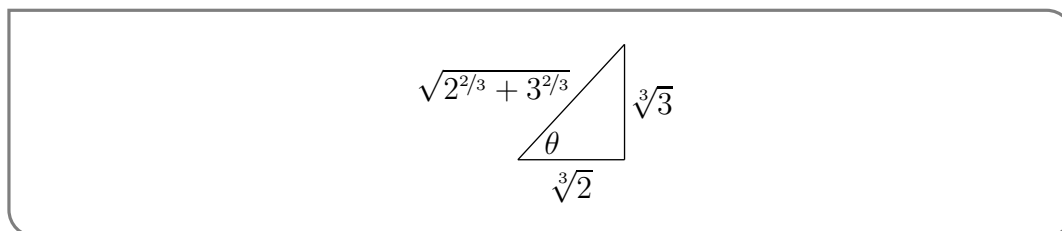
This does not exist at $\theta = 0, \frac{\pi}{2}$ (which we have already analysed) but does exist at every $0 < \theta < \frac{\pi}{2}$ and is equal to zero when the numerator is zero. Namely when

$$2\sin^3\theta = 3\cos^3\theta \quad \text{divide by } \cos^3\theta$$

$$2\tan^3\theta = 3$$

$$\tan\theta = \sqrt[3]{\frac{3}{2}}$$

- From this we can recover $\sin\theta$ and $\cos\theta$, without having to compute θ itself. We can, for example, construct a right-angle triangle with adjacent length $\sqrt[3]{2}$ and opposite length $\sqrt[3]{3}$ (so that $\tan\theta = \sqrt[3]{3/2}$):



It has hypotenuse $\sqrt{3^{2/3} + 2^{2/3}}$, and so

$$\sin\theta = \frac{3^{1/3}}{\sqrt{3^{2/3} + 2^{2/3}}}$$

$$\cos\theta = \frac{2^{1/3}}{\sqrt{3^{2/3} + 2^{2/3}}}$$

Alternatively could use the identities:

$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2\theta = \csc^2\theta$$

to obtain expressions for $1/\cos\theta$ and $1/\sin\theta$.

- Using the above expressions for $\sin \theta, \cos \theta$ we find the minimum of ℓ (which is the longest rod that we can move):

$$\begin{aligned}\ell &= \frac{3}{\sin \theta} + \frac{2}{\cos \theta} = \frac{3}{\frac{\sqrt[3]{3}}{\sqrt{2^{2/3}+3^{2/3}}}} + \frac{2}{\frac{\sqrt[3]{2}}{\sqrt{2^{2/3}+3^{2/3}}}} \\ &= \sqrt{2^{2/3}+3^{2/3}} [3^{2/3} + 2^{2/3}] \\ &= [2^{2/3} + 3^{2/3}]^{3/2} \approx 7.02\text{m}\end{aligned}$$

Example 3.5.25

3.6 ▲ Sketching Graphs

One of the most obvious applications of derivatives is to help us understand the shape of the graph of a function. In this section we will use our accumulated knowledge of derivatives to identify the most important qualitative features of graphs $y = f(x)$. The goal of this section is to highlight features of the graph $y = f(x)$ that are easily

- determined from $f(x)$ itself, and
- deduced from $f'(x)$, and
- read from $f''(x)$.

We will then use the ideas to sketch several examples.

3.6.1 ► Domain, Intercepts and Asymptotes

Given a function $f(x)$, there are several important features that we can determine from that expression before examining its derivatives.

- The domain of the function — take note of values where f does not exist. If the function is rational, look for where the denominator is zero. Similarly be careful to look for roots of negative numbers or other possible sources of discontinuities.
- Intercepts — examine where the function crosses the x -axis and the y -axis by solving $f(x) = 0$ and computing $f(0)$.
- Vertical asymptotes — look for values of x at which $f(x)$ blows up. If $f(x)$ approaches either $+\infty$ or $-\infty$ as x approaches a (or possibly as x approaches a from one side) then $x = a$ is a vertical asymptote to $y = f(x)$. When $f(x)$ is a rational function (written so that common factors are cancelled), then $y = f(x)$ has vertical asymptotes at the zeroes of the denominator.
- Horizontal asymptotes — examine the limits of $f(x)$ as $x \rightarrow +\infty$ and $x \rightarrow -\infty$. Often $f(x)$ will tend to $+\infty$ or to $-\infty$ or to a finite limit L . If, for example, $\lim_{x \rightarrow +\infty} f(x) = L$, then $y = L$ is a horizontal asymptote to $y = f(x)$ as $x \rightarrow \infty$.

Example 3.6.1

Consider the function

$$f(x) = \frac{x+1}{(x+3)(x-2)}$$

- We see that it is defined on all real numbers except $x = -3, +2$.
- Since $f(0) = -1/6$ and $f(x) = 0$ only when $x = -1$, the graph has y -intercept $(0, -1/6)$ and x -intercept $(-1, 0)$.
- Since the function is rational and its denominator is zero at $x = -3, +2$ it will have vertical asymptotes at $x = -3, +2$. To determine the shape around those asymptotes we need to examine the limits

$$\lim_{x \rightarrow -3} f(x)$$

$$\lim_{x \rightarrow 2} f(x)$$

Notice that when x is close to -3 , the factors $(x+1)$ and $(x-2)$ are both negative, so the sign of $f(x) = \frac{x+1}{x-2} \cdot \frac{1}{x+3}$ is the same as the sign of $x+3$. Hence

$$\lim_{x \rightarrow -3^+} f(x) = +\infty$$

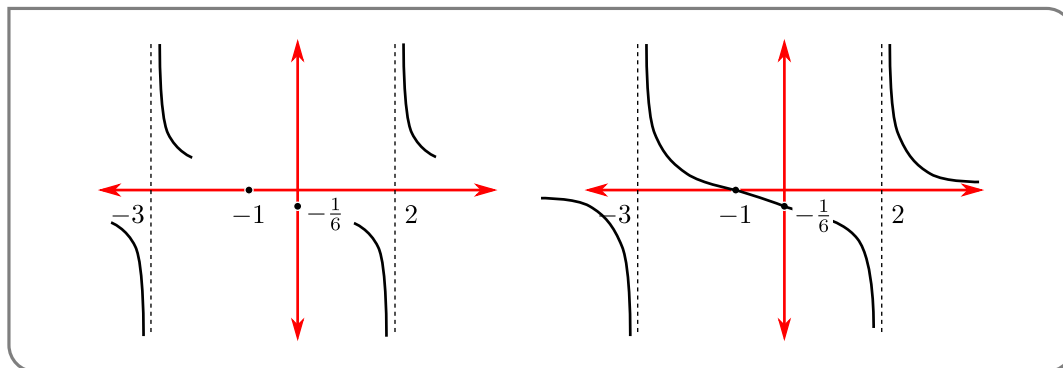
$$\lim_{x \rightarrow -3^-} f(x) = -\infty$$

A similar analysis when x is near 2 gives

$$\lim_{x \rightarrow 2^+} f(x) = +\infty$$

$$\lim_{x \rightarrow 2^-} f(x) = -\infty$$

- Finally since the numerator has degree 1 and the denominator has degree 2, we see that as $x \rightarrow \pm\infty$, $f(x) \rightarrow 0$. So $y = 0$ is a horizontal asymptote.
- Since we know the behaviour around the asymptotes and we know the locations of the intercepts (as shown in the left graph below), we can then join up the pieces and smooth them out to get the a good sketch of this function (below right).



Example 3.6.1

3.6.2 ▶ First Derivative — Increasing or Decreasing

Now we move on to the first derivative, $f'(x)$. This is a good time to revisit the mean-value theorem (Theorem 2.13.4) and some of its consequences (Corollary 2.13.11). There we considered any function $f(x)$ that is continuous on an interval $A \leq x \leq B$ and is differentiable on $A < x < B$. Then

- if $f'(x) > 0$ for all $A < x < B$, then $f(x)$ is increasing on $[A, B]$ — that is, for all $A \leq a < b \leq B$, $f(a) < f(b)$.
- if $f'(x) < 0$ for all $A < x < B$, then $f(x)$ is decreasing on $[A, B]$ — that is, for all $A \leq a < b \leq B$, $f(a) > f(b)$.

Thus the sign of the derivative indicates to us whether the function is increasing or decreasing. Further, as we discussed in Section 3.5.1, we should also examine points at which the derivative is zero — critical points — and points where the derivative does not exist. These points may indicate a local maximum or minimum.

We will now consider a function $f(x)$ that is defined on an interval I , *except possibly* at finitely many points of I . If f or its derivative f' is not defined at a point a of I , then we call a a *singular point*⁶² of f .

After studying the function $f(x)$ as described above, we should compute its derivative $f'(x)$.

- Critical points — determine where $f'(x) = 0$. At a critical point, f has a horizontal tangent.
- Singular points — determine where $f'(x)$ is not defined. If $f'(x)$ approaches $\pm\infty$ as x approaches a singular point a , then f has a vertical tangent there when f approaches a finite value as x approaches a (or possibly approaches a from one side) and a vertical asymptote when $f(x)$ approaches $\pm\infty$ as x approaches a (or possibly approaches a from one side).
- Increasing and decreasing — where is the derivative positive and where is it negative. Notice that in order for the derivative to change sign, it must either pass through zero (a critical point) or have a singular point. Thus neighbouring regions of increase and decrease will be separated by critical and singular points.

Example 3.6.2

Consider the function

$$f(x) = x^4 - 6x^3$$

- Before we move on to derivatives, let us first examine the function itself as we did above.
 - As $f(x)$ is a polynomial its domain is all real numbers.

⁶² This is the extension of the definition of “singular point” that was mentioned in the footnote in Definition 3.5.6.

- Its y -intercept is at $(0,0)$. We find its x -intercepts by factoring

$$f(x) = x^4 - 6x^3 = x^3(x - 6)$$

So it crosses the x -axis at $x = 0, 6$.

- Again, since the function is a polynomial it does not have any vertical asymptotes. And since

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} x^4(1 - 6/x) = +\infty$$

it does not have horizontal asymptotes — it blows up to $+\infty$ as x goes to $\pm\infty$.

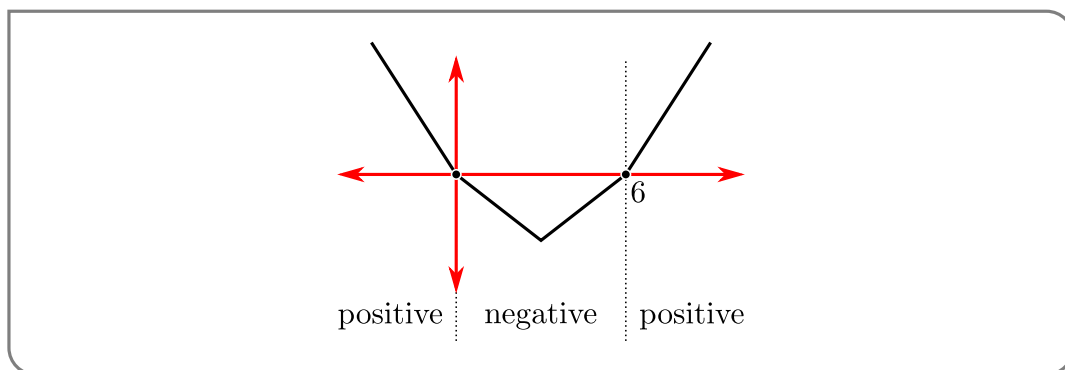
- We can also determine where the function is positive or negative since we know it is continuous everywhere and zero at $x = 0, 6$. Thus we must examine the intervals

$$(-\infty, 0) \qquad (0, 6) \qquad (6, \infty)$$

When $x < 0$, $x^3 < 0$ and $x - 6 < 0$ so $f(x) = x^3(x - 6) = (\text{negative})(\text{negative}) > 0$. Similarly when $x > 6$, $x^3 > 0$, $x - 6 > 0$ we must have $f(x) > 0$. Finally when $0 < x < 6$, $x^3 > 0$ but $x - 6 < 0$ so $f(x) < 0$. Thus

interval	$(-\infty, 0)$	0	$(0, 6)$	6	$(6, \infty)$
$f(x)$	positive	0	negative	0	positive

- Based on this information we can already construct a rough sketch.



- Now we compute its derivative

$$f'(x) = 4x^3 - 18x^2 = 2x^2(2x - 9)$$

- Since the function is a polynomial, it does not have any singular points, but it does have two critical points at $x = 0, 9/2$. These two critical points split the real line into 3 open intervals

$$(-\infty, 0) \qquad (0, 9/2) \qquad (9/2, \infty)$$

We need to determine the sign of the derivative in each intervals.

- When $x < 0$, $x^2 > 0$ but $(2x - 9) < 0$, so $f'(x) < 0$ and the function is decreasing.
- When $0 < x < 9/2$, $x^2 > 0$ but $(2x - 9) < 0$, so $f'(x) < 0$ and the function is still decreasing.
- When $x > 9/2$, $x^2 > 0$ and $(2x - 9) > 0$, so $f'(x) > 0$ and the function is increasing.

We can then summarise this in the following table

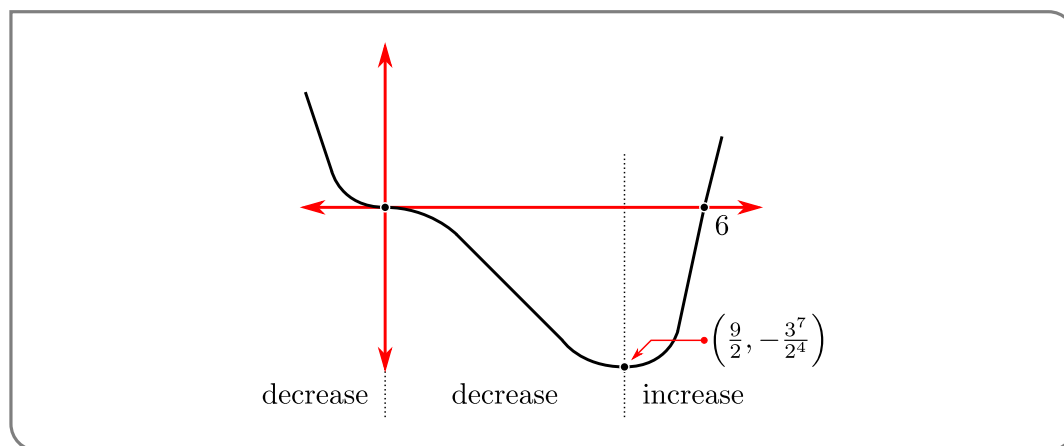
interval	$(-\infty, 0)$	0	$(0, 9/2)$	$9/2$	$(9/2, \infty)$
$f'(x)$	negative	0	negative	0	positive
	decreasing	horizontal tangent	decreasing	minimum	increasing

Since the derivative changes sign from negative to positive at the critical point $x = 9/2$, this point is a minimum. Its y -value is

$$\begin{aligned}
 y = f(9/2) &= \frac{9^3}{2^3} \left(\frac{9}{2} - 6 \right) \\
 &= \frac{3^6}{2^3} \cdot \left(\frac{-3}{2} \right) = -\frac{3^7}{2^4}
 \end{aligned}$$

On the other hand, at $x = 0$ the derivative does not change sign; while this point has a horizontal tangent line it is not a minimum or maximum.

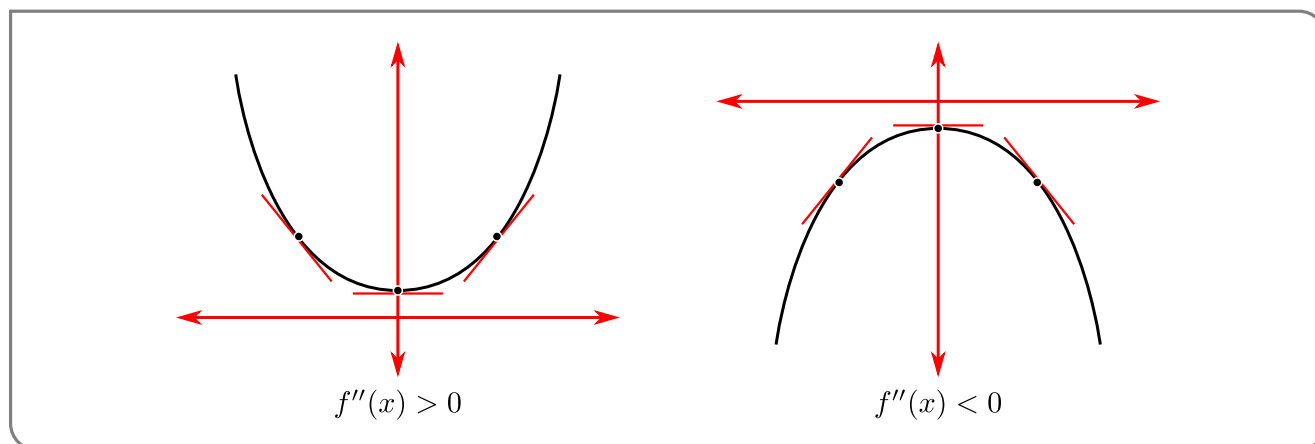
- Putting this information together we arrive at a quite reasonable sketch.



To improve upon this further we will examine the second derivative.

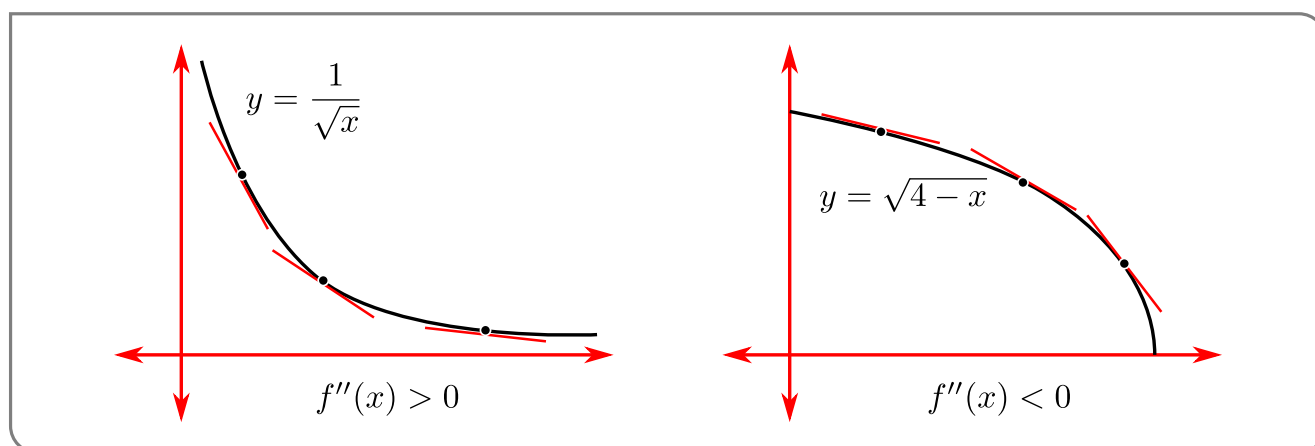
3.6.3 ► Second Derivative — Concavity

The second derivative $f''(x)$ tells us the rate at which the derivative changes. Perhaps the easiest way to understand how to interpret the sign of the second derivative is to think about what it implies about the slope of the tangent line to the graph of the function. Consider the following sketches of $y = 1 + x^2$ and $y = -1 - x^2$.



- In the case of $y = f(x) = 1 + x^2$, $f''(x) = 2 > 0$. Notice that this means the slope, $f'(x)$, of the line tangent to the graph at x increases as x increases. Looking at the figure on the left above, we see that the graph always lies above the tangent lines.
- For $y = f(x) = -1 - x^2$, $f''(x) = -2 < 0$. The slope, $f'(x)$, of the line tangent to the graph at x decreases as x increases. Looking at the figure on the right above, we see that the graph always lies below the tangent lines.

Similarly consider the following sketches of $y = x^{-1/2}$ and $y = \sqrt{4 - x}$:



Both of their derivatives, $-\frac{1}{2}x^{-3/2}$ and $-\frac{1}{2}(4 - x)^{-1/2}$, are negative, so they are decreasing functions. Examining second derivatives shows some differences.

- For the first function, $y''(x) = \frac{3}{4}x^{-5/2} > 0$, so the slopes of tangent lines are increasing with x and the graph lies above its tangent lines.

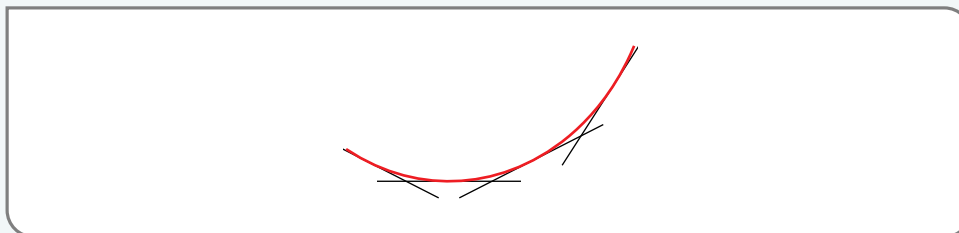
- However, the second function has $y''(x) = -\frac{1}{4}(4-x)^{-3/2} < 0$ so the slopes of the tangent lines are decreasing with x and the graph lies below its tangent lines.

More generally

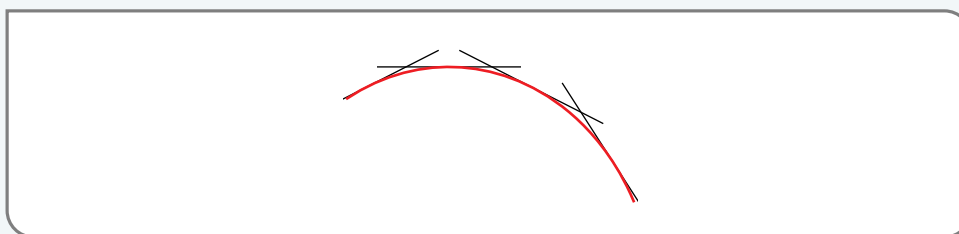
Definition 3.6.3.

Let $f(x)$ be a continuous function on the interval $[a, b]$ and suppose its first and second derivatives exist on that interval.

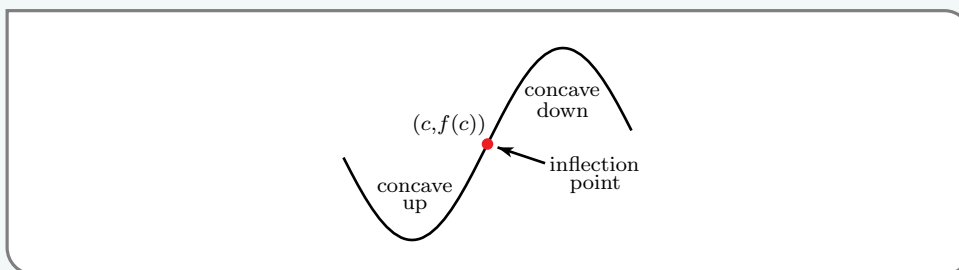
- If $f''(x) > 0$ for all $a < x < b$, then the graph of f lies above its tangent lines for $a < x < b$ and it is said to be concave up.



- If $f''(x) < 0$ for all $a < x < b$, then the graph of f lies below its tangent lines for $a < x < b$ and it is said to be concave down.



- If $f''(c) = 0$ for some $a < c < b$, and the concavity of f changes across $x = c$, then we call $(c, f(c))$ an inflection point.



Note that one might also see the terms

- “convex” or “convex up” used in place of “concave up”, and
- “concave” or “convex down” used to mean “concave down”.

To avoid confusion we recommend the reader stick with the terms “concave up” and “concave down”.

Let’s now continue Example 3.6.2 by discussing the concavity of the curve.

Example 3.6.4 (Continuation of Example 3.6.2)

Consider again the function

$$f(x) = x^4 - 6x^3$$

- Its first derivative is $f'(x) = 4x^3 - 18x^2$, so

$$f''(x) = 12x^2 - 36x = 12x(x - 3)$$

- Thus the second derivative is zero (and potentially changes sign) at $x = 0, 3$. Thus we should consider the sign of the second derivative on the following intervals

$(-\infty, 0)$

$(0, 3)$

$(3, \infty)$

A little algebra gives us

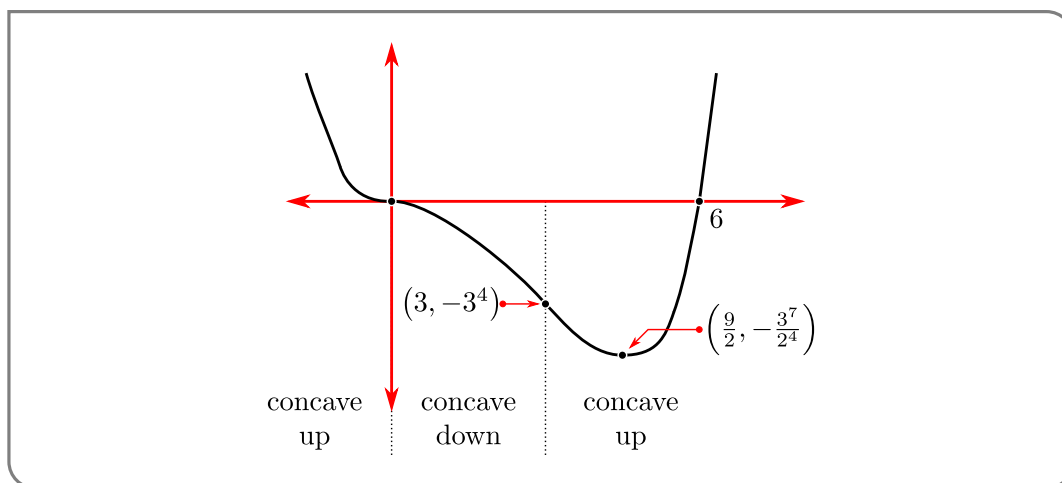
interval	$(-\infty, 0)$	0	$(0, 3)$	3	$(3, \infty)$
$f''(x)$	positive	0	negative	0	positive
concavity	up	inflection	down	inflection	up

Since the concavity changes at both $x = 0$ and $x = 3$, the following are inflection points

$(0, 0)$

$(3, 3^4 - 6 \times 3^3) = (3, -3^4)$

- Putting this together with the information we obtained earlier gives us the following sketch



Example 3.6.4

Example 3.6.5 (Optional — $y = x^{1/3}$ and $y = x^{2/3}$)

In our Definition 3.6.3, concerning concavity and inflection points, we considered only functions having first and second derivatives on the entire interval of interest. In this example, we will consider the functions

$$f(x) = x^{1/3} \quad g(x) = x^{2/3}$$

We shall see that $x = 0$ is a singular point for both of those functions. There is no universal agreement as to precisely when a singular point should also be called an inflection point. We choose to extend our definition of inflection point in Definition 3.6.3 as follows. If

- the function $f(x)$ is defined and continuous on an interval $a < x < b$ and if
- the first and second derivatives $f'(x)$ and $f''(x)$ exist on $a < x < b$ except possibly at the single point $a < c < b$ and if
- f is concave up on one side of c and is concave down on the other side of c

then we say that $(c, f(c))$ is an inflection point of $y = f(x)$. Now let's check out $y = f(x)$ and $y = g(x)$ from this point of view.

(1) Features of $y = f(x)$ and $y = g(x)$ that are read off of $f(x)$ and $g(x)$:

- Since $f(0) = 0^{1/3} = 0$ and $g(0) = 0^{2/3} = 0$, the origin $(0, 0)$ lies on both $y = f(x)$ and $y = g(x)$.
- For example, $1^3 = 1$ and $(-1)^3 = -1$ so that the cube root of 1 is $1^{1/3} = 1$ and the cube root of -1 is $(-1)^{1/3} = -1$. In general,

$$x^{1/3} \begin{cases} < 0 & \text{if } x < 0 \\ = 0 & \text{if } x = 0 \\ > 0 & \text{if } x > 0 \end{cases}$$

Consequently the graph $y = f(x) = x^{1/3}$ lies below the x -axis when $x < 0$ and lies above the x -axis when $x > 0$. On the other hand, the graph $y = g(x) = x^{2/3} = [x^{1/3}]^2$ lies on or above the x -axis for all x .

- As $x \rightarrow +\infty$, both $y = f(x) = x^{1/3}$ and $y = g(x) = x^{2/3}$ tend to $+\infty$.
- As $x \rightarrow -\infty$, $y = f(x) = x^{1/3}$ tends to $-\infty$ and $y = g(x) = x^{2/3}$ tends to $+\infty$.

(2) Features of $y = f(x)$ and $y = g(x)$ that are read off of $f'(x)$ and $g'(x)$:

$$f'(x) = \begin{cases} \frac{1}{3}x^{-2/3} & \text{if } x \neq 0 \\ \text{undefined} & \text{if } x = 0 \end{cases} \implies f'(x) > 0 \text{ for all } x \neq 0$$

$$g'(x) = \begin{cases} \frac{2}{3}x^{-1/3} & \text{if } x \neq 0 \\ \text{undefined} & \text{if } x = 0 \end{cases} \implies g'(x) \begin{cases} < 0 & \text{if } x < 0 \\ > 0 & \text{if } x > 0 \end{cases}$$

So the graph $y = f(x)$ is increasing on both sides of the singular point $x = 0$, while the graph $y = g(x)$ is decreasing to the left of $x = 0$ and is increasing to the right of $x = 0$. As $x \rightarrow 0$, $f'(x)$ and $g'(x)$ become infinite. That is, the slopes of the tangent lines at $(x, f(x))$ and $(x, g(x))$ become infinite and the tangent lines become vertical.

(3) Features of $y = f(x)$ and $y = g(x)$ that are read off of $f''(x)$ and $g''(x)$:

$$f''(x) = \begin{cases} -\frac{2}{9}x^{-5/3} = -\frac{2}{9}[x^{-1/3}]^5 & \text{if } x \neq 0 \\ \text{undefined} & \text{if } x = 0 \end{cases} \Rightarrow f''(x) \begin{cases} > 0 & \text{if } x < 0 \\ < 0 & \text{if } x > 0 \end{cases}$$

$$g''(x) = \begin{cases} -\frac{2}{9}x^{-4/3} = -\frac{2}{9}[x^{-1/3}]^4 & \text{if } x \neq 0 \\ \text{undefined} & \text{if } x = 0 \end{cases} \Rightarrow g''(x) < 0 \text{ for all } x \neq 0$$

So the graph $y = g(x)$ is concave down on both sides of the singular point $x = 0$, while the graph $y = f(x)$ is concave up to the left of $x = 0$ and is concave down to the right of $x = 0$.

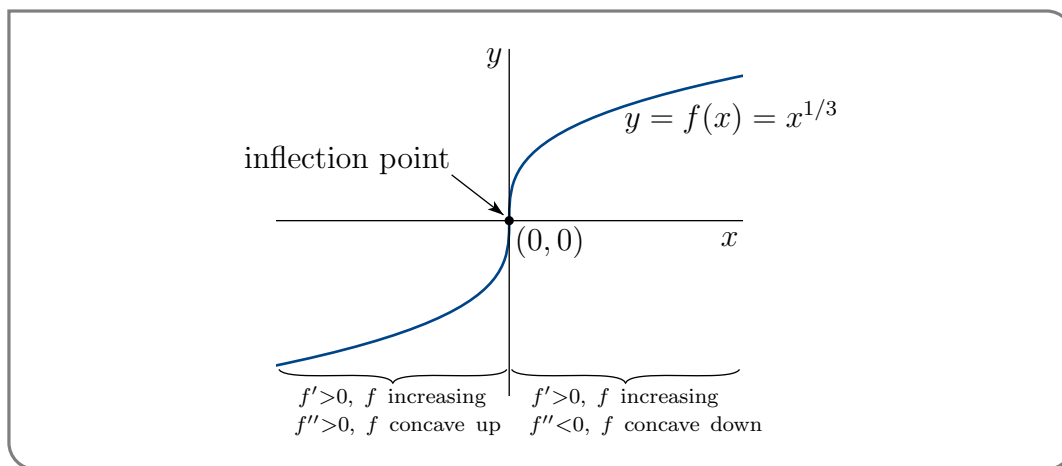
By way of summary, we have, for $f(x)$,

interval	$(-\infty, 0)$	0	$(0, \infty)$
$f(x)$	negative	0	positive
$f'(x)$	positive	undefined	positive
	increasing		increasing
$f''(x)$	positive	undefined	negative
	concave up	inflection	concave down

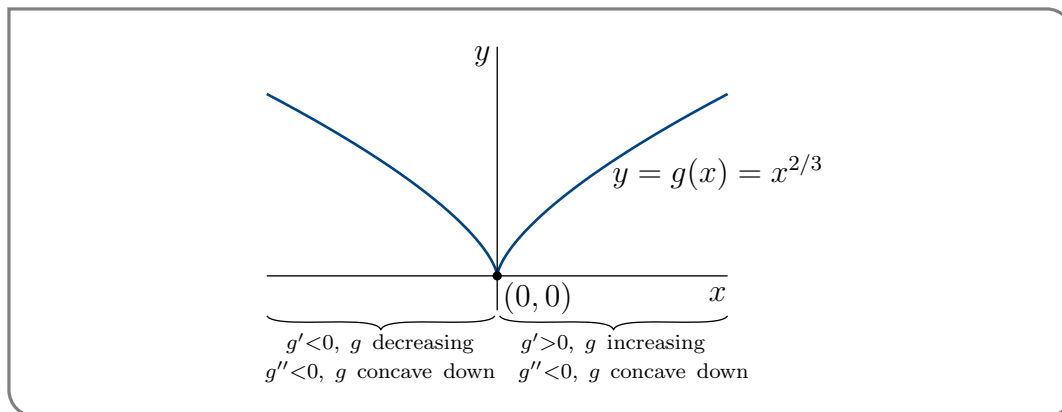
and for $g(x)$,

interval	$(-\infty, 0)$	0	$(0, \infty)$
$g(x)$	positive	0	positive
$g'(x)$	negative	undefined	positive
	decreasing		increasing
$g''(x)$	negative	undefined	negative
	concave down		concave down

Since the concavity changes at $x = 0$ for $y = f(x)$, but not for $y = g(x)$, $(0, 0)$ is an inflection point for $y = f(x)$, but not for $y = g(x)$. We have the following sketch for $y = f(x) = x^{1/3}$,



and the following sketch for $y = g(x) = x^{2/3}$.



Note that the curve $y = f(x) = x^{1/3}$ looks perfectly smooth, even though $f'(x) \rightarrow \infty$ as $x \rightarrow 0$. There is no kink or discontinuity at $(0,0)$. The singularity at $x = 0$ has caused the y -axis to be a vertical tangent to the curve, but has not prevented the curve from looking smooth.

Example 3.6.5

3.6.4 ► Symmetries

Before we proceed to some examples, we should examine some simple symmetries possessed by some functions. We'll look at three symmetries — evenness, oddness and periodicity. If a function possesses one of these symmetries then it can be exploited to reduce the amount of work required to sketch the graph of the function.

Let us start with even and odd functions.

Definition 3.6.6.

A function $f(x)$ is said to be even if $f(-x) = f(x)$ for all x .

Definition 3.6.7.

A function $f(x)$ is said to be odd if $f(-x) = -f(x)$ for all x .

Example 3.6.8

Let $f(x) = x^2$ and $g(x) = x^3$. Then

$$f(-x) = (-x)^2 = x^2 = f(x)$$

$$g(-x) = (-x)^3 = -x^3 = -g(x)$$

Hence $f(x)$ is even and $g(x)$ is odd.

Notice any polynomial involving only even powers of x will be even

$$\begin{aligned} f(x) &= 7x^6 + 2x^4 - 3x^2 + 5 && \text{remember that } 5 = 5x^0 \\ f(-x) &= 7(-x)^6 + 2(-x)^4 - 3(-x)^2 + 5 \\ &= 7x^6 + 2x^4 - 3x^2 + 5 = f(x) \end{aligned}$$

Similarly any polynomial involving only odd powers of x will be odd

$$\begin{aligned} g(x) &= 2x^5 - 8x^3 - 3x \\ g(-x) &= 2(-x)^5 - 8(-x)^3 - 3(-x) \\ &= -2x^5 + 8x^3 + 3x = -g(x) \end{aligned}$$

Example 3.6.8

Not all even and odd functions are polynomials. For example

$$|x| \qquad \cos x \qquad \text{and } (e^x + e^{-x})$$

are all even, while

$$\sin x \qquad \tan x \qquad \text{and } (e^x - e^{-x})$$

are all odd. Indeed, given any function $f(x)$, the function

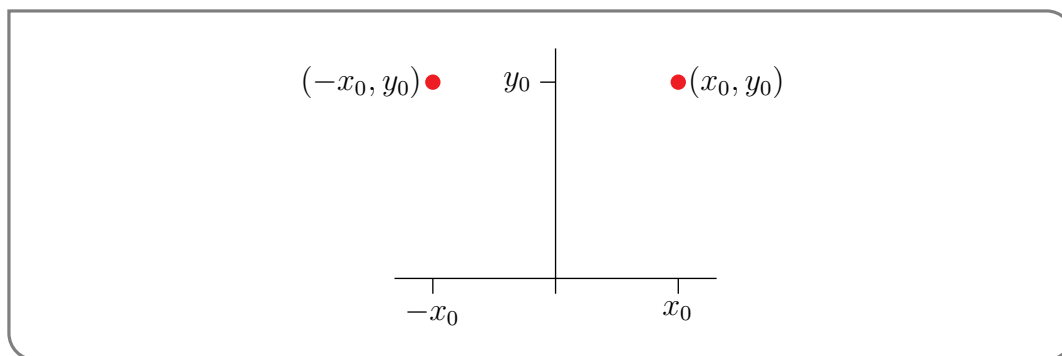
$$\begin{aligned} g(x) &= f(x) + f(-x) && \text{will be even, and} \\ h(x) &= f(x) - f(-x) && \text{will be odd.} \end{aligned}$$

Now let us see how we can make use of these symmetries to make graph sketching easier. Let $f(x)$ be an even function. Then

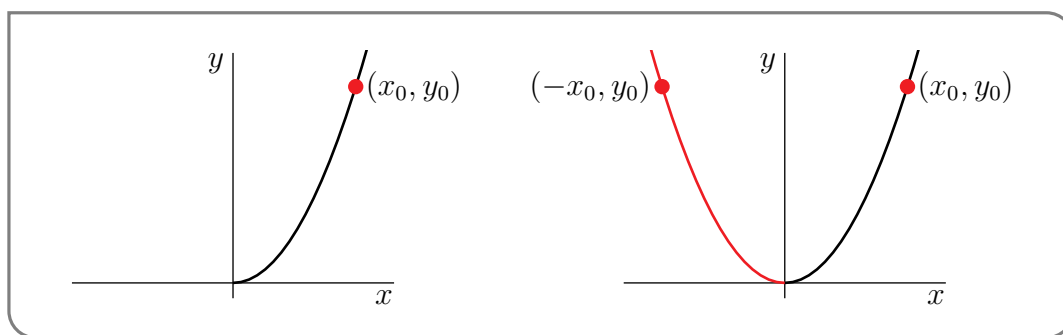
the point (x_0, y_0) lies on the graph of $y = f(x)$

if and only if $y_0 = f(x_0) = f(-x_0)$ which is the case if and only if

the point $(-x_0, y_0)$ lies on the graph of $y = f(x)$.



Notice that the points (x_0, y_0) and $(-x_0, y_0)$ are just reflections of each other across the y -axis. Consequently, to draw the graph $y = f(x)$, it suffices to draw the part of the graph with $x \geq 0$ and then reflect it in the y -axis. Here is an example. The part with $x \geq 0$ is on the left and the full graph is on the right.

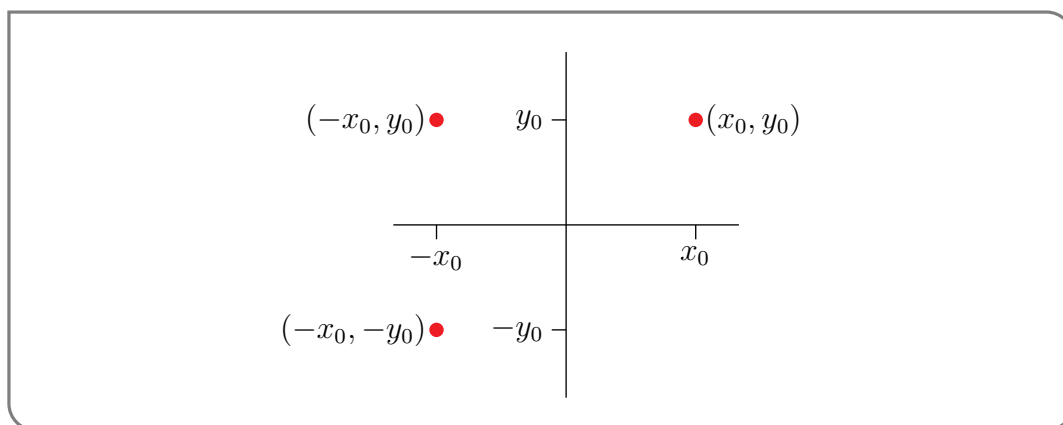


Very similarly, when $f(x)$ is an odd function then

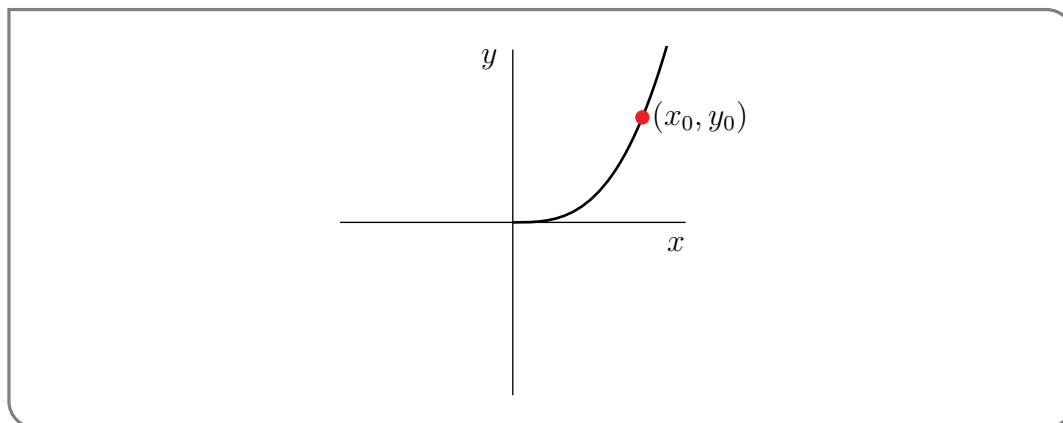
(x_0, y_0) lies on the graph of $y = f(x)$

if and only if

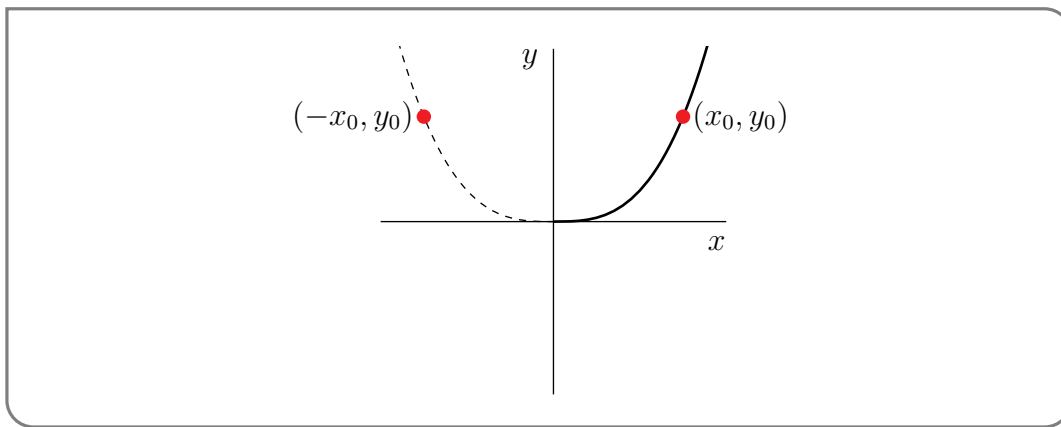
$(-x_0, -y_0)$ lies on the graph of $y = f(x)$



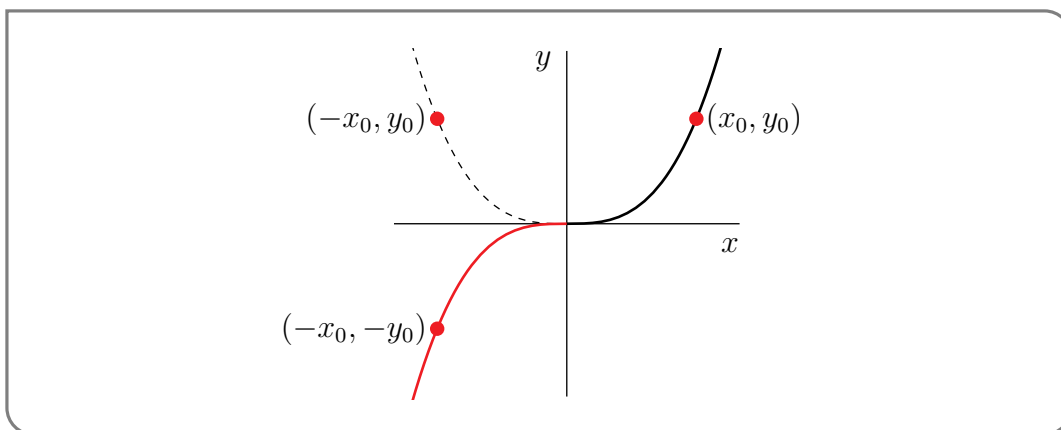
Now the symmetry is a little harder to interpret pictorially. To get from (x_0, y_0) to $(-x_0, -y_0)$ one can first reflect (x_0, y_0) in the y -axis to get to $(-x_0, y_0)$ and then reflect the result in the x -axis to get to $(-x_0, -y_0)$. Consequently, to draw the graph $y = f(x)$, it suffices to draw the part of the graph with $x \geq 0$ and then reflect it first in the y -axis and then in the x -axis. Here is an example. First, here is the part of the graph with $x \geq 0$.



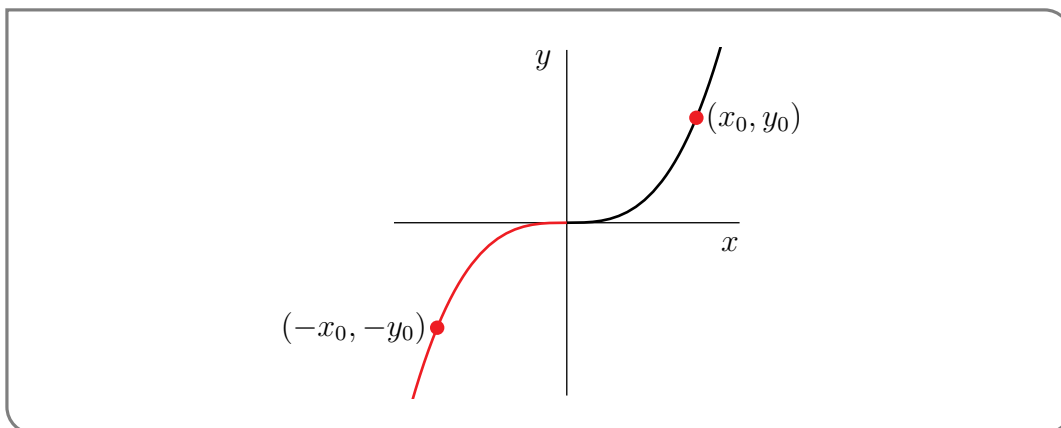
Next, as an intermediate step (usually done in our heads rather than on paper), we add in the reflection in the y -axis.



Finally to get the full graph, we reflect the dashed line in the x -axis



and then remove the dashed line.



Let's do a more substantial example of an even function

Example 3.6.9

Consider the function

$$g(x) = \frac{x^2 - 9}{x^2 + 3}$$

- The function is even since

$$g(-x) = \frac{(-x)^2 - 9}{(-x)^2 + 3} = \frac{x^2 - 9}{x^2 + 3} = g(x)$$

Thus it suffices to study the function for $x \geq 0$ because we can then use the even symmetry to understand what happens for $x < 0$.

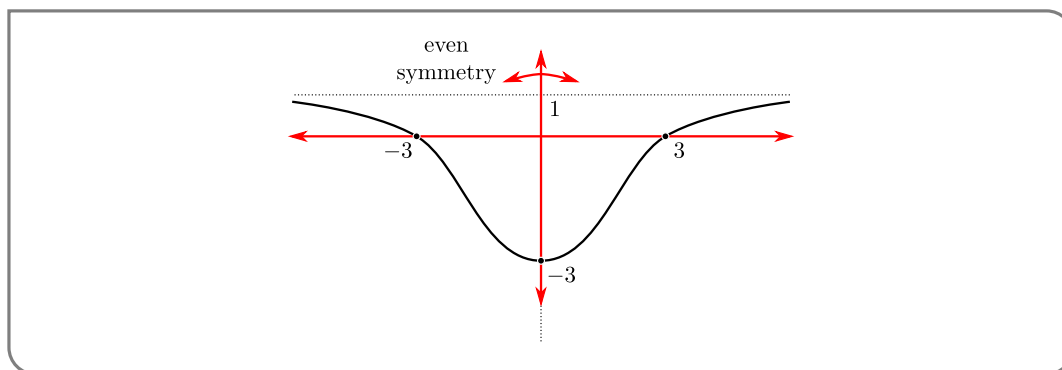
- The function is defined on all real numbers since its denominator $x^2 + 3$ is never zero. Hence it has no vertical asymptotes.
- The y -intercept is $g(0) = \frac{-9}{3} = -3$. And x -intercepts are given by the solution of $x^2 - 9 = 0$, namely $x = \pm 3$. Note that we only need to establish $x = 3$ as an intercept. Then since g is even, we know that $x = -3$ is also an intercept.
- To find the horizontal asymptotes we compute the limit as $x \rightarrow +\infty$

$$\begin{aligned} \lim_{x \rightarrow \infty} g(x) &= \lim_{x \rightarrow \infty} \frac{x^2 - 9}{x^2 + 3} \\ &= \lim_{x \rightarrow \infty} \frac{x^2(1 - 9/x^2)}{x^2(1 + 3/x^2)} \\ &= \lim_{x \rightarrow \infty} \frac{1 - 9/x^2}{1 + 3/x^2} = 1 \end{aligned}$$

Thus $y = 1$ is a horizontal asymptote. Indeed, this is also the asymptote as $x \rightarrow -\infty$ since by the even symmetry

$$\lim_{x \rightarrow -\infty} g(x) = \lim_{x \rightarrow \infty} g(-x) = \lim_{x \rightarrow \infty} g(x).$$

- We can already produce a quite reasonable sketch just by putting in the horizontal asymptote and the intercepts and drawing a smooth curve between them.



Note that we have drawn the function as never crossing the asymptote $y = 1$, however we have not yet proved that. We could by trying to solve $g(x) = 1$.

$$\begin{aligned} \frac{x^2 - 9}{x^2 + 3} &= 1 \\ x^2 - 9 &= x^2 + 3 \\ -9 &= 3 \text{ so no solutions.} \end{aligned}$$

Alternatively we could analyse the first derivative to see how the function approaches the asymptote.

- Now we turn to the first derivative:

$$\begin{aligned} g'(x) &= \frac{(x^2 + 3)(2x) - (x^2 - 9)(2x)}{(x^2 + 3)^2} \\ &= \frac{24x}{(x^2 + 3)^2} \end{aligned}$$

There are no singular points since the denominator is nowhere zero. The only critical point is at $x = 0$. Thus we must find the sign of $g'(x)$ on the intervals

$$(-\infty, 0) \qquad (0, \infty)$$

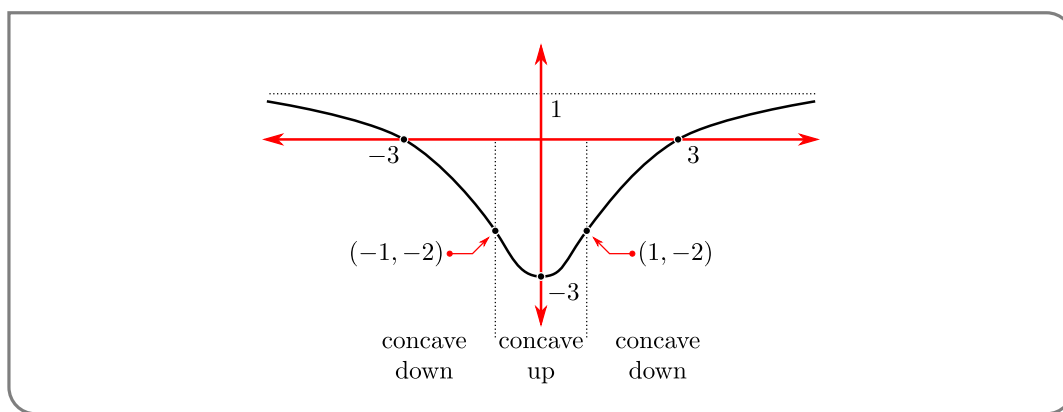
- When $x > 0$, $24x > 0$ and $(x^2 + 3) > 0$, so $g'(x) > 0$ and the function is increasing. By even symmetry we know that when $x < 0$ the function must be decreasing. Hence the critical point $x = 0$ is a local minimum of the function.
- Notice that since the function is increasing for $x > 0$ and the function must approach the horizontal asymptote $y = 1$ from below. Thus the sketch above is quite accurate.
- Now consider the second derivative:

$$\begin{aligned} g''(x) &= \frac{d}{dx} \frac{24x}{(x^2 + 3)^2} \\ &= \frac{(x^2 + 3)^2 \cdot 24 - 24x \cdot 2(x^2 + 3) \cdot 2x}{(x^2 + 3)^4} && \text{cancel a factor of } (x^2 + 3) \\ &= \frac{(x^2 + 3) \cdot 24 - 96x^2}{(x^2 + 3)^3} \\ &= \frac{72(1 - x^2)}{(x^2 + 3)^3} \end{aligned}$$

- It is clear that $g''(x) = 0$ when $x = \pm 1$. Note that, again, we can infer the zero at $x = -1$ from the zero at $x = 1$ by the even symmetry. Thus we need to examine the sign of $g''(x)$ the intervals

$$(-\infty, -1) \qquad (-1, 1) \qquad (1, \infty)$$

- When $|x| < 1$ we have $(1 - x^2) > 0$ so that $g''(x) > 0$ and the function is concave up. When $|x| > 1$ we have $(1 - x^2) < 0$ so that $g''(x) < 0$ and the function is concave down. Thus the points $x = \pm 1$ are inflection points. Their coordinates are $(\pm 1, g(\pm 1)) = (\pm 1, -2)$.
- Putting this together gives the following sketch:



Example 3.6.9

Another symmetry we should consider is periodicity.

Definition 3.6.10.

A function $f(x)$ is said to be periodic, with period $P > 0$, if $f(x + P) = f(x)$ for all x .

Note that if $f(x + P) = f(x)$ for all x , then replacing x by $x + P$, we have

$$f(x + 2P) = f(x + P + P) = f(x + P) = f(x).$$

More generally $f(x + kP) = f(x)$ for all integers k . Thus if f has period P , then it also has period nP for all natural numbers n . The smallest period is called the fundamental period.

Example 3.6.11

The classic example of a periodic function is $f(x) = \sin x$, which has period 2π since $f(x + 2\pi) = \sin(x + 2\pi) = \sin x = f(x)$.

Example 3.6.11

If $f(x)$ has period P then

$$(x_0, y_0) \text{ lies on the graph of } y = f(x)$$

if and only if $y_0 = f(x_0) = f(x_0 + P)$ which is the case if and only if

$$(x_0 + P, y_0) \text{ lies on the graph of } y = f(x)$$

and, more generally,

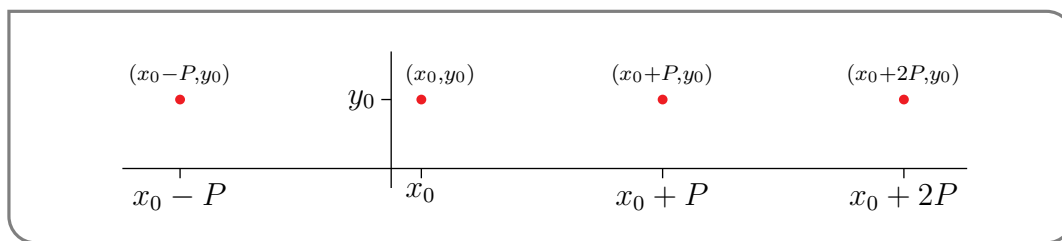
$$(x_0, y_0) \text{ lies on the graph of } y = f(x)$$

if and only if

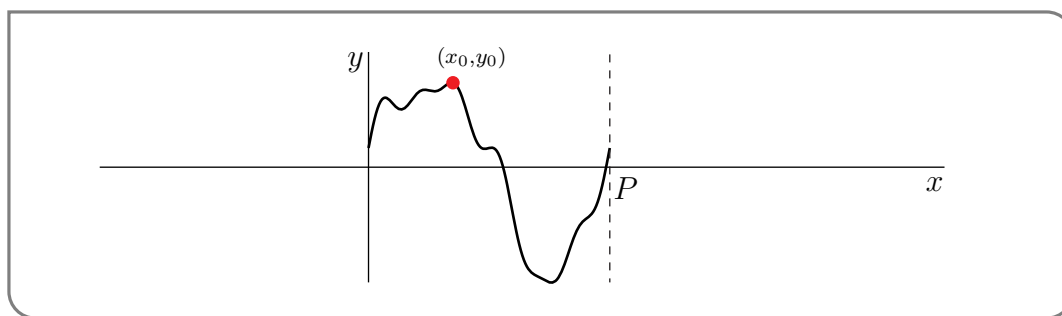
$$(x_0 + nP, y_0) \text{ lies on the graph of } y = f(x)$$

for all integers n .

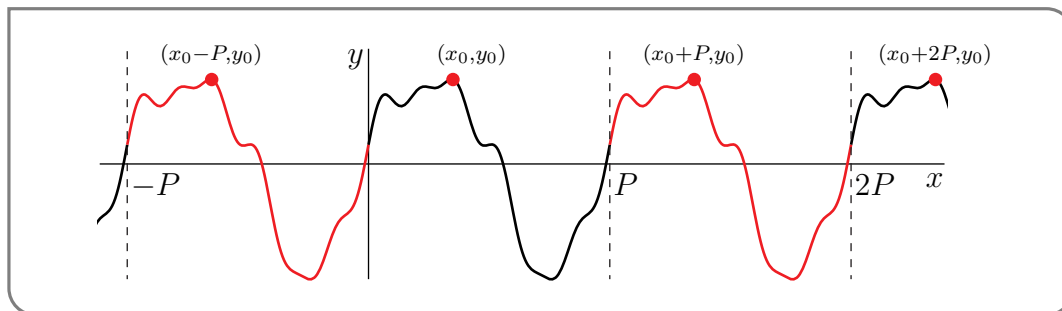
Note that the point $(x_0 + P, y_0)$ can be obtained by translating (x_0, y_0) horizontally by P . Similarly the point $(x_0 + nP, y_0)$ can be found by repeatedly translating (x_0, y_0) horizontally by P .



Consequently, to draw the graph $y = f(x)$, it suffices to draw one period of the graph, say the part with $0 \leq x \leq P$, and then translate it repeatedly. Here is an example. Here is a sketch of one period



and here is the full sketch.



3.6.5 ▶ A Checklist for Sketching

Above we have described how we can use our accumulated knowledge of derivatives to quickly identify the most important qualitative features of graphs $y = f(x)$. Here we give the reader a quick checklist of things to examine in order to produce an accurate sketch based on properties that are easily read off from $f(x)$, $f'(x)$ and $f''(x)$.

▶▶▶ A Sketching Checklist.

(1) Features of $y = f(x)$ that are read off of $f(x)$:

- First check where $f(x)$ is defined. Then

- $y = f(x)$ is plotted only for x 's in the domain of $f(x)$, i.e. where $f(x)$ is defined.
- $y = f(x)$ has vertical asymptotes at the points where $f(x)$ blows up to $\pm\infty$.
- Next determine whether the function is even, odd, or periodic.
- $y = f(x)$ is first plotted for $x \geq 0$ if the function is even or odd. The rest of the sketch is then created by reflections.
- $y = f(x)$ is first plotted for a single period if the function is periodic. The rest of the sketch is then created by translations.
- Next compute $f(0)$, $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$ and look for solutions to $f(x) = 0$ that you can easily find. Then
- $y = f(x)$ has y -intercept $(0, f(0))$.
- $y = f(x)$ has x -intercept $(a, 0)$ whenever $f(a) = 0$
- $y = f(x)$ has horizontal asymptote $y = Y$ if $\lim_{x \rightarrow \infty} f(x) = L$ or $\lim_{x \rightarrow -\infty} f(x) = L$.

(2) Features of $y = f(x)$ that are read off of $f'(x)$:

- Compute $f'(x)$ and determine its critical points and singular points, then
- $y = f(x)$ has a horizontal tangent at the points where $f'(x) = 0$.
- $y = f(x)$ is increasing at points where $f'(x) > 0$.
- $y = f(x)$ is decreasing at points where $f'(x) < 0$.
- $y = f(x)$ has vertical tangents or vertical asymptotes at the points where $f'(x) = \pm\infty$.

(3) Features of $y = f(x)$ that are read off of $f''(x)$:

- Compute $f''(x)$ and determine where $f''(x) = 0$ or does not exist, then
- $y = f(x)$ is concave up at points where $f''(x) > 0$.
- $y = f(x)$ is concave down at points where $f''(x) < 0$.
- $y = f(x)$ may or may not have inflection points where $f''(x) = 0$.

3.6.6 ► Sketching Examples

Example 3.6.12 (Sketch $f(x) = x^3 - 3x + 1$)

(1) Reading from $f(x)$:

- The function is a polynomial so it is defined everywhere.
- Since $f(-x) = -x^3 + 3x + 1 \neq \pm f(x)$, it is not even or odd. Nor is it periodic.

- The y -intercept is $y = 1$. The x -intercepts are not easily computed since it is a cubic polynomial that does not factor nicely⁶³. So for this example we don't worry about finding them.
- Since it is a polynomial it has no vertical asymptotes.
- For very large x , both positive and negative, the x^3 term in $f(x)$ dominates the other two terms so that

$$f(x) \rightarrow \begin{cases} +\infty & \text{as } x \rightarrow +\infty \\ -\infty & \text{as } x \rightarrow -\infty \end{cases}$$

and there are no horizontal asymptotes.

(2) We now compute the derivative:

$$f'(x) = 3x^2 - 3 = 3(x^2 - 1) = 3(x + 1)(x - 1)$$

- The critical points (where $f'(x) = 0$) are at $x = \pm 1$. Further since the derivative is a polynomial it is defined everywhere and there are no singular points. The critical points split the real line into the intervals $(-\infty, -1)$, $(-1, 1)$ and $(1, \infty)$.
- When $x < -1$, both factors $(x + 1)$, $(x - 1) < 0$ so $f'(x) > 0$.
- Similarly when $x > 1$, both factors $(x + 1)$, $(x - 1) > 0$ so $f'(x) > 0$.
- When $-1 < x < 1$, $(x - 1) < 0$ but $(x + 1) > 0$ so $f'(x) < 0$.
- Summarising all this

	$(-\infty, -1)$	-1	$(-1, 1)$	1	$(1, \infty)$
$f'(x)$	positive	0	negative	0	positive
	increasing	maximum	decreasing	minimum	increasing

So $(-1, f(-1)) = (-1, 3)$ is a local maximum and $(1, f(1)) = (1, -1)$ is a local minimum.

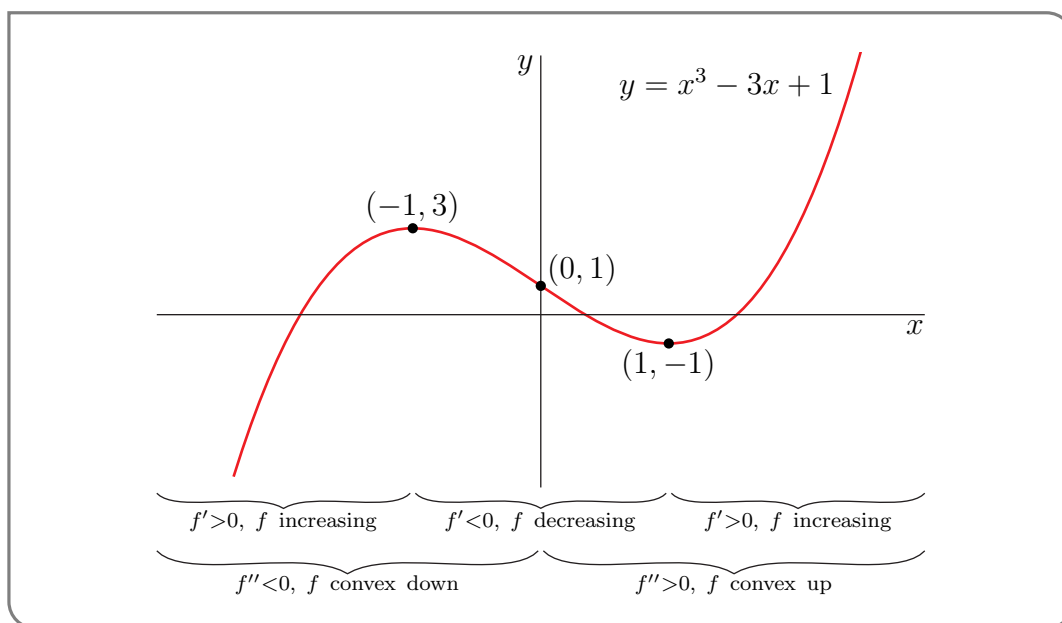
(3) Compute the second derivative:

$$f''(x) = 6x$$

- The second derivative is zero when $x = 0$, and the problem is quite easy to analyse. Clearly, $f''(x) < 0$ when $x < 0$ and $f''(x) > 0$ when $x > 0$.
- Thus f is concave down for $x < 0$, concave up for $x > 0$ and has an inflection point at $x = 0$.

Putting this all together gives:

⁶³ With the aid of a computer we can find the x -intercepts numerically: $x \approx -1.879385242, 0.3472963553$, and 1.532088886 . If you are interested in more details check out Appendix C



Example 3.6.12

Example 3.6.13 (Sketch $f(x) = x^4 - 4x^3$)

(1) Reading from $f(x)$:

- The function is a polynomial so it is defined everywhere.
- Since $f(-x) = x^4 + 4x^3 \neq \pm f(x)$, it is not even or odd. Nor is it periodic.
- The y -intercept is $y = f(0) = 0$, while the x -intercepts are given by the solution of

$$\begin{aligned} f(x) &= x^4 - 4x^3 = 0 \\ x^3(x - 4) &= 0 \end{aligned}$$

Hence the x -intercepts are 0, 4.

- Since f is a polynomial it does not have any vertical asymptotes.
- For very large x , both positive and negative, the x^4 term in $f(x)$ dominates the other term so that

$$f(x) \rightarrow \begin{cases} +\infty & \text{as } x \rightarrow +\infty \\ +\infty & \text{as } x \rightarrow -\infty \end{cases}$$

and the function has no horizontal asymptotes.

(2) Now compute the derivative $f'(x)$:

$$f'(x) = 4x^3 - 12x^2 = 4(x - 3)x^2$$

- The critical points are at $x = 0, 3$. Since the function is a polynomial there are no singular points. The critical points split the real line into the intervals $(-\infty, 0)$, $(0, 3)$ and $(3, \infty)$.
- When $x < 0$, $x^2 > 0$ and $x - 3 < 0$, so $f'(x) < 0$.
- When $0 < x < 3$, $x^2 > 0$ and $x - 3 < 0$, so $f'(x) < 0$.
- When $3 < x$, $x^2 > 0$ and $x - 3 > 0$, so $f'(x) > 0$.
- Summarising all this

	$(-\infty, 0)$	0	$(0, 3)$	3	$(3, \infty)$
$f'(x)$	negative	0	negative	0	positive
	decreasing	horizontal tangent	decreasing	minimum	increasing

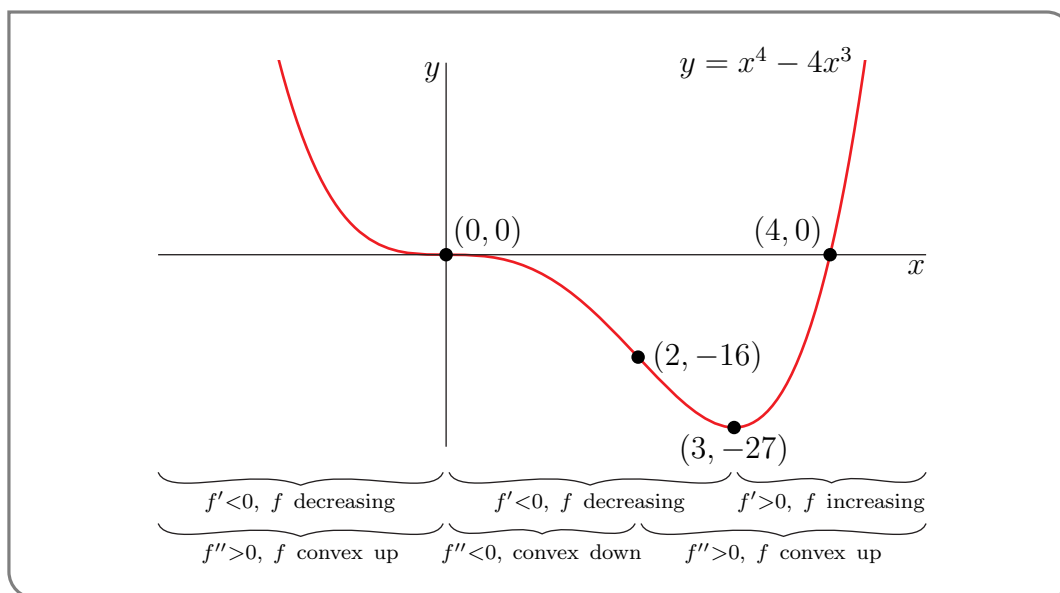
So the point $(3, f(3)) = (3, -27)$ is a local minimum. The point $(0, f(0)) = (0, 0)$ is neither a minimum nor a maximum, even though $f'(0) = 0$.

(3) Now examine $f''(x)$:

$$f''(x) = 12x^2 - 24x = 12x(x - 2)$$

- So $f''(x) = 0$ when $x = 0, 2$. This splits the real line into the intervals $(-\infty, 0)$, $(0, 2)$ and $(2, \infty)$.
- When $x < 0$, $x - 2 < 0$ and so $f''(x) > 0$.
- When $0 < x < 2$, $x > 0$ and $x - 2 < 0$ and so $f''(x) < 0$.
- When $2 < x$, $x > 0$ and $x - 2 > 0$ and so $f''(x) > 0$.
- Thus the function is convex up for $x < 0$, then convex down for $0 < x < 2$, and finally convex up again for $x > 2$. Hence $(0, f(0)) = (0, 0)$ and $(2, f(2)) = (2, -16)$ are inflection points.

Putting all this information together gives us the following sketch.



Example 3.6.13

Example 3.6.14 ($f(x) = x^3 - 6x^2 + 9x - 54$)(1) Reading from $f(x)$:

- The function is a polynomial so it is defined everywhere.
- Since $f(-x) = -x^3 - 6x^2 - 9x - 54 \neq \pm f(x)$, it is not even or odd. Nor is it periodic.
- The y -intercept is $y = f(0) = -54$, while the x -intercepts are given by the solution of

$$\begin{aligned} f(x) &= x^3 - 6x^2 + 9x - 54 = 0 \\ x^2(x - 6) + 9(x - 6) &= 0 \\ (x^2 + 9)(x - 6) &= 0 \end{aligned}$$

Hence the only x -intercept is 6.

- Since f is a polynomial it does not have any vertical asymptotes.
- For very large x , both positive and negative, the x^3 term in $f(x)$ dominates the other term so that

$$f(x) \rightarrow \begin{cases} +\infty & \text{as } x \rightarrow +\infty \\ -\infty & \text{as } x \rightarrow -\infty \end{cases}$$

and the function has no horizontal asymptotes.

(2) Now compute the derivative $f'(x)$:

$$\begin{aligned} f'(x) &= 3x^2 - 12x + 9 \\ &= 3(x^2 - 4x + 3) = 3(x - 3)(x - 1) \end{aligned}$$

- The critical points are at $x = 1, 3$. Since the function is a polynomial there are no singular points. The critical points split the real line into the intervals $(-\infty, 1)$, $(1, 3)$ and $(3, \infty)$.
- When $x < 1$, $(x - 1) < 0$ and $(x - 3) < 0$, so $f'(x) > 0$.
- When $1 < x < 3$, $(x - 1) > 0$ and $(x - 3) < 0$, so $f'(x) < 0$.
- When $3 < x$, $(x - 1) > 0$ and $(x - 3) > 0$, so $f'(x) > 0$.
- Summarising all this

	$(-\infty, 1)$	1	$(1, 3)$	3	$(3, \infty)$
$f'(x)$	positive	0	negative	0	positive
	increasing	maximum	decreasing	minimum	increasing

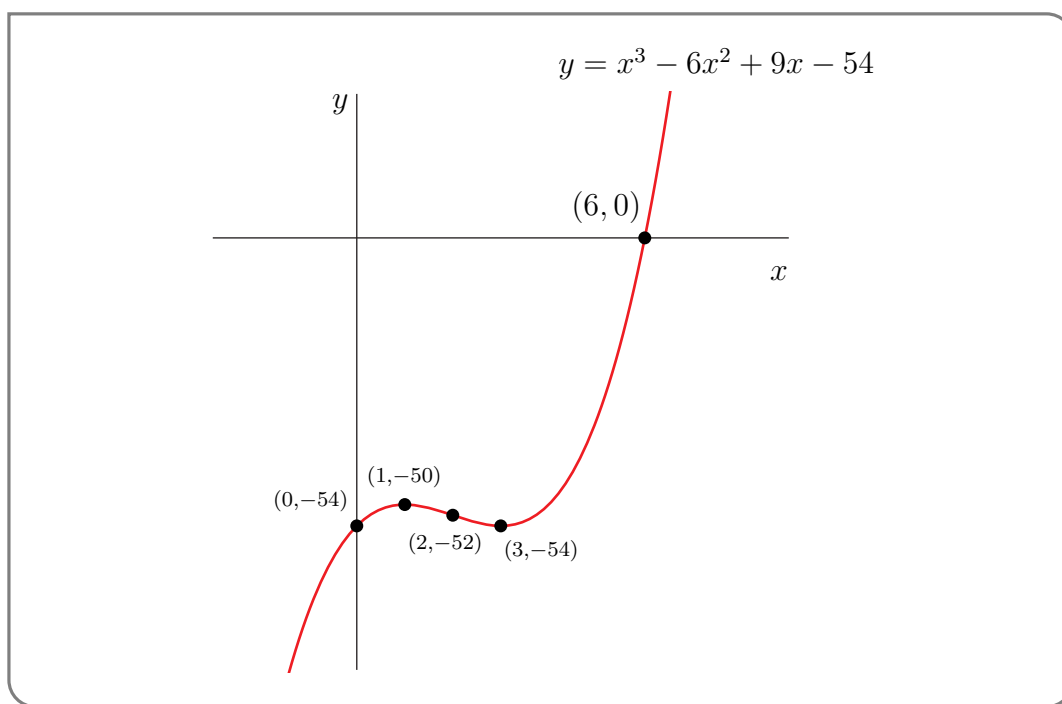
So the point $(1, f(1)) = (1, -50)$ is a local maximum. The point $(3, f(3)) = (3, -54)$ is a local minimum.

(3) Now examine $f''(x)$:

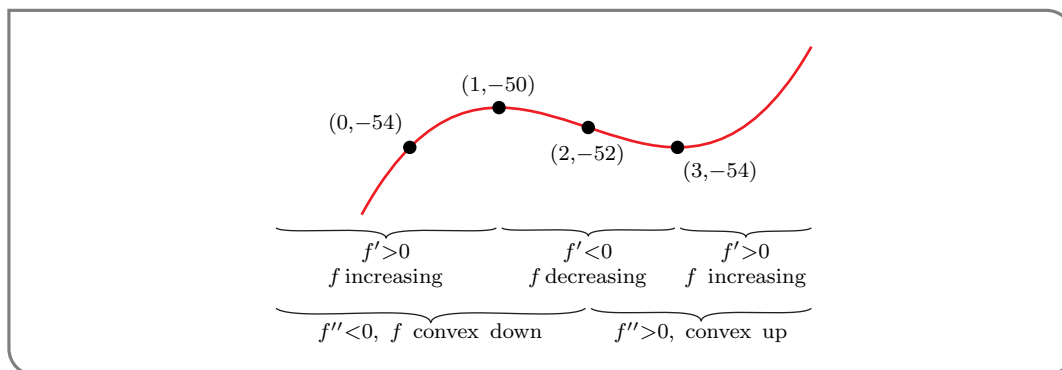
$$f''(x) = 6x - 12$$

- So $f''(x) = 0$ when $x = 2$. This splits the real line into the intervals $(-\infty, 2)$ and $(2, \infty)$.
- When $x < 2$, $f''(x) < 0$.
- When $x > 2$, $f''(x) > 0$.
- Thus the function is convex down for $x < 2$, then convex up for $x > 2$. Hence $(2, f(2)) = (2, -52)$ is an inflection point.

Putting all this information together gives us the following sketch.



and if we zoom in around the interesting points (minimum, maximum and inflection point), we have



Example 3.6.14

An example of sketching a simple rational function.

Example 3.6.15 $\left(f(x) = \frac{x}{x^2 - 4}\right)$

(1) Reading from $f(x)$:

- The function is rational so it is defined except where its denominator is zero — namely at $x = \pm 2$.
- Since $f(-x) = \frac{-x}{x^2 - 4} = -f(x)$, it is odd. Indeed this means that we only need to examine what happens to the function for $x \geq 0$ and we can then infer what happens for $x \leq 0$ using $f(-x) = -f(x)$. In practice we will sketch the graph for $x \geq 0$ and then infer the rest from this symmetry.
- The y -intercept is $y = f(0) = 0$, while the x -intercepts are given by the solution of $f(x) = 0$. So the only x -intercept is 0.
- Since f is rational, it may have vertical asymptotes where its denominator is zero — at $x = \pm 2$. Since the function is odd, we only have to analyse the asymptote at $x = 2$ and we can then infer what happens at $x = -2$ by symmetry.

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} \frac{x}{(x-2)(x+2)} = +\infty$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{x}{(x-2)(x+2)} = -\infty$$

- We now check for horizontal asymptotes:

$$\begin{aligned} \lim_{x \rightarrow +\infty} f(x) &= \lim_{x \rightarrow +\infty} \frac{x}{x^2 - 4} \\ &= \lim_{x \rightarrow +\infty} \frac{1}{x - 4/x} = 0 \end{aligned}$$

(2) Now compute the derivative $f'(x)$:

$$\begin{aligned} f'(x) &= \frac{(x^2 - 4) \cdot 1 - x \cdot 2x}{(x^2 - 4)^2} \\ &= \frac{-(x^2 + 4)}{(x^2 - 4)^2} \end{aligned}$$

- Hence there are no critical points. There are singular points where the denominator is zero, namely $x = \pm 2$. Before we proceed, notice that the numerator is always negative and the denominator is always positive. Hence $f'(x) < 0$ except at $x = \pm 2$ where it is undefined.

- The function is decreasing except at $x = \pm 2$.
- We already know that at $x = 2$ we have a vertical asymptote and that $f'(x) < 0$ for all x . So

$$\lim_{x \rightarrow 2} f'(x) = -\infty$$

- Summarising all this

	$[0, 2)$	2	$(2, \infty)$
$f'(x)$	negative	DNE	negative
	decreasing	vertical asymptote	decreasing

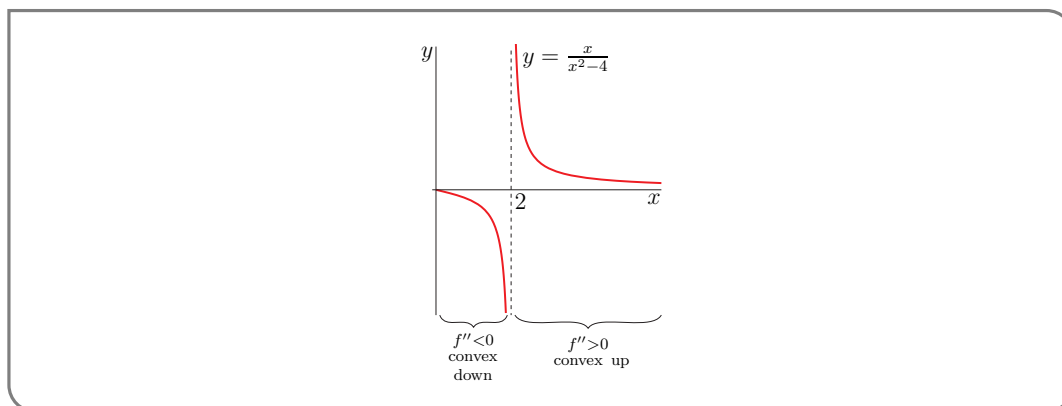
Remember — we will draw the graph for $x \geq 0$ and then use the odd symmetry to infer the graph for $x < 0$.

(3) Now examine $f''(x)$:

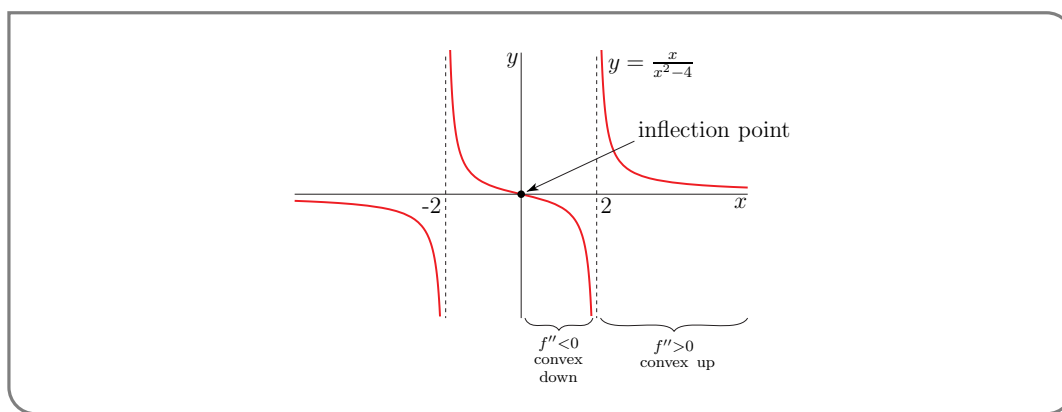
$$\begin{aligned}
 f''(x) &= -\frac{(x^2 - 4)^2 \cdot (2x) - (x^2 + 4) \cdot 2 \cdot 2x \cdot (x^2 - 4)}{(x^2 - 4)^4} \\
 &= -\frac{(x^2 - 4) \cdot (2x) - (x^2 + 4) \cdot 4x}{(x^2 - 4)^3} \\
 &= -\frac{2x^3 - 8x - 4x^3 - 16x}{(x^2 - 4)^3} \\
 &= \frac{2x(x^2 + 12)}{(x^2 - 4)^3}
 \end{aligned}$$

- So $f''(x) = 0$ when $x = 0$ and does not exist when $x = \pm 2$. This splits the real line into the intervals $(-\infty, -2)$, $(-2, 0)$, $(0, 2)$ and $(2, \infty)$. However we only need to consider $x \geq 0$ (because of the odd symmetry).
- When $0 < x < 2$, $x > 0$, $(x^2 + 12) > 0$ and $(x^2 - 4) < 0$ so $f''(x) < 0$.
- When $x > 2$, $x > 0$, $(x^2 + 12) > 0$ and $(x^2 - 4) > 0$ so $f''(x) > 0$.

Putting all this information together gives the following sketch for $x \geq 0$:



We can then draw in the graph for $x < 0$ using $f(-x) = -f(x)$:



Notice that this means that the concavity changes at $x = 0$, so the point $(0, f(0)) = (0, 0)$ is a point of inflection (as indicated).

Example 3.6.15

This final example is more substantial since the function has singular points (points where the derivative is undefined). The analysis is more involved.

Example 3.6.16 $\left(f(x) = \sqrt[3]{\frac{x^2}{(x-6)^2}}\right)$

(1) Reading from $f(x)$:

- First notice that we can rewrite

$$f(x) = \sqrt[3]{\frac{x^2}{(x-6)^2}} = \sqrt[3]{\frac{x^2}{x^2 \cdot (1 - 6/x)^2}} = \sqrt[3]{\frac{1}{(1 - 6/x)^2}}$$

- The function is the cube root of a rational function. The rational function is defined except at $x = 6$, so the domain of f is all reals except $x = 6$.
- Clearly the function is not periodic, and examining

$$\begin{aligned} f(-x) &= \sqrt[3]{\frac{1}{(1 - 6/(-x))^2}} \\ &= \sqrt[3]{\frac{1}{(1 + 6/x)^2}} \neq \pm f(x) \end{aligned}$$

shows the function is neither even nor odd.

- To compute horizontal asymptotes we examine the limit of the portion of the function inside the cube-root

$$\lim_{x \rightarrow \pm\infty} \frac{1}{(1 - \frac{6}{x})^2} = 1$$

This means we have

$$\lim_{x \rightarrow \pm\infty} f(x) = 1$$

That is, the line $y = 1$ will be a horizontal asymptote to the graph $y = f(x)$ both for $x \rightarrow +\infty$ and for $x \rightarrow -\infty$.

- Our function $f(x) \rightarrow +\infty$ as $x \rightarrow 6$, because of the $(1 - 6/x)^2$ in its denominator. So $y = f(x)$ has $x = 6$ as a vertical asymptote.

(2) Now compute $f'(x)$. Since we rewrote

$$f(x) = \sqrt[3]{\frac{1}{(1 - 6/x)^2}} = \left(1 - \frac{6}{x}\right)^{-2/3}$$

we can use the chain rule

$$\begin{aligned} f'(x) &= -\frac{2}{3} \left(1 - \frac{6}{x}\right)^{-5/3} \frac{6}{x^2} \\ &= -4 \left(\frac{x-6}{x}\right)^{-5/3} \frac{1}{x^2} \\ &= -4 \left(\frac{1}{x-6}\right)^{5/3} \frac{1}{x^{1/3}} \end{aligned}$$

- Notice that the derivative is nowhere equal to zero, so the function has no critical points. However there are two places the derivative is undefined. The terms

$$\left(\frac{1}{x-6}\right)^{5/3} \qquad \frac{1}{x^{1/3}}$$

are undefined at $x = 6, 0$ respectively. Hence $x = 0, 6$ are singular points. These split the real line into the intervals $(-\infty, 0)$, $(0, 6)$ and $(6, \infty)$.

- When $x < 0$, $(x - 6) < 0$, we have that $(x - 6)^{-5/3} < 0$ and $x^{-1/3} < 0$ and so $f'(x) = -4 \cdot (\text{negative}) \cdot (\text{negative}) < 0$.
- When $0 < x < 6$, $(x - 6) < 0$, we have that $(x - 6)^{-5/3} < 0$ and $x^{-1/3} > 0$ and so $f'(x) > 0$.
- When $x > 6$, $(x - 6) > 0$, we have that $(x - 6)^{-5/3} > 0$ and $x^{-1/3} > 0$ and so $f'(x) < 0$.
- We should also examine the behaviour of the derivative as $x \rightarrow 0$ and $x \rightarrow 6$.

$$\lim_{x \rightarrow 0^-} f'(x) = -4 \left(\lim_{x \rightarrow 0^-} (x - 6)^{-5/3} \right) \left(\lim_{x \rightarrow 0^-} x^{-1/3} \right) = -\infty$$

$$\lim_{x \rightarrow 0^+} f'(x) = -4 \left(\lim_{x \rightarrow 0^+} (x - 6)^{-5/3} \right) \left(\lim_{x \rightarrow 0^+} x^{-1/3} \right) = +\infty$$

$$\lim_{x \rightarrow 6^-} f'(x) = -4 \left(\lim_{x \rightarrow 6^-} (x - 6)^{-5/3} \right) \left(\lim_{x \rightarrow 6^-} x^{-1/3} \right) = +\infty$$

$$\lim_{x \rightarrow 6^+} f'(x) = -4 \left(\lim_{x \rightarrow 6^+} (x - 6)^{-5/3} \right) \left(\lim_{x \rightarrow 6^+} x^{-1/3} \right) = -\infty$$

We already know that $x = 6$ is a vertical asymptote of the function, so it is not surprising that the lines tangent to the graph become vertical as we approach 6.

The behavior around $x = 0$ is less standard, since the lines tangent to the graph become vertical, but $x = 0$ is not a vertical asymptote of the function. Indeed the function takes a finite value $y = f(0) = 0$.

- Summarising all this

	$(-\infty, 0)$	0	$(0, 6)$	6	$(6, \infty)$
$f'(x)$	negative	DNE	positive	DNE	negative
	decreasing	vertical tangents	increasing	vertical asymptote	decreasing

(3) Now look at $f''(x)$:

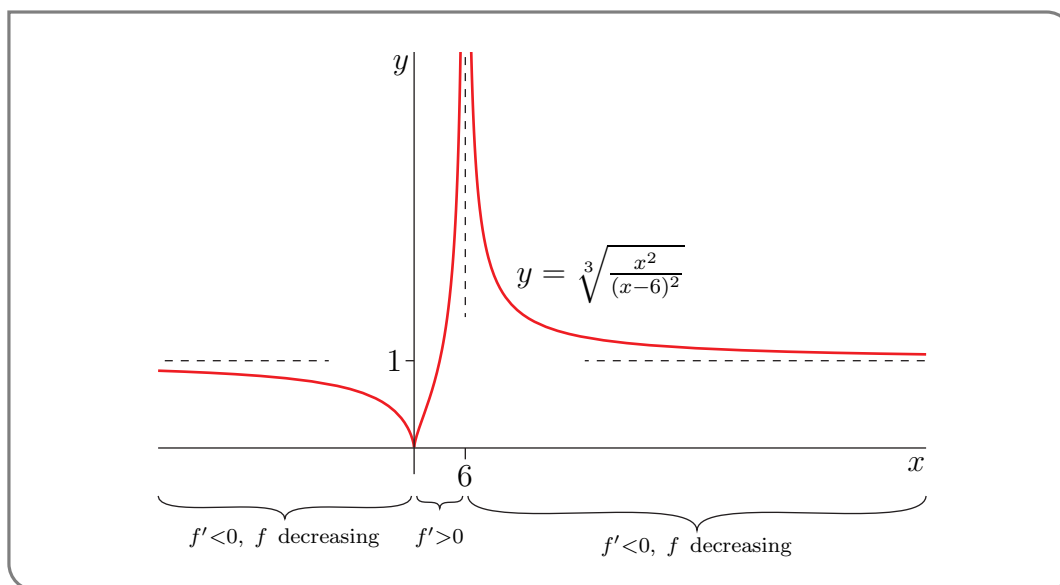
$$\begin{aligned}
 f''(x) &= -4 \frac{d}{dx} \left[\left(\frac{1}{x-6} \right)^{5/3} \frac{1}{x^{1/3}} \right] = -4 \left[-\frac{5}{3} \left(\frac{1}{x-6} \right)^{8/3} \frac{1}{x^{1/3}} - \frac{1}{3} \left(\frac{1}{x-6} \right)^{5/3} \frac{1}{x^{4/3}} \right] \\
 &= \frac{4}{3} \left(\frac{1}{x-6} \right)^{8/3} \frac{1}{x^{4/3}} [5x + (x-6)] \\
 &= 8 \left(\frac{1}{x-6} \right)^{8/3} \frac{1}{x^{4/3}} [x-1]
 \end{aligned}$$

Oof!

- Both of the factors $\left(\frac{1}{x-6} \right)^{8/3} = \left(\frac{1}{\sqrt[3]{x-6}} \right)^8$ and $\frac{1}{x^{4/3}} = \left(\frac{1}{\sqrt[3]{x}} \right)^4$ are even powers and so are positive (though possibly infinite). So the sign of $f''(x)$ is the same as the sign of the factor $x-1$. Thus

	$(-\infty, 1)$	1	$(1, \infty)$
$f''(x)$	negative	0	positive
	concave down	inflection point	concave up

Here is a sketch of the graph $y = f(x)$.



What is the third order Maclaurin polynomial for $f(x)$?

Q[19](*) : The equation $y^4 + xy = x^2 - 1$ defines y implicitly as a function of x near the point $x = 2, y = 1$.

- (a) Use the tangent line approximation at the given point to estimate the value of y when $x = 2.1$.
- (b) Use the quadratic approximation at the given point to estimate the value of y when $x = 2.1$.
- (c) Make a sketch showing how the curve relates to the tangent line at the given point.

Q[20](*) : The equation $x^4 + y + xy^4 = 1$ defines y implicitly as a function of x near the point $x = -1, y = 1$.

- (a) Use the tangent line approximation at the given point to estimate the value of y when $x = -0.9$.
- (b) Use the quadratic approximation at the given point to get another estimate of y when $x = -0.9$.
- (c) Make a sketch showing how the curve relates to the tangent line at the given point.

Q[21](*) : Given that $\log 10 \approx 2.30259$, estimate $\log 10.3$ using a suitable tangent line approximation. Give an upper and lower bound for the error in your approximation by using a suitable error estimate.

Q[22](*) : Consider $f(x) = e^{e^x}$.

- (a) Give the linear approximation for f near $x = 0$ (call this $L(x)$).
- (b) Give the quadratic approximation for f near $x = 0$ (call this $Q(x)$).
- (c) Prove that $L(x) < Q(x) < f(x)$ for all $x > 0$.
- (d) Find an interval of length at most 0.01 that is guaranteed to contain the number $e^{0.1}$.

3.5▲ Optimization

3.5.1 ► Local and global maxima and minima

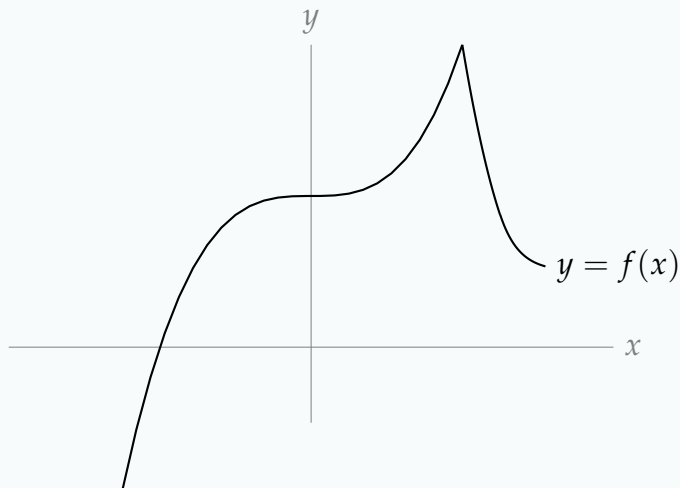
Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

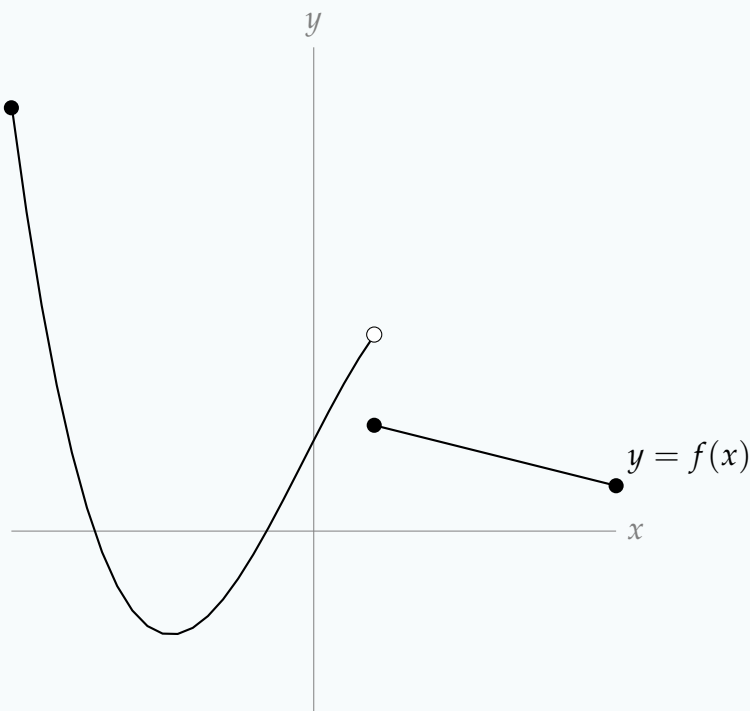
► Stage 1

Q[1]: Identify every critical point and every singular point of $f(x)$ shown on the graph below.

Which correspond to local extrema?



Q[2]: Identify every critical point and every singular point of $f(x)$ on the graph below. Which correspond to local extrema? Which correspond to global extrema over the interval shown?



Q[3]: Draw a graph $y = f(x)$ where $f(2)$ is a local maximum, but it is not a global maximum.

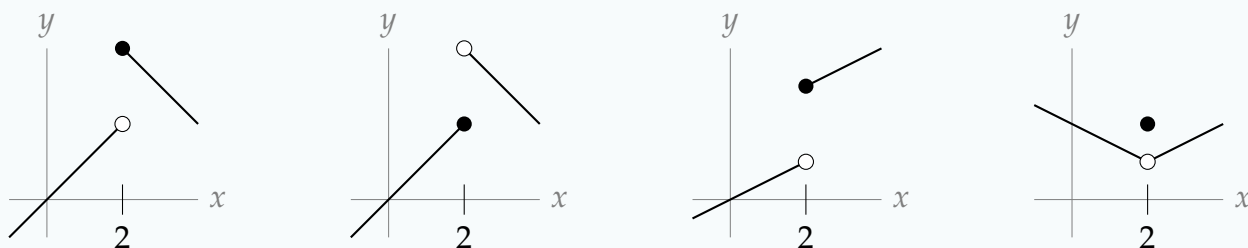
►► Stage 2

Q[4]: Suppose $f(x) = \frac{x-1}{x^2+3}$.

- Find all critical points.
- Find all singular points.
- What are the possible points where local extrema of $f(x)$ may exist?

►► Stage 3

Q[5]: Below are a number of curves, all of which have a singular point at $x = 2$. For each, label whether $x = 2$ is a local maximum, a local minimum, or neither.



Q[6]: Draw a graph $y = f(x)$ where $f(2)$ is a local maximum, but $x = 2$ is not a critical point and is not an endpoint.

Q[7]:

$$f(x) = \sqrt{|(x-5)(x+7)|}$$

Find all critical points and all singular points of $f(x)$. You do not have to specify whether a point is critical or singular.

Q[8]: Suppose $f(x)$ is the constant function $f(x) = 4$. What are the critical points and singular points of $f(x)$? What are its local and global maxima and minima?

3.5.2 ►► Finding global maxima and minima

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 1

Q[1]: Sketch a function $f(x)$ such that:

- $f(x)$ is defined over all real numbers
- $f(x)$ has a global max but no global min.

Q[2]: Sketch a function $f(x)$ such that:

- $f(x)$ is defined over all real numbers
- $f(x)$ is always positive
- $f(x)$ has no global max and no global min.

Q[3]: Sketch a function $f(x)$ such that:

- $f(x)$ is defined over all real numbers
- $f(x)$ has a global minimum at $x = 5$
- $f(x)$ has a global minimum at $x = -5$, too.

►► Stage 2

Q[4]: $f(x) = x^2 + 6x - 10$. Find all global extrema on the interval $[-5, 5]$

Q[5]: $f(x) = \frac{2}{3}x^3 - 2x^2 - 30x + 7$. Find all global extrema on the interval $[-4, 0]$.

3.5.3 ►► Max/min examples

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 2

For Questions [1](#) through [3](#), the quantity to optimize is already given to you as a function of a single variable.

Q[1](*) : Find the global maximum and the global minimum for $f(x) = x^5 - 5x + 2$ on the interval $[-2, 0]$.

Q[2](*) : Find the global maximum and the global minimum for $f(x) = x^5 - 5x - 10$ on the interval $[0, 2]$.

Q[3](*) : Find the global maximum and the global minimum for $f(x) = 2x^3 - 6x^2 - 2$ on the interval $[1, 4]$.

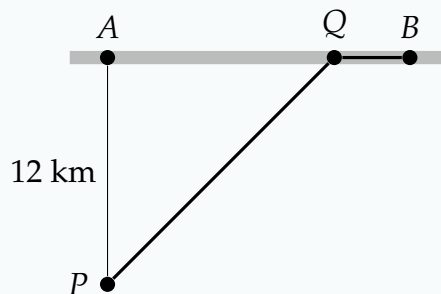
For Questions [4](#) and [5](#), you can decide whether a critical point is a local extremum by considering the derivative of the function.

Q[4](*) : Consider the function $h(x) = x^3 - 12x + 4$. What are the coordinates of the local maximum of $h(x)$? What are the coordinates of the local minimum of $h(x)$?

Q[5](*) : Consider the function $h(x) = 2x^3 - 24x + 1$. What are the coordinates of the local maximum of $h(x)$? What are the coordinates of the local minimum of $h(x)$?

For Questions 6 through 13, you will have to find an expression for the quantity you want to optimize as a function of a single variable.

Q[6](*) : You are in a dune buggy at a point P in the desert, 12 km due south of the nearest point A on a straight east-west road. You want to get to a town B on the road 18 km east of A . If your dune buggy can travel at an average speed of 15 km/hr through the desert and 30 km/hr along the road, towards what point Q on the road should you head to minimize your travel time from P to B ?



Q[7](*) : A closed three dimensional box is to be constructed in such a way that its volume is 4500 cm^3 . It is also specified that the length of the base is 3 times the width of the base. Find the dimensions of the box that satisfies these conditions and has the minimum possible surface area. Justify your answer.

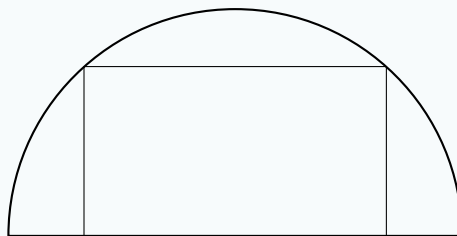
Q[8](*) : A closed rectangular container with a square base is to be made from two different materials. The material for the base costs \$5 per square metre, while the material for the other five sides costs \$1 per square metre. Find the dimensions of the container which has the largest possible volume if the total cost of materials is \$72.

Q[9](*) : Find a point X on the positive x -axis and a point Y on the positive y -axis such that (taking $O = (0,0)$)

- (i) The triangle XOY contains the first quadrant portion of the unit circle $x^2 + y^2 = 1$ and
- (ii) the area of the triangle XOY is as small as possible.

A complete and careful mathematical justification of property (i) is required.

Q[10](*) : A rectangle is inscribed in a semicircle of radius R so that one side of the rectangle lies along a diameter of the semicircle. Find the largest possible perimeter of such a rectangle, if it exists, or explain why it does not. Do the same for the smallest possible perimeter.



Q[11](*) : Find the maximal possible volume of a cylinder with surface area A .

- * Food is often packaged in cylinders, and companies wouldn't want to waste the metal they are made out of. So, you might expect the dimensions you find in this problem to describe a tin of, say, cat food. [Read here](#) about why this *isn't* the case.

Q[12](*) : What is the largest possible area of a window, with perimeter P , in the shape of a rectangle with a semicircle on top (so the diameter of the semicircle equals the width of the rectangle)?

Q[13](*) : Consider an open-top rectangular baking pan with base dimensions x centimetres by y centimetres and height z centimetres that is made from A square centimetres of tin plate. Suppose $y = px$ for some fixed constant p .

- Find the dimensions of the baking pan with the maximum capacity (i.e., maximum volume). Prove that your answer yields the baking pan with maximum capacity. Your answer will depend on the value of p .
- Find the value of the constant p that yields the baking pan with maximum capacity and give the dimensions of the resulting baking pan. Prove that your answer yields the baking pan with maximum capacity.

► Stage 3

Q[14](*) : Let $f(x) = x^x$ for $x > 0$.

- Find $f'(x)$.
- At what value of x does the curve $y = f(x)$ have a horizontal tangent line?
- Does the function f have a local maximum, a local minimum, or neither of these at the point x found in part (b)?

Q[15](*) : A length of wire is cut into two pieces, one of which is bent to form a circle, the other to form a square. How should the wire be cut if the area enclosed by the two curves is maximized? How should the wire be cut if the area enclosed by the two curves is minimized? Justify your answers.

3.6▲ Sketching Graphs

3.6.1 ► Domain, intercepts and asymptotes

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 1

Q[1]: Suppose $f(x)$ is a function given by

$$f(x) = \frac{g(x)}{x^2 - 9}$$

where $g(x)$ is also a function. True or false: $f(x)$ has a vertical asymptote at $x = -3$.

►► Stage 2

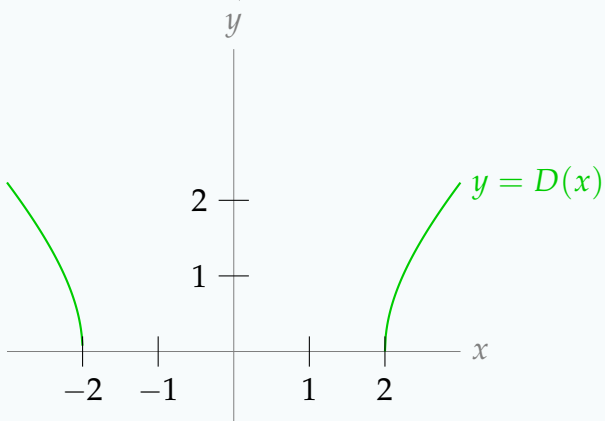
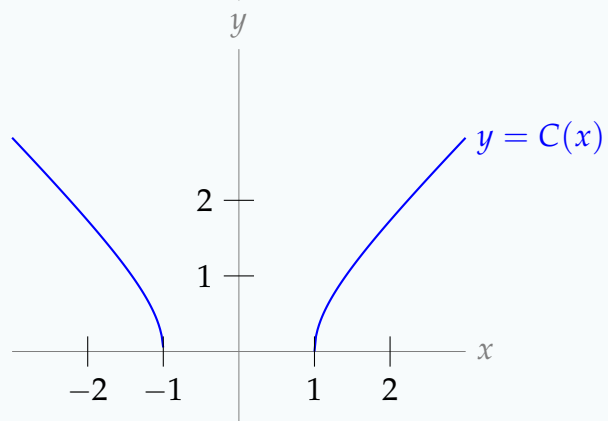
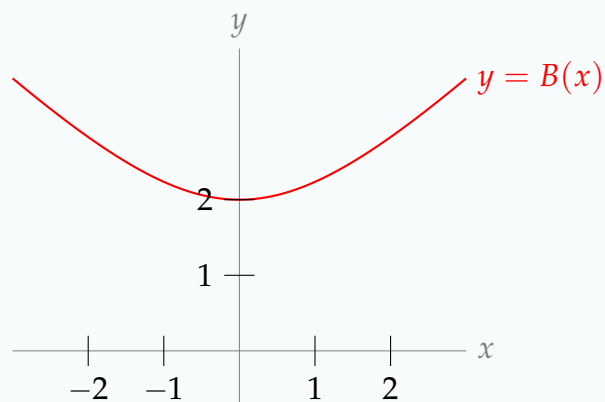
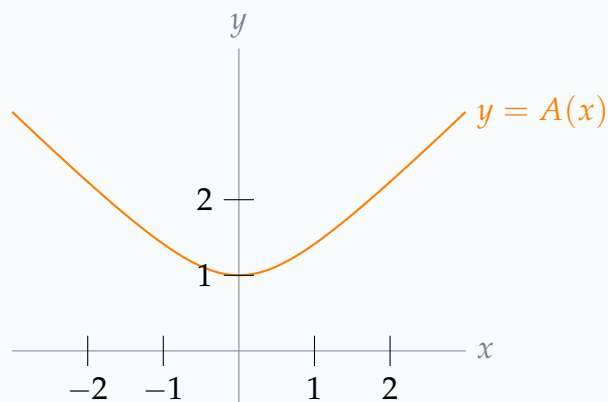
Q[2]: Match the functions $f(x)$, $g(x)$, $h(x)$, and $k(x)$ to the curves $y = A(x)$ through $y = D(x)$.

$$f(x) = \sqrt{x^2 + 1}$$

$$g(x) = \sqrt{x^2 - 1}$$

$$h(x) = \sqrt{x^2 + 4}$$

$$k(x) = \sqrt{x^2 - 4}$$

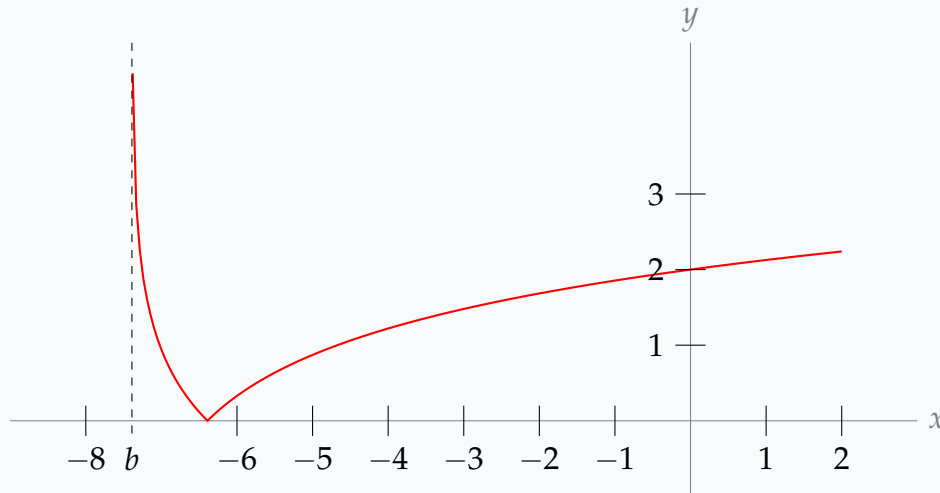


Q[3]: Below is the graph of

$$y = f(x) = \sqrt{\log^2(x + p)}$$

- (a) What is p ?
- (b) What is b (marked on the graph)?
- (c) What is the x -intercept of $f(x)$?

Remember $\log(x + p)$ is the natural logarithm of $x + p$, $\log_e(x + p)$.



Q[4]: Find all asymptotes of $f(x) = \frac{x(2x + 1)(x - 7)}{3x^3 - 81}$.

Q[5]: Find all asymptotes of $f(x) = 10^{3x-7}$.

3.6.2 ► First derivative - increasing or decreasing

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

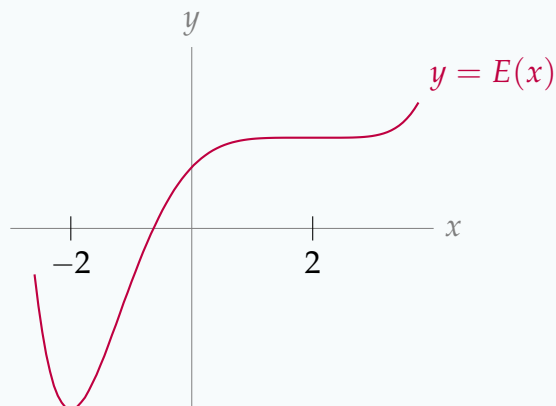
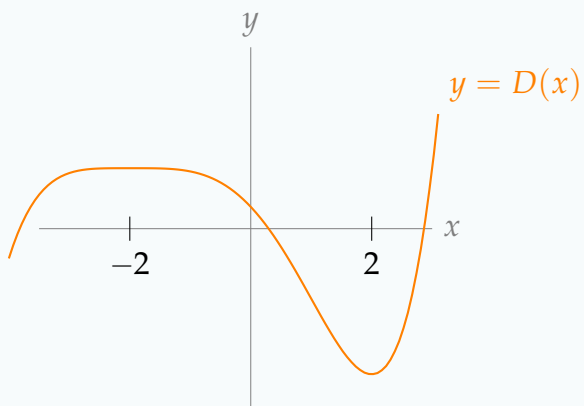
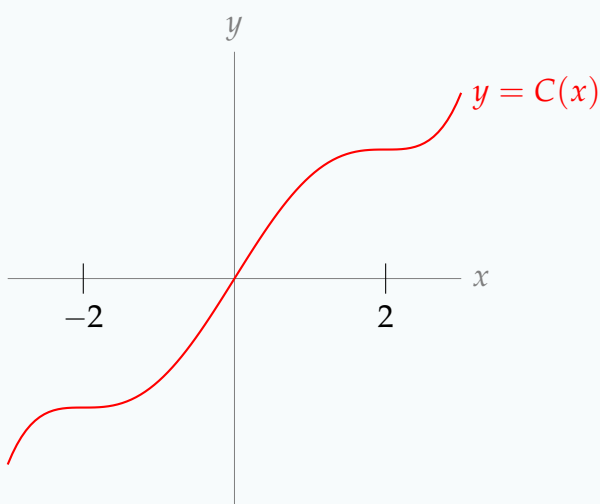
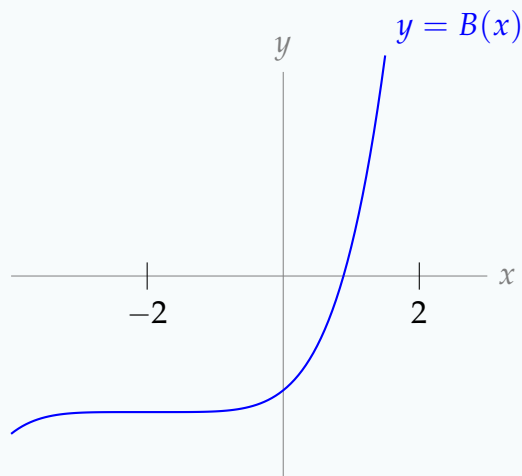
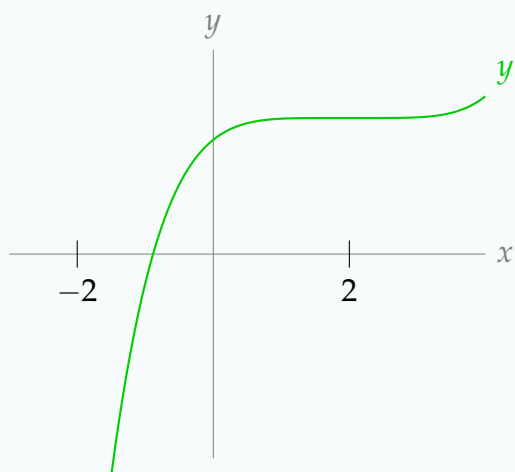
► Stage 1

Q[1]: Match each function graphed below to its *derivative* from the list. (For example, which function on the list corresponds to $A'(x)$?)

The y -axes have been scaled to make the curve's behaviour clear, so the vertical scales differ from graph to graph.

$$l(x) = (x-2)^4 \quad m(x) = (x-2)^4(x+2) \quad n(x) = (x-2)^2(x+2)^2$$

$$o(x) = (x-2)(x+2)^3 \quad p(x) = (x+2)^4$$



►► Stage 2

Q[2](*) : Find the largest open interval on which $f(x) = \frac{e^x}{x+3}$ is increasing.

Q[3](*) : Find the largest open interval on which $f(x) = \frac{\sqrt{x-1}}{2x+4}$ is increasing.

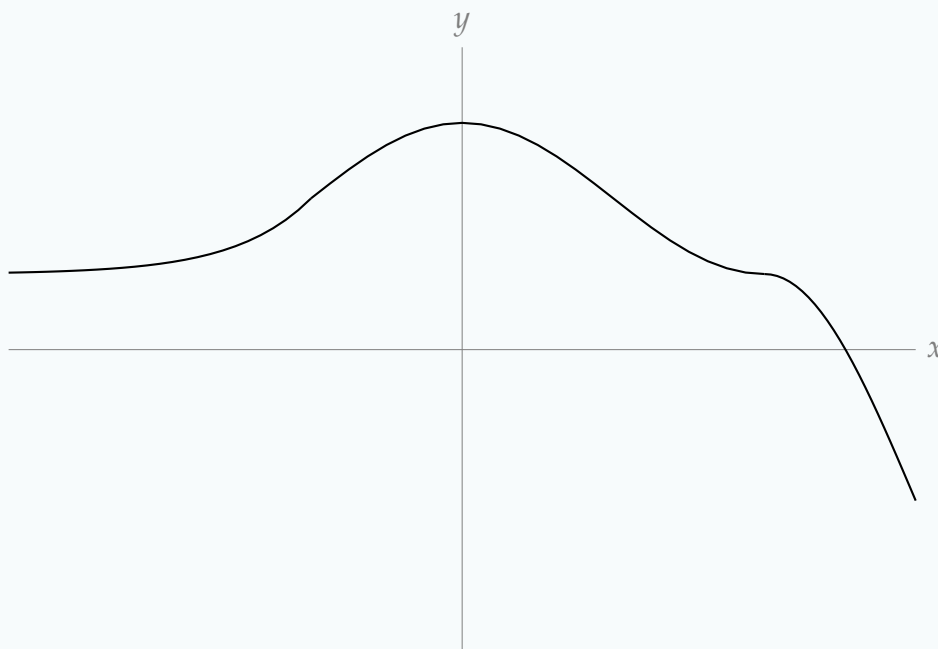
Q[4](*) : Find the largest open interval on which $f(x) = 2 \arctan(x) - \log(1+x^2)$ is increasing.

3.6.3 ►► Second derivative - concavity**Exercises**

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 1

Q[1]: On the graph below, mark the intervals where $f''(x) > 0$ (i.e. $f(x)$ is concave up) and where $f''(x) < 0$ (i.e. $f(x)$ is concave down).



Q[2]: Sketch a curve that is:

- concave up when $|x| > 5$,
- concave down when $|x| < 5$,
- increasing when $x < 0$, and
- decreasing when $x > 0$.

Q[3]: Suppose $f(x)$ is a function whose second derivative exists and is continuous for all real numbers.

True or false: if $f''(3) = 0$, then $x = 3$ is an inflection point of $f(x)$.

Remark: compare to Question [7](#)

►► Stage 2

Q[4](*) : Find all inflection points for the graph of $f(x) = 3x^5 - 5x^4 + 13x$.

►► Stage 3

Questions [5](#) through [7](#) ask you to show that certain things are true. Give a clear explanation using concepts and theorems from the CLP-1 text.

Q[5](*) : Let

$$f(x) = \frac{x^5}{20} + \frac{5x^3}{6} - 10x^2 + 500x + 1000$$

Show that $f(x)$ has exactly one inflection point.

Q[6](*) : Let $f(x)$ be a function whose first two derivatives exist everywhere, and $f''(x) > 0$ for all x .

- Show that $f(x)$ has at most one critical point and that any critical point is an absolute minimum for $f(x)$.
- Show that the maximum value of $f(x)$ on any finite interval occurs at one of the endpoints of the interval.

Q[7]: Suppose $f(x)$ is a function whose second derivative exists and is continuous for all real numbers, and $x = 3$ is an inflection point of $f(x)$. Use the Intermediate Value Theorem to show that $f''(3) = 0$.

Remark: compare to Question [3](#).

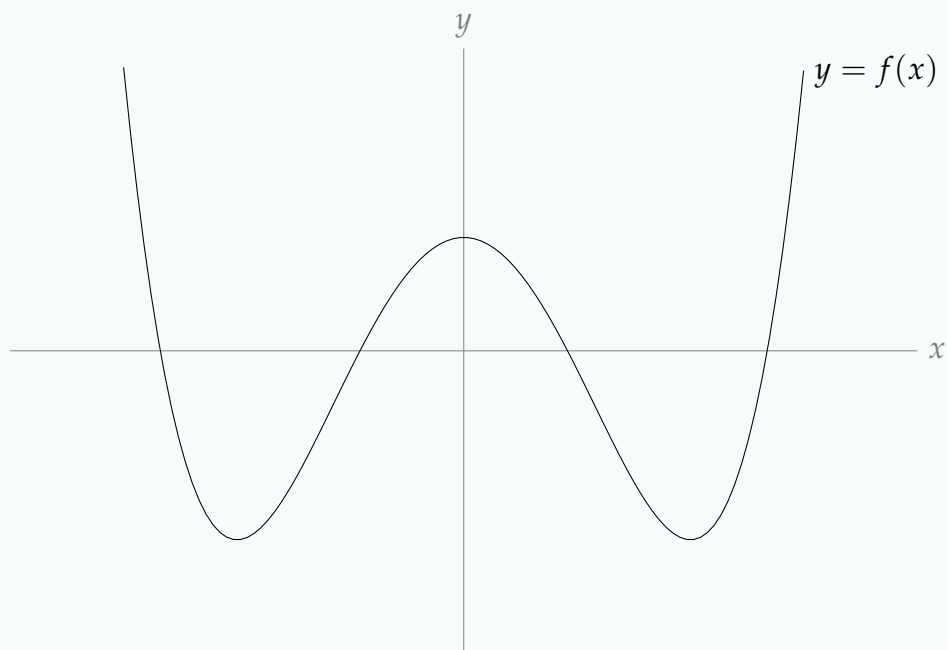
3.6.4 ► Symmetries

Exercises

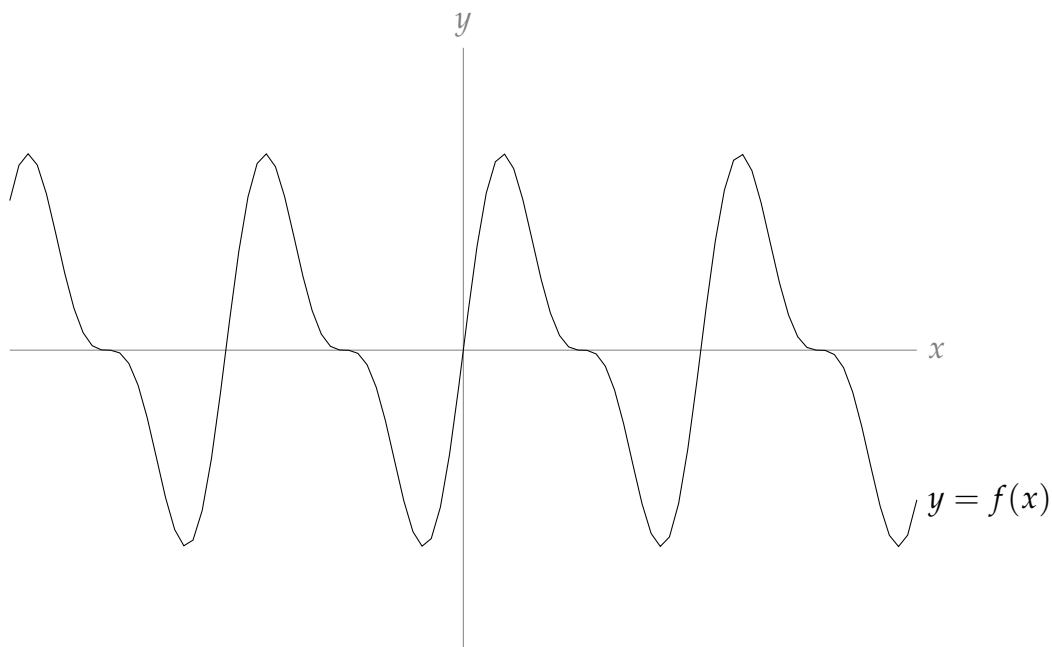
Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

► Stage 1

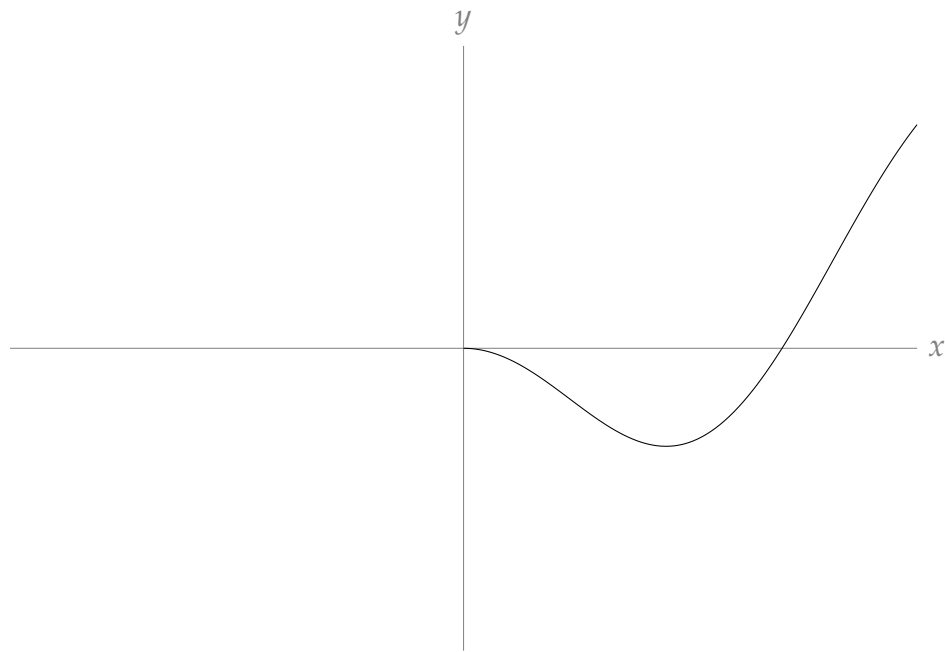
Q[1]: What symmetries (even, odd, periodic) does the function graphed below have?



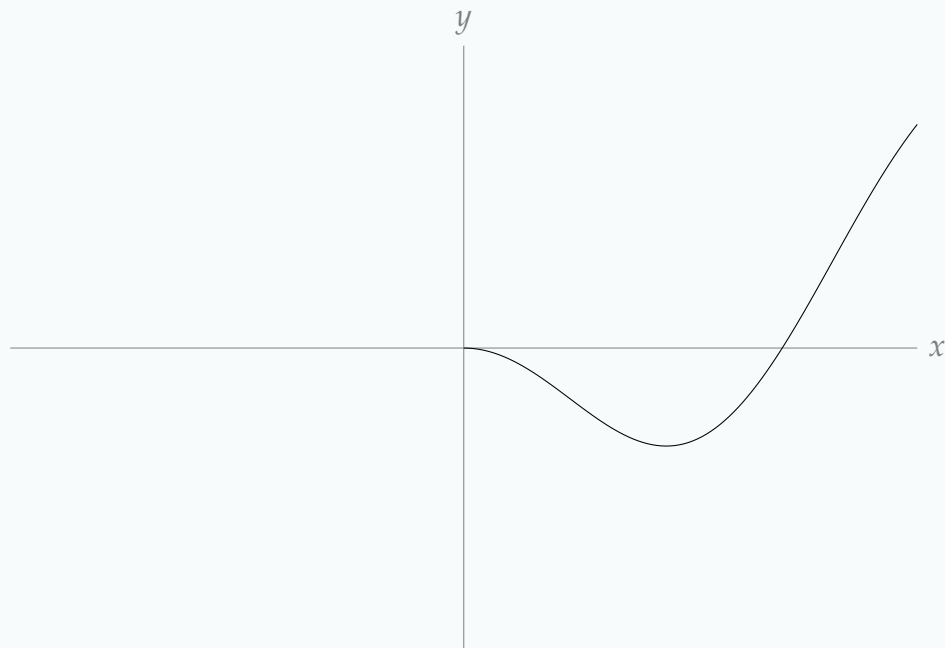
Q[2]: What symmetries (even, odd, periodic) does the function graphed below have?



Q[3]: Suppose $f(x)$ is an even function defined for all real numbers. Below is the curve $y = f(x)$ when $x > 0$. Complete the sketch of the curve.



Q[4]: Suppose $f(x)$ is an odd function defined for all real numbers. Below is the curve $y = f(x)$ when $x > 0$. Complete the sketch of the curve.



► Stage 2

Q[5]:

$$f(x) = \frac{x^4 - x^6}{e^{x^2}}$$

Show that $f(x)$ is even.

Q[6]:

$$f(x) = \sin(x) + \cos\left(\frac{x}{2}\right)$$

Show that $f(x)$ is periodic.*In Questions [7](#) through [10](#), find the symmetries of a function from its equation.*

Q[7]:

$$f(x) = x^4 + 5x^2 + \cos(x^3)$$

What symmetries (even, odd, periodic) does $f(x)$ have?

Q[8]:

$$f(x) = x^5 + 5x^4$$

What symmetries (even, odd, periodic) does $f(x)$ have?

Q[9]:

$$f(x) = \tan(\pi x)$$

What is the period of $f(x)$?

► Stage 3

Q[10]:

$$f(x) = \tan(3x) + \sin(4x)$$

What is the period of $f(x)$?

3.6.5 ► A checklist for sketching

No exercises for Section [3.6.5](#)Jump to [TABLE OF CONTENTS](#).

3.6.6 ► Sketching examples

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

► Stage 2

Q[1](*) : Let $f(x) = x\sqrt{3-x}$.

- Find the domain of $f(x)$.
- Determine the x -coordinates of the local maxima and minima (if any) and intervals where $f(x)$ is increasing or decreasing.
- Determine intervals where $f(x)$ is concave upwards or downwards, and the x coordinates of inflection points (if any). You may use, without verifying it, the formula $f''(x) = (3x - 12)(3 - x)^{-3/2}/4$.
- There is a point at which the tangent line to the curve $y = f(x)$ is vertical. Find this point.
- Sketch the graph $y = f(x)$, showing the features given in items (a) to (d) above and giving the (x, y) coordinates for all points occurring above.

In Questions 2 through 4, you will sketch the graphs of rational functions.

Q[2](*) : Sketch the graph of

$$f(x) = \frac{x^3 - 2}{x^4}.$$

Indicate the critical points, local and absolute maxima and minima, vertical and horizontal asymptotes, inflection points and regions where the curve is concave upward or downward.

Q[3](*) : The first and second derivatives of the function $f(x) = \frac{x^4}{1+x^3}$ are:

$$f'(x) = \frac{4x^3 + x^6}{(1+x^3)^2} \quad \text{and} \quad f''(x) = \frac{12x^2 - 6x^5}{(1+x^3)^3}$$

Graph $f(x)$. Include local and absolute maxima and minima, regions where $f(x)$ is increasing or decreasing, regions where the curve is concave upward or downward, and any asymptotes.

Q[4](*) : The first and second derivatives of the function $f(x) = \frac{x^3}{1-x^2}$ are:

$$f'(x) = \frac{3x^2 - x^4}{(1-x^2)^2} \quad \text{and} \quad f''(x) = \frac{6x + 2x^3}{(1-x^2)^3}$$

Graph $f(x)$. Include local and absolute maxima and minima, regions where the curve is concave upward or downward, and any asymptotes.

Q[5](*) : The function $f(x)$ is defined by

$$f(x) = \begin{cases} e^x & x < 0 \\ \frac{x^2+3}{3(x+1)} & x \geq 0 \end{cases}$$

- (a) Explain why $f(x)$ is continuous everywhere.
- (b) Determine all of the following if they are present:
 - i. x -coordinates of local maxima and minima, intervals where $f(x)$ is increasing or decreasing;
 - ii. intervals where $f(x)$ is concave upwards or downwards;
 - iii. equations of any horizontal or vertical asymptotes.
- (c) Sketch the graph of $y = f(x)$, giving the (x, y) coordinates for all points of interest above.

In Questions [6](#) and [7](#), you will sketch the graphs of functions with an exponential component. In the next section, you will learn how to find their horizontal asymptotes, but for now these are given to you.

Q[6](*) : The function $f(x)$ and its derivative are given below:

$$f(x) = (1 + 2x)e^{-x^2} \quad \text{and} \quad f'(x) = 2(1 - x - 2x^2)e^{-x^2}$$

Sketch the graph of $f(x)$. Indicate the critical points, local and/or absolute maxima and minima, and asymptotes. Without actually calculating the inflection points, indicate on the graph their approximate location.

Note: $\lim_{x \rightarrow \pm\infty} f(x) = 0$.

Q[7](*) : Consider the function $f(x) = xe^{-x^2/2}$.

Note: $\lim_{x \rightarrow \pm\infty} f(x) = 0$.

- (a) Find all inflection points and intervals of increase, decrease, convexity up, and convexity down.

You may use without proof the formula $f''(x) = (x^3 - 3x)e^{-x^2/2}$.

- (b) Find local and global minima and maxima.
- (c) Use all the above to draw a graph for f . Indicate all special points on the graph.

In Questions [8](#) and [9](#), you will sketch the graphs of functions that have a trigonometric component.

Q[8]: Use the techniques from this section to sketch the graph of $f(x) = x + 2 \sin x$.

Q[9](*) :

$$f(x) = 4 \sin x - 2 \cos 2x$$

Graph the equation $y = f(x)$, including all important features. (In particular, find all local maxima and minima and all inflection points.) Additionally, find the maximum and minimum values of $f(x)$ on the interval $[0, \pi]$.

Q[10]: Sketch the curve $y = \sqrt[3]{\frac{x+1}{x^2}}$.

You may use the facts $y'(x) = \frac{-(x+2)}{3x^{5/3}(x+1)^{2/3}}$ and $y''(x) = \frac{4x^2 + 16x + 10}{9x^{8/3}(x+1)^{5/3}}$.

► Stage 3

Q[11](*) : A function $f(x)$ defined on the whole real number line satisfies the following conditions

$$f(0) = 0 \quad f(2) = 2 \quad \lim_{x \rightarrow +\infty} f(x) = 0 \quad f'(x) = K(2x - x^2)e^{-x}$$

for some positive constant K . (Read carefully: you are given the *derivative* of $f(x)$, not $f(x)$ itself.)

- Determine the intervals on which f is increasing and decreasing and the location of any local maximum and minimum values of f .
- Determine the intervals on which f is concave up or down and the x -coordinates of any inflection points of f .
- Determine $\lim_{x \rightarrow -\infty} f(x)$.
- Sketch the graph of $y = f(x)$, showing any asymptotes and the information determined in parts (a) and (b).

Q[12](*) : Let $f(x) = e^{-x}$, $x \geq 0$.

- Sketch the graph of the equation $y = f(x)$. Indicate any local extrema and inflection points.
- Sketch the graph of the inverse function $y = g(x) = f^{-1}(x)$.
- Find the domain and range of the inverse function $g(x) = f^{-1}(x)$.
- Evaluate $g'(\frac{1}{2})$.

Q[13](*) :

- Sketch the graph of $y = f(x) = x^5 - x$, indicating asymptotes, local maxima and minima, inflection points, and where the graph is concave up/concave down.
- Consider the function $f(x) = x^5 - x + k$, where k is a constant, $-\infty < k < \infty$. How many roots does the function have? (Your answer might depend on the value of k .)

Q[14](*) : The hyperbolic trigonometric functions $\sinh(x)$ and $\cosh(x)$ are defined by

$$\sinh(x) = \frac{e^x - e^{-x}}{2} \quad \cosh(x) = \frac{e^x + e^{-x}}{2}$$

They have many properties that are similar to corresponding properties of $\sin(x)$ and $\cos(x)$. In particular, it is easy to see that

$$\frac{d}{dx} \sinh(x) = \cosh(x) \quad \frac{d}{dx} \cosh(x) = \sinh(x) \quad \cosh^2(x) - \sinh^2(x) = 1$$

S-21: Let $f(x) = \log x$ and $x_0 = 10$. Then

$$\begin{array}{lll} f(x) = \log x & f'(x) = \frac{1}{x} & f''(x) = -\frac{1}{x^2} \\ f(10) = \log 10 \approx 2.30259 & f'(10) = \frac{1}{10} & f''(10) = -\frac{1}{100} \end{array}$$

so that, with $x = 10.3$,

$$\log 10.3 = f(10.3) \approx f(10) + f'(10)(10.3 - 10) = 2.30259 + \frac{0.3}{10} = 2.33259$$

The error in this approximation (excluding the error in the given data $\log 10 \approx 2.30259$) is

$\frac{1}{2}f''(z)(10.3 - 10)^2$ for some z between 10 and 10.3. Because $f''(z) = -\frac{1}{z^2}$ increases as z increases, it must be between $-\frac{1}{10^2}$ and $-\frac{1}{10.3^2}$. This forces $\frac{1}{2}f''(z)(10.3 - 10)^2$ to be between $-\frac{1}{2} \cdot \frac{1}{10^2}(0.3)^2 = -0.00045$ and $-\frac{1}{2} \cdot \frac{1}{10.3^2}(0.3)^2 < -0.00042$.

S-22: We begin by finding the values of the derivatives of f at $x = 0$. We can use the chain rule, or the formula we found in Question **19**, Section 2.14.

$$\begin{array}{ll} f(x) = e^{e^x} & f(0) = e \\ f'(x) = e^x e^{e^x} & f'(0) = e \\ f''(x) = (e^x + e^{2x}) e^{e^x} & f''(0) = 2e \\ f'''(x) = (e^x + 3e^{2x} + e^{3x}) e^{e^x} & \end{array}$$

(a) $L(x) = f(0) + f'(0)(x - 0) = e + ex$

(b) $Q(x) = f(0) + f'(0)(x - 0) + \frac{1}{2}f''(0)(x - 0)^2 = e + ex + ex^2$

(c) Since $ex^2 > 0$ for all $x > 0$, $L(x) < Q(x)$ for all $x > 0$.

From the error formula, we know that

$$\begin{aligned} f(x) &= f(0) + f'(0)x + \frac{1}{2}f''(0)x^2 + \frac{1}{3!}f'''(c)x^3 \\ &= Q(x) + \frac{1}{6}(e^c + 3e^{2c} + e^{3c})e^{e^c}x^3 \end{aligned}$$

for some c between 0 and x . Since $\frac{1}{6}(e^c + 3e^{2c} + e^{3c})e^{e^c}$ is positive for any c , for all $x > 0$, $\frac{1}{6}(e^c + 3e^{2c} + e^{3c})e^{e^c}x^3 > 0$, so $Q(x) < f(x)$.

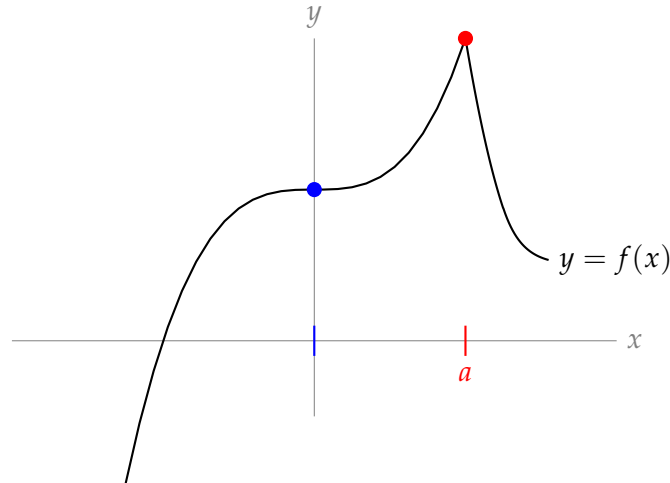
(d) Write $g(x) = e^x = 1 + x + \frac{1}{2!}e^c x^2$, for some c between 0 and x . For $x = 0.1$ we have $0 < c < 0.1$ and $1 < e^c < e^{0.1} < e < 3$. So

$$\begin{array}{llllllll} e^{0.1} & = & f(0.1) & = & 1 + 0.1 + \frac{1}{2}e^c(0.1)^2 & > & 1 + 0.1 + \frac{1}{2}(1)(0.1)^2 & = & 1.105 \\ e^{0.1} & = & f(0.1) & = & 1 + 0.1 + \frac{1}{2}e^c(0.1)^2 & < & 1 + 0.1 + \frac{1}{2}(3)(0.1)^2 & = & 1.115 \end{array}$$

That is, $1.105 < e^{0.1} < 1.115$.

Solutions to Exercises 3.5.1 — Jump to [TABLE OF CONTENTS](#)

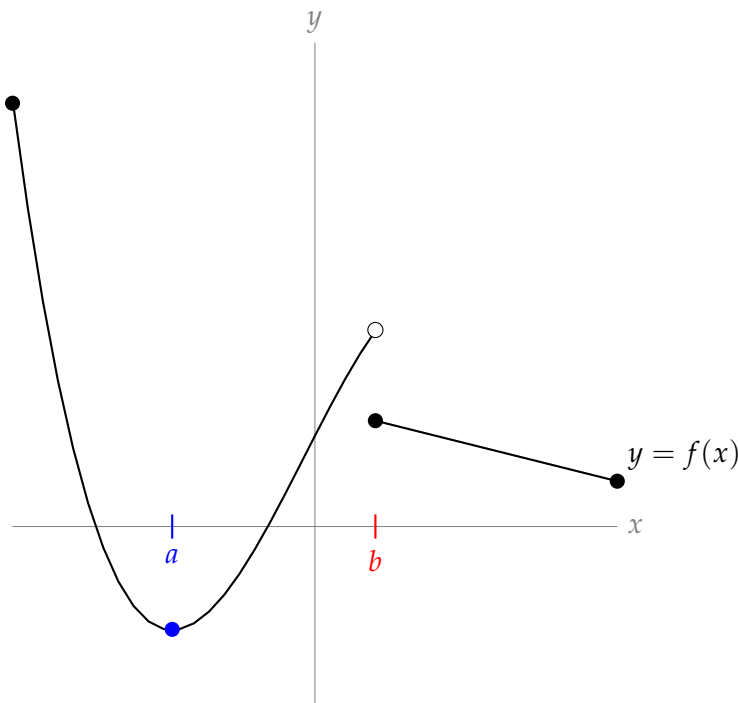
S-1:



When $x = 0$, the curve $y = f(x)$ appears to have a flat tangent line, so the $x = 0$ is a critical point. However, it is not a local extremum: it is not true that $f(0) \geq f(x)$ for all x near 0, and it is not true that $f(0) \leq f(x)$ for all x near 0.

To the right of the x -axis, there is a spike where the derivative of $f(x)$ does not exist. The x -value corresponding to this spike (call it a) is a singular point, and $f(x)$ has a local maximum at $x = a$.

S-2:

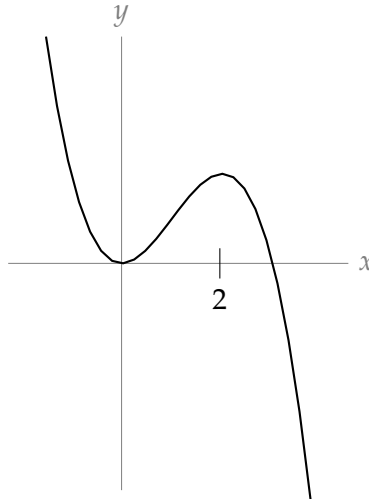


The x -coordinate corresponding to the blue dot (let's call it a) is a critical point, because the tangent line to $f(x)$ at $x = a$ is horizontal. There is no lower point nearby, and actually no lower point on the whole interval shown, so $f(x)$ has both a local minimum and a global minimum at $x = a$.

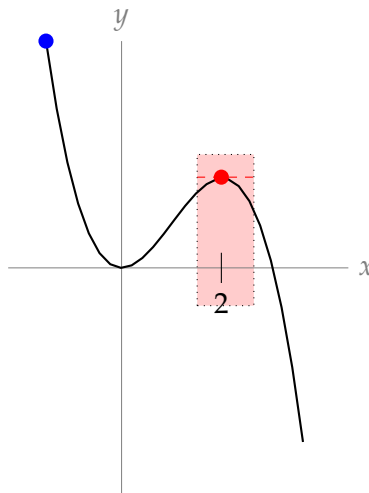
If a function is not continuous at a point, then it is not differentiable at that point. So, the

x -coordinate corresponding to the discontinuity (let's call it b) is a singular point. Values of $f(x)$ immediately to the right of b are lower, and values immediately to the left of b are higher, so $f(x)$ has no local (or global) extremum at $x = b$.

S-3: One possible answer is shown below.



For every x in the red interval shown below, $f(2) \geq f(x)$, so $f(2)$ is a local maximum. However, the point marked with a blue dot shows that $f(x) > f(2)$ for some x , so $f(2)$ is not a global maximum.

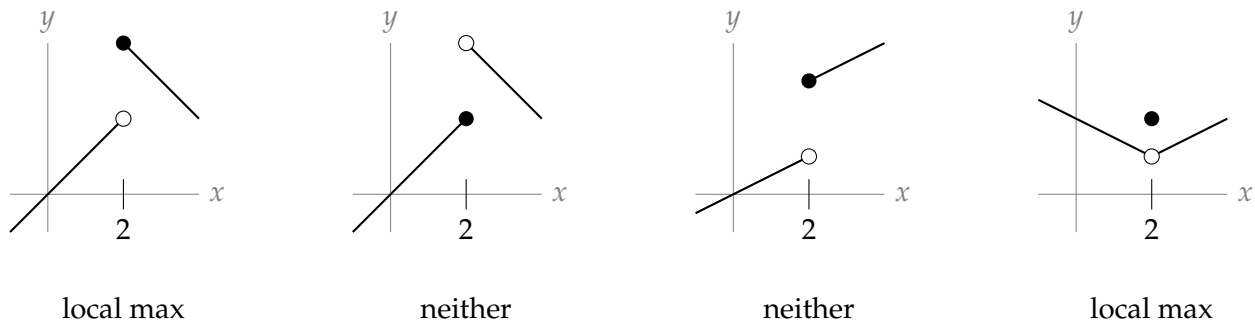


S-4: Critical points are those values of x for which $f'(x) = 0$, and singular points are those values of x for which $f(x)$ is not differentiable. So, we ought to find $f'(x)$. Using the quotient rule,

$$\begin{aligned} f'(x) &= \frac{(1)(x^2 + 3) - (x - 1)(2x)}{(x^2 + 3)^2} \\ &= \frac{-x^2 + 2x + 3}{(x^2 + 3)^2} \\ &= -\frac{(x - 3)(x + 1)}{(x^2 + 3)^2} \end{aligned}$$

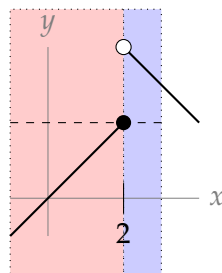
- (a) The derivative $f'(x)$ is zero when $x = 3$ and when $x = -1$, so those are the critical points.
- (b) The denominator of $f'(x)$ is never zero, so the derivative $f'(x)$ exists for all x and $f(x)$ has no singular points.
- (c) Theorem 3.5.4 tells us that local extrema of $f(x)$ can only occur at critical points and singular points. So, the possible points where extrema of $f(x)$ may exist are $x = 3$ and $x = -1$.

S-5:



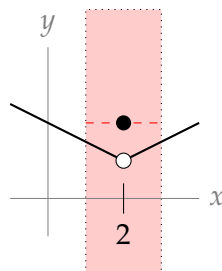
For the first curve, the function's value at $x = 2$ (that is, the y -value of the solid dot) is higher than anything around it. So, it's a local maximum.

For the second curve, the function's value at $x = 2$ (that is, the y -value of the solid dot) is higher than everything to the left, but lower than values immediately to the right. (On the graph reproduced below, $f(x)$ is higher than everything in the red section, and lower than everything in the blue section.) So, it is neither a local max nor a local min.



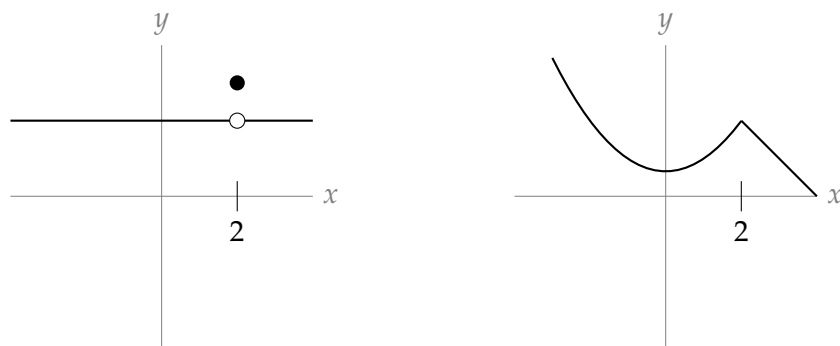
Similarly, for the third curve, $f(2)$ is lower than the values to the right of it, and higher than values to the left of it, so it is neither a local minimum nor a local maximum.

In the final curve, $f(2)$ (remember—this is the y -value of the solid dot) is higher than everything immediately to the left or right of it (for instance, over the interval marked in red below), so it is a local maximum.



S-6: The question specifies that $x = 2$ must not be an endpoint. By Theorem 3.5.4, if $x = 2$ not a

critical point, then it must be a singular point. That is, $f(x)$ is not differentiable at $x = 2$. Two possibilities are shown below, but there are infinitely many possible answers.



S-7: Critical points are those values of x for which $f'(x) = 0$, and singular points are those values of x for which $f(x)$ is not differentiable. So, we ought to find $f'(x)$. Since $f(x)$ has an absolute value sign, let's re-write it in a version that is friendlier to differentiation. Remember that $|X| = X$ when $X \geq 0$, and $|X| = -X$ when $X < 0$.

$$\begin{aligned} f(x) &= \sqrt{|(x-5)(x+7)|} \\ &= \begin{cases} \sqrt{(x-5)(x+7)} & \text{if } (x-5)(x+7) \geq 0 \\ \sqrt{-(x-5)(x+7)} & \text{if } (x-5)(x+7) < 0 \end{cases} \end{aligned}$$

The product $(x-5)(x+7)$ is positive when $(x-5)$ and $(x+7)$ have the same sign, and negative when they have opposite signs, so

$$f(x) = \begin{cases} \sqrt{(x-5)(x+7)} & \text{if } x \in (-\infty, -7] \cup [5, \infty) \\ \sqrt{-(x-5)(x+7)} & \text{if } x \in (-7, 5) \end{cases}$$

Now, when $x \neq -7, 5$, we can differentiate, using the chain rule.

$$\begin{aligned} f'(x) &= \begin{cases} \frac{\frac{d}{dx}\{(x-5)(x+7)\}}{2\sqrt{(x-5)(x+7)}} & \text{if } x \in (-\infty, -7) \cup (5, \infty) \\ \frac{\frac{d}{dx}\{-(x-5)(x+7)\}}{2\sqrt{-(x-5)(x+7)}} & \text{if } x \in (-7, 5) \\ ?? & \text{if } x = -7, x = 5 \end{cases} \\ &= \begin{cases} \frac{2x+2}{2\sqrt{(x-5)(x+7)}} & \text{if } x \in (-\infty, -7) \cup (5, \infty) \\ \frac{-2x-2}{2\sqrt{-(x-5)(x+7)}} & \text{if } x \in (-7, 5) \\ ?? & \text{if } x = -7, x = 5 \end{cases} \end{aligned}$$

We are tempted to say that the derivative doesn't exist when $x = -7$ and $x = 5$, but be careful—we don't actually know that yet. The formulas we have for the $f'(x)$ are only good when x is *not* -7 or 5 .

The middle formula $\frac{-2x-2}{2\sqrt{-(x-5)(x+7)}}$ tells us $x = -1$ is a critical point: when $x = -1$, $f'(x)$ is given by the middle line, and it is 0. Note that $x = -1$ also makes the top formula 0, but $f'(-1)$ is not given by the top formula, so that doesn't matter.

What we've concluded so far is that $x = -1$ is a critical point of $f(x)$, and $f(x)$ has no other critical points or singular points when $x \neq -7, 5$. It remains to figure out what's going on at -7 and 5 . One way to do this is to use the definition of the derivative to figure out what $f'(-7)$ and $f'(5)$ are, if they exist. This is somewhat laborious. Let's look for a better way.

-
- First, let's notice that $f(x)$ is defined for all values of x , thanks to that handy absolute value sign.
 - Next, notice $f(x) \geq 0$ for all x , since square roots never give a negative value.
 - Then if there is some value of x that gives $f(x) = 0$, that x gives a global minimum, and therefore a local minimum.
 - $f(x) = 0$ exactly when $(x - 5)(x + 7) = 0$, which occurs at $x = -7$ and $x = 5$
 - Therefore, $f(x)$ has global and local minima at $x = -7$ and $x = 5$
 - So, $x = -7$ and $x = 5$ are critical points or singular points by Theorem 3.5.4.

So, all together:

$x = -1$ is a critical point, and $x = -7$ and $x = 5$ are critical points or singular points (but we don't know which).

Remark: if you would like a review of how to use the definition of the derivative, below we show that $f(x)$ is not differentiable at $x = -7$. (In fact, $x = -7$ and $x = 5$ are both singular points.)

$$\begin{aligned}
 f'(-7) &= \lim_{h \rightarrow 0} \frac{f(-7+h) - f(-7)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sqrt{|(-13+h)(h)|} - \sqrt{|0|}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sqrt{|(-13+h)(h)|}}{h}
 \end{aligned}$$

Let's first consider the case $h > 0$.

$$\begin{aligned}
 \lim_{h \rightarrow 0^+} \frac{\sqrt{|(-13+h)(h)|}}{h} &= \lim_{h \rightarrow 0^+} \frac{\sqrt{(13-h)(h)}}{h} \\
 &= \lim_{h \rightarrow 0^+} \frac{\sqrt{13h - h^2}}{\sqrt{h^2}} \\
 &= \lim_{h \rightarrow 0^+} \sqrt{\frac{13h - h^2}{h^2}} \\
 &= \lim_{h \rightarrow 0^+} \sqrt{\frac{13}{h} - 1} \\
 &= \infty
 \end{aligned}$$

Since one side of the limit doesn't exist,

$$\lim_{h \rightarrow 0} \frac{f(-7+h) - f(-7)}{h} = DNE$$

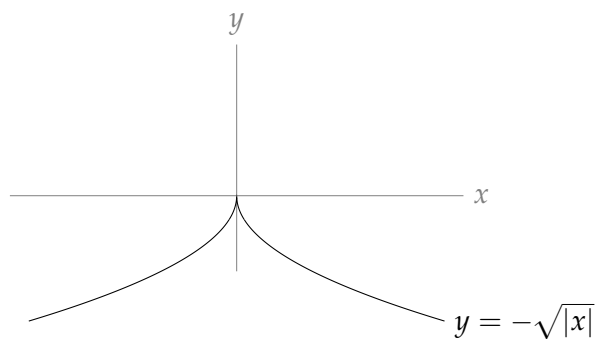
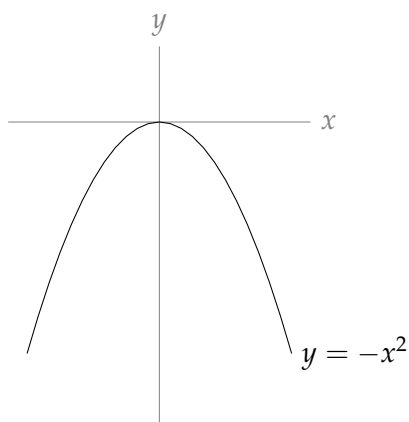
so $f'(x)$ is not differentiable at $x = -7$. Therefore, $x = -7$ is a singular point.

S-8: For any real number c , c is in the domain of $f(x)$ and $f'(c)$ exists and is equal to zero. So, following Definition 3.5.6, every real number is a critical point of $f(x)$, and $f(x)$ has no singular points.

For every number c , let $a = c - 1$ and $b = c + 1$, so $a < c < b$. Then $f(x)$ is defined for every x in the interval $[a, b]$, and $f(x) = f(c)$ for every $a \leq x \leq b$. That means $f(x) \leq f(c)$ and $f(x) \geq f(c)$. So, comparing with Definition 3.5.3, we see that $f(x)$ has a global and local maximum AND minimum at every real number $x = c$.

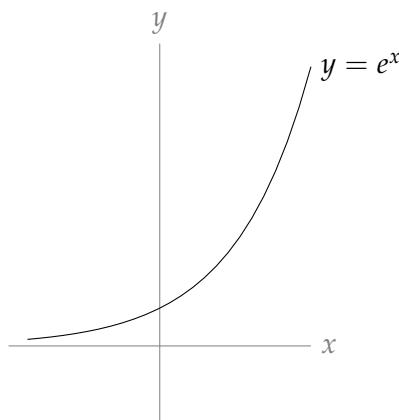
Solutions to Exercises 3.5.2 — Jump to [TABLE OF CONTENTS](#)

S-1: Two examples are given below, but many are possible.

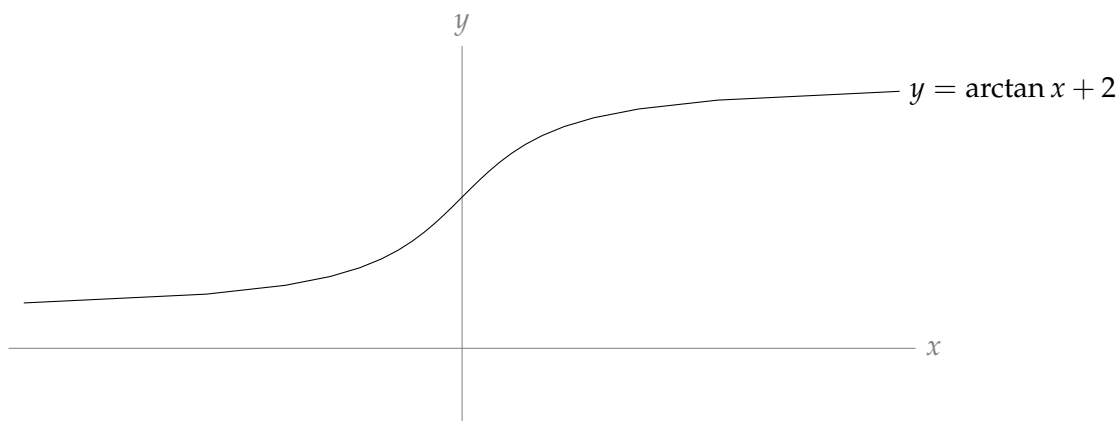


If $f(x) = -x^2$ or $f(x) = -\sqrt{|x|}$, then $f(x)$ has a global maximum at $x = 0$. Since $f(x)$ keeps getting more and more strongly negative as x gets farther and farther from 0, $f(x)$ has no global minimum.

S-2: Two examples are given below, but many are possible.



If $f(x) = e^x$, then $f(x) > 0$ for all x . As we move left along the x -axis, $f(x)$ gets smaller and smaller, approaching 0 but never reaching it. Since $f(x)$ gets smaller and smaller as we move left, there is no global minimum. Likewise, $f(x)$ increases more and more as we move right, so there is no maximum.



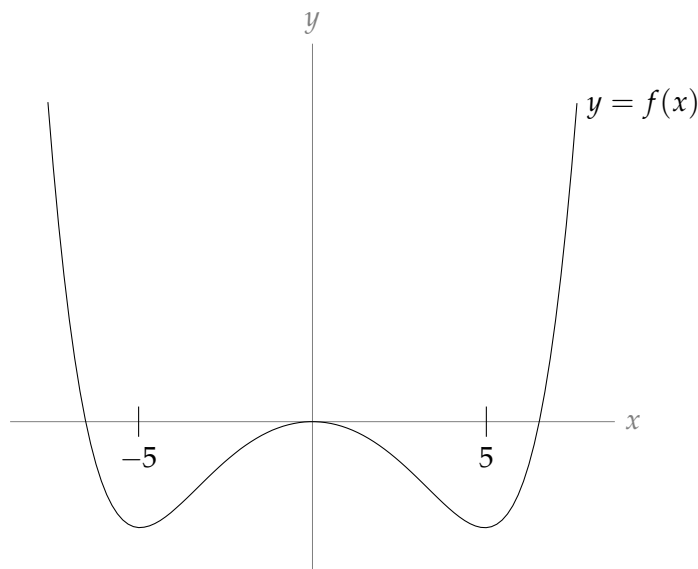
If $f(x) = \arctan(x) + 2$, then $f(x) > (-\frac{\pi}{2}) + 2 > 0$ for all x .

As we move left along the x -axis, $f(x)$ gets smaller and smaller, approaching $(-\frac{\pi}{2} + 2)$ but never reaching it. Since $f(x)$ gets smaller and smaller as we move left, there is no global minimum.

Likewise, as we move right along the x -axis, $f(x)$ gets bigger and bigger, approaching $(\frac{\pi}{2} + 2)$ but never reaching it. Since $f(x)$ gets bigger and bigger as we move right, there is no global maximum.

S-3: Since $f(5)$ is a global minimum, $f(5) \leq f(x)$ for all x , and so in particular $f(5) \leq f(-5)$. Similarly, $f(-5) \leq f(x)$ for all x , so in particular $f(-5) \leq f(5)$. Since $f(-5) \leq f(5)$ AND $f(5) \leq f(-5)$, it must be true that $f(-5) = f(5)$.

A sketch of one such graph is below.



S-4: Global extrema will occur at critical or singular points in the interval $(-5, 5)$ or at the endpoints $x = 5$, $x = -5$.

$f'(x) = 2x + 6$. Since this is defined for all real numbers, there are no singular points. The only time $f'(x) = 0$ is when $x = -3$. This is inside the interval $[-5, 5]$. So, our points to check are $x = -3$, $x = -5$, and $x = 5$.

c	-3	-5	5
type	critical point	endpoint	endpoint
$f(c)$	-19	-15	45

The global maximum is 45 at $x = 5$ and the global minimum is -19 at $x = -3$.

S-5: Global extrema will occur at the endpoints of the interval, $x = -4$ and $x = 0$, or at singular or critical points inside the interval. Since $f(x)$ is a polynomial, it is differentiable everywhere, so there are no singular points. To find the critical points, we set the derivative equal to zero.

$$\begin{aligned}
 f'(x) &= 2x^2 - 4x - 30 \\
 0 &= 2x^2 - 4x - 30 &= (2x - 10)(x + 3) \\
 x &= 5, -3
 \end{aligned}$$

The only critical point inside the interval is $x = -3$.

c	-3	-4	0
type	critical point	endpoint	endpoint
$f(c)$	61	$\frac{157}{3} = 52 + \frac{1}{3}$	7

The global maximum over the interval is 61 at $x = -3$, and the global minimum is 7 at $x = 0$.

Solutions to Exercises **3.5.3** — Jump to [TABLE OF CONTENTS](#)

S-1: We compute $f'(x) = 5x^4 - 5$, which means that $f(x)$ has no singular points (i.e., it is differentiable for all values of x), but it has two critical points:

$$\begin{aligned}
 0 &= 5x^4 - 5 \\
 0 &= x^4 - 1 = (x^2 + 1)(x^2 - 1) \\
 0 &= x^2 - 1 \\
 x &= \pm 1
 \end{aligned}$$

Note, however, that 1 is not in the interval $[-2, 0]$.

The global maximum and the global minimum for $f(x)$ on the interval $[-2, 0]$ will occur at $x = -2$, $x = 0$, or $x = -1$.

c	-2	0	-1
type	endpoint	endpoint	critical point
$f(c)$	-20	2	6

So, the global maximum is $f(-1) = 6$ while the global minimum is $f(-2) = -20$.

S-2: We compute $f'(x) = 5x^4 - 5$, which means that $f(x)$ has no singular points (i.e., it is differentiable for all values of x), but it has two critical points:

$$\begin{aligned}
 0 &= 5x^4 - 5 \\
 0 &= x^4 - 1 = (x^2 + 1)(x^2 - 1) \\
 0 &= x^2 - 1 \\
 x &= \pm 1
 \end{aligned}$$

Note, however, that -1 is not in the interval $[0, 2]$.

The global maximum and the global minimum for $f(x)$ on the interval $[0, 2]$ will occur at $x = 2$, $x = 0$, or $x = 1$.

c	2	0	1
type	endpoint	endpoint	critical point
$f(c)$	12	-10	-14

So, the global maximum is $f(2) = 12$ while the global minimum is $f(1) = -14$.

S-3: We compute $f'(x) = 6x^2 - 12x = 6x(x - 2)$, which means that $f(x)$ has no singular points (i.e., it is differentiable for all values of x), but it has the two critical points: $x = 0$ and $x = 2$. Note, however, 0 is not in the interval $[1, 4]$.

c	1	4	2
type	endpoint	endpoint	critical point
$f(c)$	-6	30	-10

So, the global maximum is $f(4) = 30$ while the global minimum is $f(2) = -10$.

S-4: Since $h(x)$ is a polynomial, it has no singular points. We compute its critical points:

$$h'(x) = 3x^2 - 12$$

$$0 = 3x^2 - 12$$

$$x = \pm 2$$

Notice as $x \rightarrow \infty$, $h(x) \rightarrow \infty$, and as $x \rightarrow -\infty$ $h(x) \rightarrow -\infty$. So Theorem 3.5.17 doesn't exactly apply. Instead, let's consider the signs of $h'(x)$.

x	$(-\infty, -2)$	$(-2, 2)$	$(2, \infty)$
$h'(x)$	> 0	< 0	> 0
$h(x)$	increasing	decreasing	increasing

So, $h(x)$ increases until $x = -2$, then decreases. That means $h(x)$ has a local maximum at $x = -2$. The function decreases from -2 until 2, after which it increases, so $h(x)$ has a local minimum at $x = 2$. We compute $f(-2) = 20$ and $f(2) = -12$.

S-5: Since $h(x)$ is a polynomial, it has no singular points. We compute its critical points:

$$h'(x) = 6x^2 - 24$$

$$0 = 6x^2 - 24$$

$$x = \pm 2$$

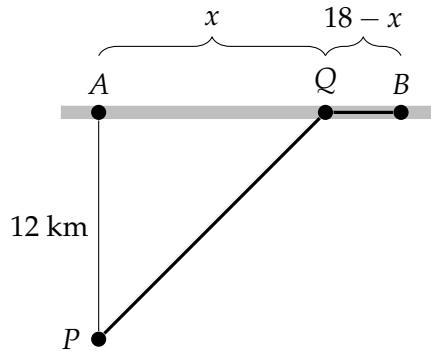
Notice as $x \rightarrow \infty$, $h(x) \rightarrow \infty$, and as $x \rightarrow -\infty$ $h(x) \rightarrow -\infty$. So Theorem 3.5.17 doesn't exactly apply. Instead, let's consider the signs of $h'(x)$.

x	$(-\infty, -2)$	$(-2, 2)$	$(2, \infty)$
$h'(x)$	> 0	< 0	> 0
$h(x)$	increasing	decreasing	increasing

So, $h(x)$ increases until $x = -2$, then decreases. That means $h(x)$ has a local maximum at $x = -2$. The function decreases from -2 until 2, after which it increases, so $h(x)$ has a local minimum at $x = 2$.

We compute $f(-2) = 33$ and $f(2) = -31$.

S-6: Suppose that Q is a distance of x from A . Then it is a distance of $18 - x$ from B .



Using the Pythagorean Theorem, the distance from P to Q is $\sqrt{12^2 + x^2}$ kilometres, and the buggy travels 15 kph over this off-road stretch. The travel time from P to Q is $\frac{\sqrt{12^2 + x^2}}{15}$ hours.

The distance from Q to B is $18 - x$ kilometres, and the dune buggy travels 30 kph along this road. The travel time from Q to B is $\frac{18 - x}{30}$ hours. So, the total travel time is

$$f(x) = \frac{\sqrt{12^2 + x^2}}{15} + \frac{18 - x}{30}.$$

We wish to minimize this for $0 \leq x \leq 18$. We will test all singular points, critical points, and endpoints to find which yields the smallest value of $f(x)$. Since there are no singular points, we begin by locating the critical points.

$$\begin{aligned} 0 = f'(x) &= \frac{1}{15} \cdot \frac{1}{2}(144 + x^2)^{-1/2}(2x) - \frac{1}{30} \\ \frac{1}{15} \cdot \frac{x}{\sqrt{144 + x^2}} &= \frac{1}{30} \\ \frac{x}{\sqrt{144 + x^2}} &= \frac{1}{2} \\ \frac{x^2}{144 + x^2} &= \frac{1}{4} \\ 4x^2 &= 144 + x^2 \\ x &= \frac{12}{\sqrt{3}} = 4\sqrt{3} \end{aligned}$$

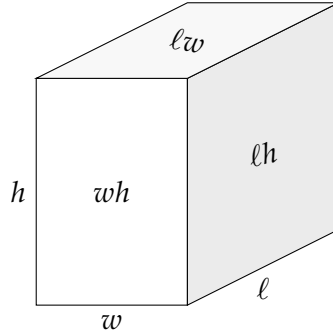
So the minimum travel times must be one of $f(0)$, $f(18)$, and $f(4\sqrt{3})$.

$$\begin{aligned} f(0) &= \frac{12}{15} + \frac{18}{30} = 1.4 \\ f(18) &= \frac{\sqrt{12^2 + 18^2}}{15} \approx 1.44 \\ f(4\sqrt{3}) &= \frac{\sqrt{144 + 144/3}}{15} + \frac{18 - 12/\sqrt{3}}{30} \approx 1.29 \end{aligned}$$

So Q should be $4\sqrt{3}$ km from A .

S-7: Let ℓ , w and h denote the length, width and height of the box respectively. We are told that $\ell wh = 4500$ and that $\ell = 3w$. Hence $h = \frac{4500}{\ell w} = \frac{4500}{3w^2} = \frac{1500}{w^2}$. The surface area of the box is

$$A = 2\ell w + 2\ell h + 2wh = 2 \left(3w^2 + 3w \frac{1500}{w^2} + w \frac{1500}{w^2} \right) = 2 \left(3w^2 + \frac{6000}{w} \right) = 6 \left(w^2 + \frac{2000}{w} \right)$$



As w tends to zero or to infinity, the surface area approaches infinity. By Theorem 3.5.17 the minimum surface area must occur at a critical point of $w^2 + \frac{2000}{w}$.

$$\begin{aligned} 0 &= \frac{d}{dw} \left\{ w^2 + \frac{2000}{w} \right\} \\ &= 2w - \frac{2000}{w^2} \\ 2w &= \frac{2000}{w^2} \\ w^3 &= 1000 \\ w &= 10 \end{aligned}$$

Therefore,

$$\begin{aligned} \ell &= 3w = 30 \\ h &= \frac{1500}{w^2} = 15. \end{aligned}$$

The dimensions of the box with minimum surface area are $10 \times 30 \times 15$.

S-8: Let the length of the sides of the square base be b metres and let the height be h metres. The area of the base is b^2 , the area of the top is b^2 and the area of each of the remaining four sides is bh so the total cost is

$$\underbrace{5(b^2)}_{\text{cost of base}} + \underbrace{1(b^2 + 4bh)}_{\text{cost of 5 sides}} = 6b^2 + 4bh = 72$$

Solving for h ,

$$\begin{aligned} h &= \frac{72 - 6b^2}{4b} \\ &= \frac{6}{4} \left(\frac{12 - b^2}{b} \right) \\ &= \frac{3}{2} \left(\frac{12 - b^2}{b} \right) \end{aligned}$$

The volume is

$$\begin{aligned} V &= b^2 h = b^2 \cdot \frac{3}{2} \left(\frac{12 - b^2}{b} \right) \\ &= 18b - \frac{3}{2} b^3. \end{aligned}$$

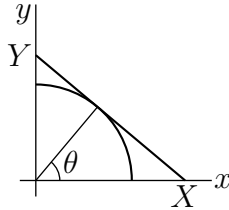
This is the function we want to maximize. Since volume is never negative, the endpoints of the functions are the values of b that make the volume 0. So, the maximum volume will not occur at an endpoint, it will occur at a critical point. The only critical point is $b = 2$:

$$\begin{aligned} 0 &= \frac{d}{db} \left\{ 18b - \frac{3}{2} b^3 \right\} \\ &= 18 - \frac{9}{2} b^2 \\ b^2 &= 4 \\ b &= 2, \quad h = \frac{3}{2} \left(\frac{12 - 4}{2} \right) = 6 \end{aligned}$$

The desired dimensions are $2 \times 2 \times 6$.

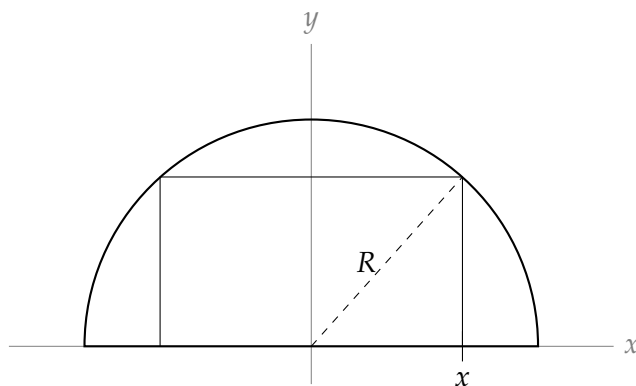
S-9: It suffices to consider X and Y such that the line XY is tangent to the circle. Otherwise we could reduce the area of the triangle by, for example, holding X fixed and reducing Y . So let X and Y be the x - and y -intercepts of the line tangent to the circle at $(\cos \theta, \sin \theta)$. Then $\frac{1}{X} = \cos \theta$ and $\frac{1}{Y} = \cos \left(\frac{\pi}{2} - \theta \right) = \sin \theta$. The area of the triangle is

$$\frac{1}{2} XY = \frac{1}{2 \cos \theta \sin \theta} = \frac{1}{\sin(2\theta)}$$



This is a minimum when $\sin(2\theta)$ is a maximum. That is when $2\theta = \frac{\pi}{2}$. Hence $X = \frac{1}{\cos(\pi/4)}$ and $Y = \frac{1}{\sin(\pi/4)}$. That is, $X = Y = \sqrt{2}$.

S-10: For ease of notation, we place the semicircle on a Cartesian plane with diameter along the x -axis and centre at the origin.



If x is the point where the rectangle touches the diameter to the right of the y -axis, then $2x$ is the width of the rectangle. The origin and the two right corners of the rectangle form a right triangle with hypotenuse R , so by the Pythagorean Theorem, the upper right hand corner of the rectangle is at $(x, \sqrt{R^2 - x^2})$. The perimeter of the rectangle is given by the function:

$$P(x) = 4x + 2\sqrt{R^2 - x^2}$$

So, this is what we optimize. The endpoints of the domain for this function are $x = 0$ and $x = R$. To find the critical points, we differentiate:

$$\begin{aligned} P'(x) &= 4 - \frac{2x}{\sqrt{R^2 - x^2}} \\ P'(x) = 0 &\iff 4 = \frac{2x}{\sqrt{R^2 - x^2}} \\ x &= 2\sqrt{R^2 - x^2} \\ x^2 &= 4(R^2 - x^2) \\ 5x^2 &= 4R^2 \\ x &= \frac{2}{\sqrt{5}}R \end{aligned}$$

Note that since our perimeter formula was defined to work only for x in $[0, R]$, we neglect the negative square root, $-\frac{2}{\sqrt{5}}R$.

Now, we find the size of the perimeter at the critical point and the endpoints:

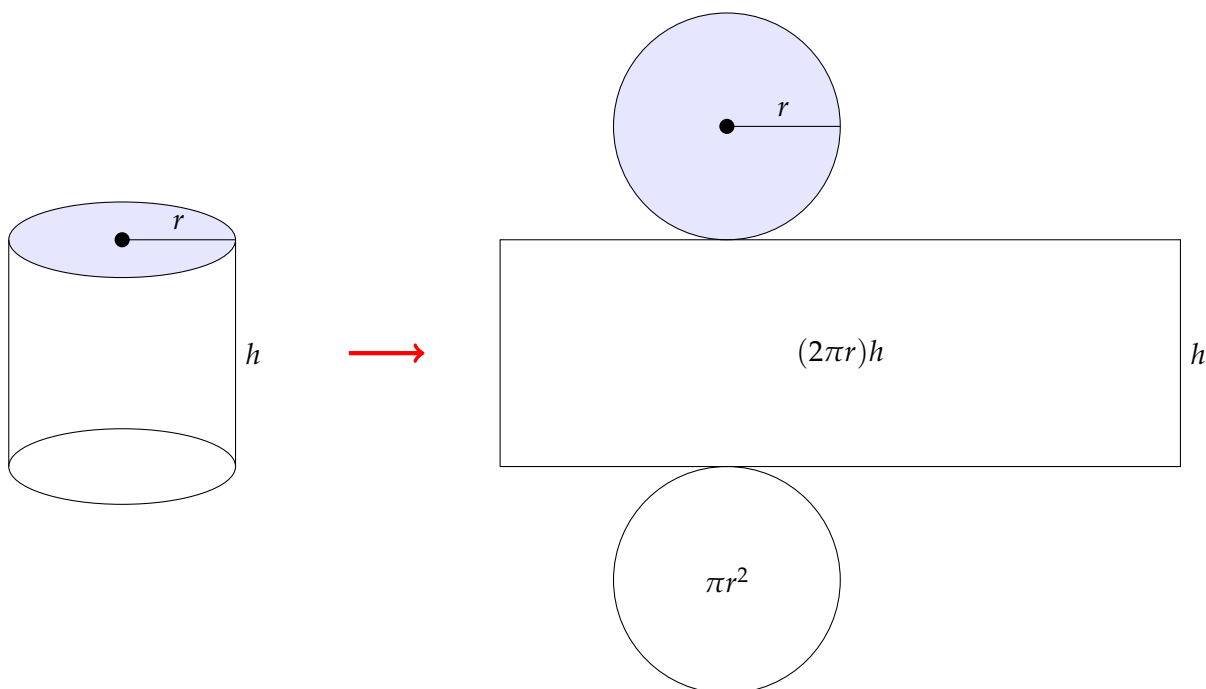
c	0	R	$\frac{2}{\sqrt{5}}R$
type	endpoint	endpoint	critical point
$P(c)$	$2R$	$4R$	$2\sqrt{5}R$

So, the largest possible perimeter is $2\sqrt{5}R$ and the smallest possible perimeter is $2R$.

Remark: as a check on the correctness of our formula for $P(x)$, when $x = 0$ the rectangle degenerates to the line segment from $(0, 0)$ to $(0, R)$. The perimeter of this “width zero rectangle” is $2R$, agreeing with $P(0)$. Similarly, when $x = R$ the rectangle degenerates to the line segment from $(R, 0)$ to $(-R, 0)$. The perimeter of this “width zero rectangle” is $4R$, agreeing with $P(R)$.

S-11: Let the cylinder have radius r and height h . If we imagine popping off the ends, they are two circular disks, each with surface area πr^2 . Then we imagine unrolling the remaining tube. It

has height h , and its other dimension is given by the circumference of the disks, which is $2\pi r$. Then the area of the “unrolled tube” is $2\pi rh$.



So, the surface area is $2\pi r^2 + 2\pi rh$. Since the area is given as A , we can solve for h :

$$\begin{aligned} A &= 2\pi r^2 + 2\pi rh \\ 2\pi rh &= A - 2\pi r^2 \\ h &= \frac{A - 2\pi r^2}{2\pi r}. \end{aligned}$$

Then we can write the volume as a function of the variable r and the constant A :

$$\begin{aligned} V(r) &= \pi r^2 h \\ &= \pi r^2 \left(\frac{A - 2\pi r^2}{2\pi r} \right) \\ &= \frac{1}{2} (Ar - 2\pi r^3) \end{aligned}$$

This is the function we want to maximize. Let's find its critical points.

$$\begin{aligned} V'(r) &= \frac{1}{2} (A - 6\pi r^2) \\ V'(r) = 0 &\iff A = 6\pi r^2 \iff r = \sqrt{\frac{A}{6\pi}} \end{aligned}$$

since negative values of r don't make sense. At this critical point,

$$\begin{aligned} V\left(\sqrt{\frac{A}{6\pi}}\right) &= \frac{1}{2} \left[A \left(\sqrt{\frac{A}{6\pi}} \right) - 2\pi \left(\sqrt{\frac{A}{6\pi}} \right)^3 \right] \\ &= \frac{1}{2} \left[\frac{A^{3/2}}{\sqrt{6\pi}} - \frac{2\pi A^{3/2}}{6\pi\sqrt{6\pi}} \right] \\ &= \frac{1}{2} \left[\frac{A^{3/2}}{\sqrt{6\pi}} - \frac{A^{3/2}}{3\sqrt{6\pi}} \right] \\ &= \frac{A^{3/2}}{3\sqrt{6\pi}}. \end{aligned}$$

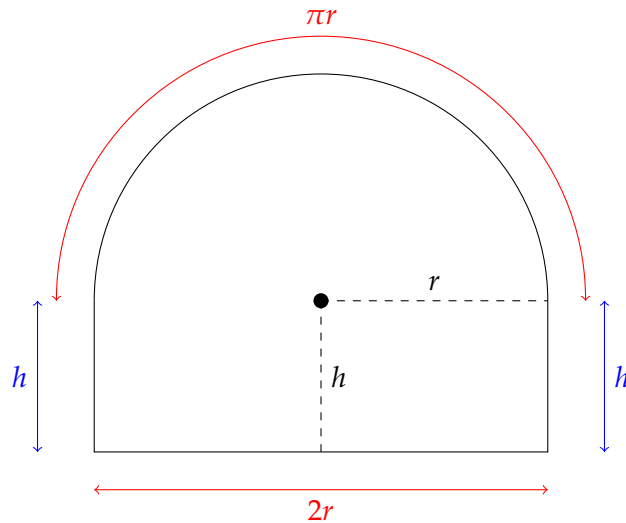
We should also check the volume of the cylinder at the endpoints of the function. Since $r \geq 0$, one endpoint is $r = 0$. Since $h \geq 0$, and r grows as h shrinks, the other endpoint is whatever value of r causes h to be 0. We could find this value of r , but it's not strictly necessary: when $r = 0$, the volume of the cylinder is zero, and when $h = 0$, the volume of the cylinder is still zero. So, the maximum volume does not occur at the endpoints.

Therefore, the maximum volume is achieved at the critical point, where

$$V_{\max} = \frac{A^{3/2}}{3\sqrt{6\pi}}.$$

Remark: as a check, A has units m^2 and, because of the $A^{3/2}$, our answer has units m^3 , which are the correct units for a volume.

S-12: Denote by r the radius of the semicircle, and let h be the height of the rectangle.



Since the perimeter is required to be P , the height, h , of the rectangle must obey

$$\begin{aligned} P &= \pi r + 2r + 2h \\ h &= \frac{1}{2}(P - \pi r - 2r) \end{aligned}$$

So the area is

$$\begin{aligned} A(r) &= \frac{1}{2}\pi r^2 + 2rh \\ &= \frac{1}{2}\pi r^2 + r(P - \pi r - 2r) \\ &= rP - \frac{1}{2}(\pi + 4)r^2 \end{aligned}$$

Finding all critical points:

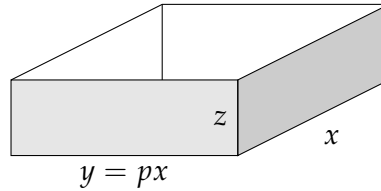
$$\begin{aligned} 0 &= A'(r) = P - (\pi + 4)r \\ r &= \frac{P}{\pi + 4} \end{aligned}$$

Now we want to know what radius yields the maximum area. We notice that $A'(r) > 0$ for $r < \frac{P}{\pi + 4}$ and $A'(r) < 0$ for $r > \frac{P}{\pi + 4}$. So, $A(r)$ is increasing until the critical point, then decreasing after it. That means *the global maximum occurs at the critical point*, $r = \frac{P}{\pi + 4}$. The maximum area is

$$\begin{aligned} rP - \frac{1}{2}(\pi + 4)r^2 &= \frac{P^2}{\pi + 4} - \frac{1}{2}(\pi + 4)\frac{P^2}{(\pi + 4)^2} \\ &= \frac{P^2}{2(\pi + 4)} \end{aligned}$$

Remark: another way to see that the global maximum occurs at the critical point is to compare the area at the critical point to the areas at the endpoints of the function. The smallest value of r is 0, while the biggest is $\frac{P}{\pi + 2}$ (when the shape is simply a half-circle). Comparing $A(0)$, $A\left(\frac{P}{\pi + 2}\right)$, and $A\left(\frac{P}{\pi + 4}\right)$ is somewhat laborious, but certainly possible.

S-13:



(a) The surface area of the pan is

$$\begin{aligned} xy + 2xz + 2yz &= px^2 + 2xz + 2pxz \\ &= px^2 + 2(1 + p)xz \end{aligned}$$

and the volume of the pan is $xyz = px^2z$. Assuming that all $A \text{ cm}^2$ is used, we have the constraint

$$px^2 + 2(1 + p)xz = A \quad \text{or} \quad z = \frac{A - px^2}{2(1 + p)x}$$

So

$$\begin{aligned} V(x) &= xyz = x(px) \left(\frac{A - px^2}{2(1 + p)x} \right) \\ &= \frac{p}{2(1 + p)} x(A - px^2) \end{aligned}$$

Using the product rule,

$$\begin{aligned} V'(x) &= \frac{p}{2(1+p)} [x(-2px) + (A - px^2)] \\ &= \frac{p}{2(1+p)} [A - 3px^2] \end{aligned}$$

The derivative $V'(x)$ is 0 when $x = \sqrt{\frac{A}{3p}}$. The derivative is positive (i.e. $V(x)$ is increasing) for $x < \sqrt{\frac{A}{3p}}$ and is negative (i.e. $V(x)$ is decreasing) for $x > \sqrt{\frac{A}{3p}}$. So the pan of maximum volume has dimensions $x = \sqrt{\frac{A}{3p}}$, $y = p\sqrt{\frac{A}{3p}} = \sqrt{\frac{Ap}{3}}$ and $z = \frac{2A/3}{2(1+p)\sqrt{A/(3p)}} = \frac{\sqrt{Ap}}{\sqrt{3}(1+p)}$.

(b) The volume of the pan from part (a) is

$$V(p) = \left(\sqrt{\frac{A}{3p}}\right) \left(p\sqrt{\frac{A}{3p}}\right) \frac{\sqrt{Ap}}{\sqrt{3}(1+p)} = \left(\frac{A}{3}\right)^{3/2} \frac{\sqrt{p}}{1+p}$$

Since

$$\frac{d}{dp} \left\{ \frac{\sqrt{p}}{1+p} \right\} = \frac{\frac{1}{2}(1+p)/\sqrt{p} - \sqrt{p}}{(1+p)^2} = \frac{\sqrt{p} \left(\frac{1}{p} - 1 \right)}{2(1+p)^2}$$

the volume is increasing with p for $p < 1$ and decreasing with p for $p > 1$. So the maximum volume is achieved for $p = 1$ (a square base).

S-14: (a) We use logarithmic differentiation.

$$\begin{aligned} f(x) &= x^x \\ \log f(x) &= \log(x^x) = x \log x \\ \frac{d}{dx} \{\log f(x)\} &= \frac{d}{dx} \{x \log x\} \\ \frac{f'(x)}{f(x)} &= x \left(\frac{1}{x} \right) + \log x = 1 + \log x \\ f'(x) &= f(x) (1 + \log x) = x^x (1 + \log x) \end{aligned}$$

(b) Since $x > 0$, $x^x > 0$. Therefore,

$$f'(x) = 0 \iff 1 + \log x = 0 \iff \log x = -1 \iff x = \frac{1}{e}$$

(c) Since $x > 0$, $x^x > 0$. So, the sign of $f'(x)$ is the same as the sign of $1 + \log x$.

For $x < \frac{1}{e}$, $\log x < -1$ and $f'(x) < 0$. That is, $f(x)$ decreases as x increases, when $x < \frac{1}{e}$. For $x > \frac{1}{e}$, $\log x > -1$ and $f'(x) > 0$. That is, $f(x)$ increases as x increases, when $x > \frac{1}{e}$. Hence $f(x)$ is a local minimum at $x = \frac{1}{e}$.

S-15: Call the length of the wire L units and suppose that it is cut ℓ units from one end. Make the square from the piece of length ℓ , and make the circle from the remaining piece of length $L - \ell$.

The square has perimeter ℓ , so its side length is $\ell/4$ and its area is $\left(\frac{\ell}{4}\right)^2$. The circle has circumference $L - \ell$, so its radius is $\frac{L - \ell}{2\pi}$ and its area is $\pi \left(\frac{L - \ell}{2\pi}\right)^2 = \frac{(L - \ell)^2}{4\pi}$.

The area enclosed by the shapes, when the square is made from a length of size ℓ , is

$$A(\ell) = \frac{\ell^2}{16} + \frac{(L - \ell)^2}{4\pi}$$

We want to find the global max and min for this function, given the constraint $0 \leq \ell \leq L$, so we find its derivative:

$$A'(\ell) = \frac{\ell}{8} - \frac{L - \ell}{2\pi} = \frac{\pi + 4}{8\pi}\ell - \frac{L}{2\pi}$$

Now, we find the critical point.

$$\begin{aligned} A'(\ell) &= 0 \\ \frac{\pi + 4}{8\pi}\ell &= \frac{L}{2\pi} \\ \ell &= \frac{4L}{\pi + 4} \end{aligned}$$

ℓ	0	L	$\frac{4L}{\pi+4}$
type	endpoint	endpoint	critical point
$A(\ell)$	$\frac{L^2}{4\pi}$	$\frac{L^2}{16}$	$A\left(\frac{4L}{\pi+4}\right)$

It seems obnoxious to evaluate $A\left(\frac{4L}{\pi+4}\right)$, and the problem doesn't ask for it—but we still have to figure out whether it is a global max or min.

When $\ell < \frac{4L}{\pi+4}$, $A'(\ell) < 0$, and when $\ell > \frac{4L}{\pi+4}$, $A'(\ell) > 0$. So, $A(\ell)$ is decreasing until $\ell = \frac{4L}{\pi+4}$, then increasing. That means our critical point $\ell = \frac{4L}{\pi+4}$ is a local minimum.

So, the minimum occurs at the only critical point, which is $\ell = \left(\frac{4}{4 + \pi}\right)L$. This corresponds to $\frac{\ell}{L} = \frac{4}{4 + \pi}$: the proportion of the wire that is cut is $\frac{4}{4 + \pi}$.

The maximum has to be either at $\ell = 0$ or at $\ell = L$. As $A(0) = \frac{L^2}{4\pi} > A(L) = \frac{L^2}{16}$, the maximum has $\ell = 0$ (that is, no square).

Solutions to Exercises 3.6.1—Jump to [TABLE OF CONTENTS](#)

S-1: In general, this is false. For example, the function $f(x) = \frac{x^2 - 9}{x^2 - 9}$ has no vertical asymptotes, because it is equal to 1 in every point in its domain (and is undefined when $x = \pm 3$).

However, it is certainly *possible* that $f(x)$ has a vertical asymptote at $x = -3$. For example, $f(x) = \frac{1}{x^2 - 9}$ has a vertical asymptote at $x = -3$. More generally, if $g(x)$ is continuous and $g(-3) \neq 0$, then $f(x)$ has a vertical asymptote at $x = -3$.

S-2: Since $x^2 + 1$ and $x^2 + 4$ are always positive, $f(x)$ and $h(x)$ are defined over all real numbers. So, $f(x)$ and $h(x)$ correspond to $A(x)$ and $B(x)$. Which is which? $A(0) = 1 = f(0)$ while $B(0) = 2 = h(0)$, so $A(x) = f(x)$ and $B(x) = h(x)$.

That leaves $g(x)$ and $k(x)$ matching to $C(x)$ and $D(x)$. The domain of $g(x)$ is all x such that $x^2 - 1 \geq 0$. That is, $|x| \geq 1$, like $C(x)$. The domain of $k(x)$ is all x such that $x^2 - 4 \geq 0$. That is, $|x| \geq 2$, like $D(x)$. So, $C(x) = g(x)$ and $D(x) = k(x)$.

S-3: (a) Since $f(0) = 2$, we solve

$$\begin{aligned} 2 &= \sqrt{\log^2(0 + p)} \\ &= \sqrt{\log^2 p} \\ &= |\log p| \\ \log p &= \pm 2 \\ p &= e^{\pm 2} \\ p &= e^2 \text{ or } p = \frac{1}{e^2} \end{aligned}$$

We know that p is e^2 or $\frac{1}{e^2}$, but we have to decide between the two. In both cases, $f(0) = 2$. Let's consider the domain of $f(x)$. Since $\log^2(x + p)$ is never negative, the square root does not restrict our domain. However, we can only take the logarithm of positive numbers. Therefore, the domain is

$$\begin{aligned} &x \text{ such that } x + p > 0 \\ &x \text{ such that } x > -p \end{aligned}$$

If $p = \frac{1}{e^2}$, then the domain of $f(x)$ is $\left(-\frac{1}{e^2}, \infty\right)$. In particular, since $-\frac{1}{e^2} > -1$, the domain of $f(x)$ does not include $x = -1$. However, it is clear from the graph that $f(-1)$ exists. So, $p = e^2$.

(b) Now, we need to figure out what b is. Notice that b is the end of the domain of $f(x)$, which we already found to be $(-p, \infty)$. So, $b = -p = -e^2$.

(As a quick check, if we take $e \approx 2.7$, then $-e^2 = -7.29$, and this looks about right on the graph.)

(c) The x -intercept is the value of x for which $f(x) = 0$:

$$\begin{aligned} 0 &= \sqrt{\log^2(x + p)} \\ 0 &= \log(x + p) \\ 1 &= x + p \\ x &= 1 - p = 1 - e^2 \end{aligned}$$

The x -intercept is $1 - e^2$.

(As another quick check, the x -intercept we found is a distance of 1 from the vertical asymptote, and this looks about right on the graph.)

S-4: Vertical asymptotes occur where the function blows up. In rational functions, this can only happen when the denominator goes to 0. In our case, the denominator is 0 when $x = 3$, and in this case the numerator is 147. That means that as x gets closer and closer to 3, the numerator gets closer and closer to 147 while the denominator gets closer and closer to 0, so $|f(x)|$ grows without bound. That is, there is a vertical asymptote at $x = 3$.

The horizontal asymptotes are found by taking the limits as x goes to infinity and negative infinity. In our case, they are the same, so we condense our work.

$$\lim_{x \rightarrow \pm\infty} \frac{x(2x+1)(x-7)}{3x^3-81} = \lim_{x \rightarrow \pm\infty} \frac{2x^3 + ax^2 + bx + c}{3x^3 - 81}$$

where a, b , and c are some constants. Remember, for rational functions, you can figure out the end behaviour by looking only at the terms with the highest degree—the others won't matter, so we don't bother finding them. From here, we divide the numerator and denominator by the highest power of x in the denominator, x^3 .

$$\begin{aligned} &= \lim_{x \rightarrow \pm\infty} \frac{2x^3 + ax^2 + bx + c}{3x^3 - 81} \left(\frac{\frac{1}{x^3}}{\frac{1}{x^3}} \right) \\ &= \lim_{x \rightarrow \pm\infty} \frac{2 + \frac{a}{x} + \frac{b}{x^2} + \frac{c}{x^3}}{3 - \frac{81}{x^3}} \\ &= \frac{2 + 0 + 0 + 0}{3 - 0} = \frac{2}{3} \end{aligned}$$

So there is a horizontal asymptote of $y = \frac{2}{3}$ both as $x \rightarrow \infty$ and as $x \rightarrow -\infty$.

S-5: Since $f(x)$ is continuous over all real numbers, it has no vertical asymptote.

To find the horizontal asymptotes, we evaluate $\lim_{x \rightarrow \pm\infty} f(x)$.

$$\lim_{x \rightarrow \infty} 10^{3x-7} = \underbrace{\lim_{X \rightarrow \infty} 10^X}_{\text{let } X=3x-7} = \infty$$

So, there's no horizontal asymptote as $x \rightarrow \infty$.

$$\begin{aligned} \lim_{x \rightarrow -\infty} 10^{3x-7} &= \underbrace{\lim_{X \rightarrow -\infty} 10^X}_{\text{let } X=3x-7} \\ &= \underbrace{\lim_{X' \rightarrow \infty} 10^{-X'}}_{\text{let } X'=-X} \\ &= \lim_{X' \rightarrow \infty} \frac{1}{10^{X'}} \\ &= 0 \end{aligned}$$

That is, $y = 0$ is a horizontal asymptote as $x \rightarrow -\infty$.

Solutions to Exercises 3.6.2 — Jump to [TABLE OF CONTENTS](#)

S-1: Functions $A(x)$ and $B(x)$ share something in common that sets them apart from the others: they have a horizontal tangent line only once. In particular, $A'(-2) \neq 0$ and $B'(2) \neq 0$. The only listed functions that do not have two distinct roots are $l(x)$ and $p(x)$. Since $l(-2) \neq 0$ and $p(2) \neq 0$, we conclude

$$A'(x) = l(x) \quad B'(x) = p(x)$$

Function $C(x)$ is never decreasing. Its tangent line is horizontal when $x = \pm 2$, but the curve never decreases, so $C'(x) \geq 0$ for all x and $C'(2) = C'(-2) = 0$. The only function that matches this is $n(x) = (x-2)^2(x+2)^2$. Since its linear terms have even powers, it is never negative, and its roots are precisely $x = \pm 2$.

$$C'(x) = n(x)$$

For the functions $D(x)$ and $E(x)$ we consider their behaviour near $x = 0$. $D(x)$ is decreasing near $x = 0$, so $D'(0) < 0$, which matches with $o(0) < 0$. Contrastingly, $E(x)$ is increasing near zero, so $E'(0) > 0$, which matches with $m(0) > 0$.

$$D'(x) = o(x) \quad E'(x) = m(x)$$

S-2: The domain of $f(x)$ is all real numbers except -3 (because when $x = -3$ the denominator is zero). For $x \neq -3$, we differentiate using the quotient rule:

$$f'(x) = \frac{e^x(x+3) - e^x(1)}{(x+3)^2} = \frac{e^x}{(x+3)^2}(x+2)$$

Since e^x and $(x+3)^2$ are positive for every x in the domain of $f(x)$, the sign of $f'(x)$ is the same as the sign of $x+2$. We conclude that $f(x)$ is increasing for every x in its domain with $x+2 > 0$. That is, over the open interval $(-2, \infty)$.

S-3: Since we can't take the square root of a negative number, $f(x)$ is only defined when $x \geq 1$. Furthermore, since we can't have zero as a denominator, $x = -2$ is not in the domain — but as long as $x \geq 1$, we also have $x \neq -2$. So, the domain of the function is $[1, \infty)$.

In order to find where $f(x)$ is increasing, we find where $f'(x)$ is positive.

$$f'(x) = \frac{\frac{2x+4}{2\sqrt{x-1}} - 2\sqrt{x-1}}{(2x+4)^2} = \frac{(x+2) - 2(x-1)}{\sqrt{x-1}(2x+4)^2} = \frac{-x+4}{\sqrt{x-1}(2x+4)^2}$$

The denominator is never negative, so $f(x)$ is increasing when the numerator of $f'(x)$ is positive, i.e. when $4-x > 0$, or $x < 4$. Recalling that the domain of definition for $f(x)$ is $[1, \infty)$, we conclude that $f(x)$ is increasing on the open interval $(1, 4)$.

S-4: The domain of arctangent is all real numbers. The domain of the logarithm function is all positive numbers, and $1+x^2$ is positive for all x . So, the domain of $f(x)$ is all real numbers.

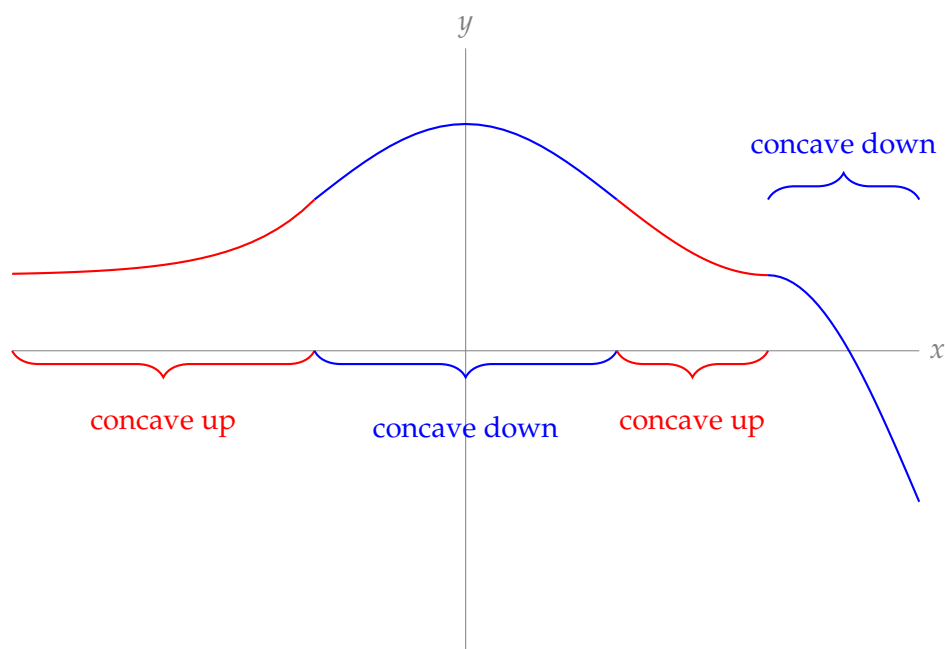
In order to find where $f(x)$ is increasing, we find where $f'(x)$ is positive.

$$f'(x) = \frac{2}{1+x^2} - \frac{2x}{1+x^2} = \frac{2-2x}{1+x^2}$$

Since the denominator is always positive, $f(x)$ is increasing when $2-2x > 0$. We conclude that $f(x)$ is increasing on the open interval $(-\infty, 1)$.

Solutions to Exercises 3.6.3 — Jump to [TABLE OF CONTENTS](#)

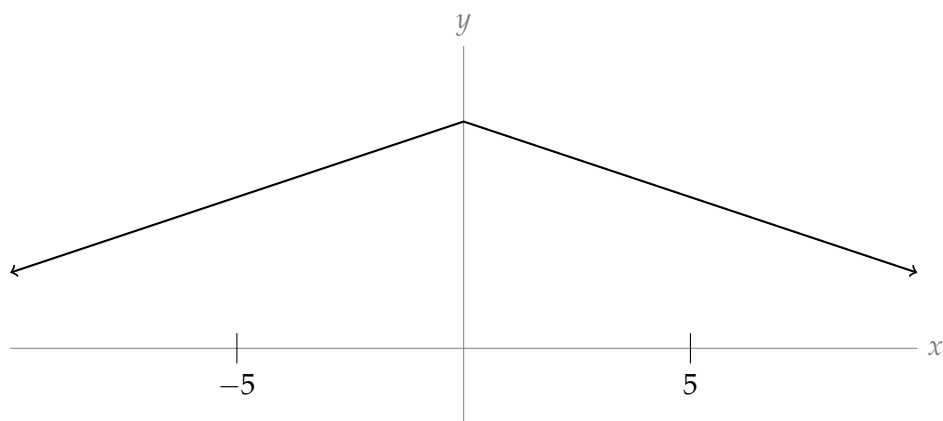
S-1:



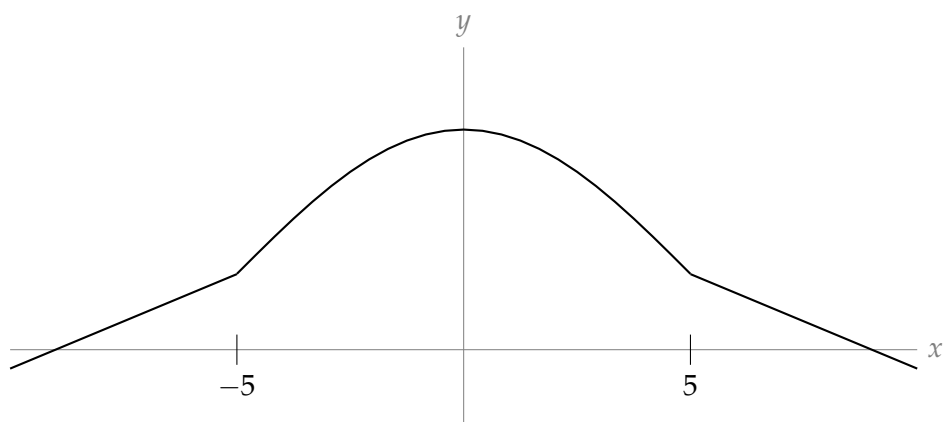
In the graph above, the concave-up sections are marked in red. These are where the graph has an increasing derivative; equivalently, where the graph lies above its tangent lines; more descriptively, where it curves like a smiley face.

Concave-down sections are marked in blue. These are where the graph has a decreasing derivative; equivalently, where the graph lies below its tangent lines; more descriptively, where it curves like a frowney face.

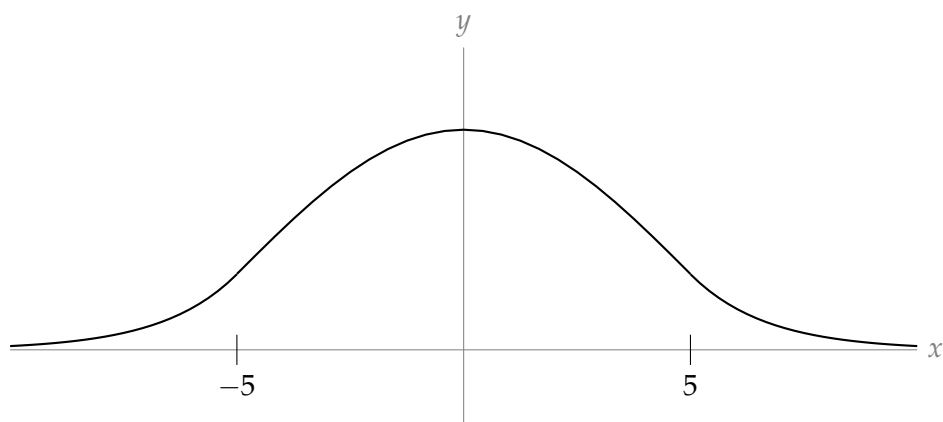
S-2: The most basic shape of the graph is given by the last two bullet points:



The curve is concave down over the interval $(-5, 5)$, so let's give it a frowney-face curvature there.



Finally, when $x > 5$ or $x < -5$, our curve should be concave up, so let's give it smiley-face curvature there, without changing its basic increasing/decreasing shape.



This finishes our sketch.

S-3: An inflection point is where the concavity of a function changes. It is possible that $x = 3$ is an inflection point, but it is also possible that it is not. So, the statement is false, in general.

For example, let $f(x) = (x - 3)^4$. Since $f(x)$ is a polynomial, all its derivatives exist and are continuous. $f''(x) = 12(x - 3)^2$, so $f''(3) = 0$. However, since $f''(x)$ is something squared, it is never negative, so $f(x)$ is never concave down. Since $f(x)$ is never concave down, it never

changes concavity, so it has no inflection points.

Remark: finding inflection points is somewhat reminiscent of finding local extrema. To find local extrema, we first find all critical and singular points, since local extrema can only occur there or at endpoints. Then, we have to figure out which critical and singular points are actually local extrema. Similarly, if you want to find inflection points, start by finding where $f''(x)$ is zero or non-existent, because inflection points can only occur there (see Question 7). Then, you still have to check whether those points are actually inflection points.

S-4: Inflection points occur where $f''(x)$ changes sign. Since $f(x)$ is a polynomial, its first and second derivatives exist everywhere, and are themselves polynomials. In particular,

$$\begin{aligned}f(x) &= 3x^5 - 5x^4 + 13x \\f'(x) &= 15x^4 - 20x^3 + 13 \\f''(x) &= 60x^3 - 60x^2 = 60x^2(x - 1)\end{aligned}$$

The second derivative is negative for $x < 1$ and positive for $x > 1$. Thus the concavity changes between concave up and concave down at $x = 1$, $y = 11$.

This is the only inflection point. It is true that $f''(0) = 0$, but for values of x both a little larger than and a little smaller than 0, $f''(x) < 0$, so the concavity does not change at $x = 0$.

S-5: In order to show that $f(x)$ has *exactly one* inflection point, we will show that it has *at least one*, and *no more than one*.

Let

$$g(x) = f''(x) = x^3 + 5x - 20.$$

Then $g'(x) = 3x^2 + 5$, which is always positive. That means $g(x)$ is strictly increasing for all x . So, $g(x)$ can change signs once, from negative to positive, but it can never change back to negative. An inflection point of $f(x)$ occurs when $g(x)$ changes signs. So, $f(x)$ has at most one inflection point. (At this point, we don't know that $f(x)$ has any inflection points: maybe $g(x)$ is always positive.)

Since $g(x)$ is continuous, we can apply the Intermediate Value Theorem to it. Notice $g(3) > 0$ while $g(0) < 0$. By the IVT, $g(x) = 0$ for at least one $x \in (0, 3)$. Since $g(x)$ is strictly increasing, at the point where $g(x) = 0$, $g(x)$ changes from negative to positive. So, the concavity of $f(x)$ changes. Therefore, $f(x)$ has at least one inflection point.

Now that we've shown that $f(x)$ has at most one inflection point, and at least one inflection point, we conclude it has exactly one inflection point.

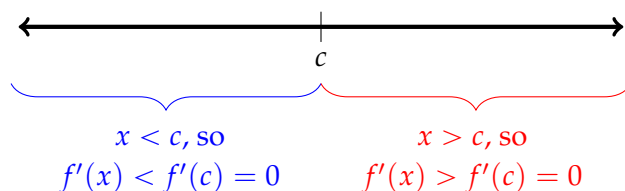
S-6: (a) Let

$$g(x) = f'(x)$$

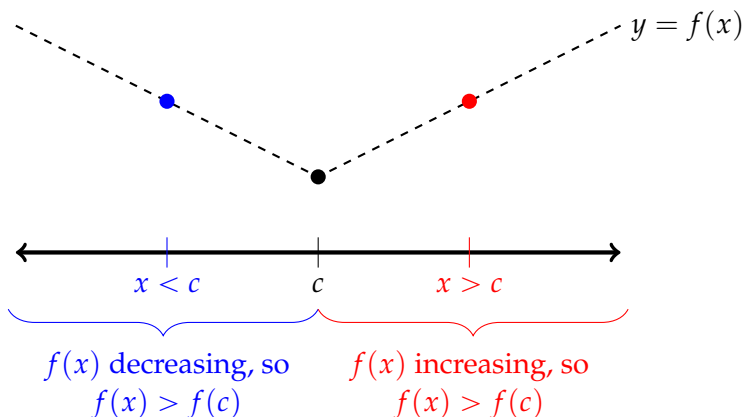
Then $f''(x)$ is the derivative of $g(x)$. Since $f''(x) > 0$ for all x , $g(x) = f'(x)$ is strictly increasing for all x . In other words, if $y > x$ then $g(y) > g(x)$.

Suppose $g(x) = 0$. Then for every y that is larger than x , $g(y) > g(x)$, so $g(y) \neq 0$. Similarly, for every y that is smaller than x , $g(y) < g(x)$, so $g(y) \neq 0$. Therefore, $g(x)$ can only be zero for at most one value of x . Since $g(x) = f'(x)$, that means $f(x)$ can have at most one critical point.

Suppose $f'(c) = 0$. Since $f'(x)$ is a strictly increasing function, $f'(x) < 0$ for all $x < c$ and $f'(x) > 0$ for all $x > c$.



Then $f(x)$ is decreasing for $x < c$ and increasing for $x > c$. So $f(x) > f(c)$ for all $x < c$ and $f(x) > f(c)$ for all $x > c$.



We have concluded that $f(x) > f(c)$ for all $x \neq c$, so c is an absolute minimum for $f(x)$.

(b) We know that the maximum over an interval occurs at an endpoint, at a critical point, or at a singular point.

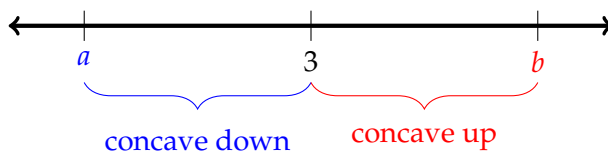
- Since $f'(x)$ exists everywhere, there are no singular points.
- If the maximum were achieved at a critical point, that critical point would have to provide both the absolute maximum and the absolute minimum (by part a). So, the function would have to be a constant and consequently could not have a nonzero second derivative. So the maximum is not at a critical point.

That leaves only the endpoints of the interval.

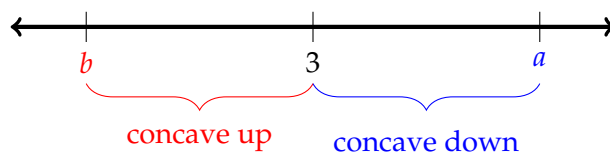
S-7: If $x = 3$ is an inflection point, then the concavity of $f(x)$ changes at $x = 3$. That is, there is some interval strictly containing 3, with endpoints a and b , such that

- $f''(a) < 0$ and $f''(x) < 0$ for every x between a and 3, and
- $f''(b) > 0$ and $f''(x) > 0$ for every x between b and 3.

Remark: we are leaving unknown whether $a < 3 < b$ or $b < 3 < a$. Since we don't know whether $f(x)$ changes from concave up to concave down, or from concave down to concave up, by remaining vague we cover both cases.



OR



Since $f''(a) < 0$ and $f''(b) > 0$, and since $f''(x)$ is continuous, the Intermediate Value Theorem tells us that there exists some x strictly between a and b with $f''(x) = 0$. So, we know $f''(x) = 0$ somewhere between a and b . The question is, where exactly could that be?

- $f''(x) < 0$ (and hence $f''(x) \neq 0$) for all x between a and 3
- $f''(x) > 0$ (and hence $f''(x) \neq 0$) for all x between b and 3
- So, any number between a and b that is not 3 has $f''(x) \neq 0$.

So, $x = 3$ is the only possible place between a and b where $f''(x)$ could be zero. Therefore, $f''(3) = 0$.

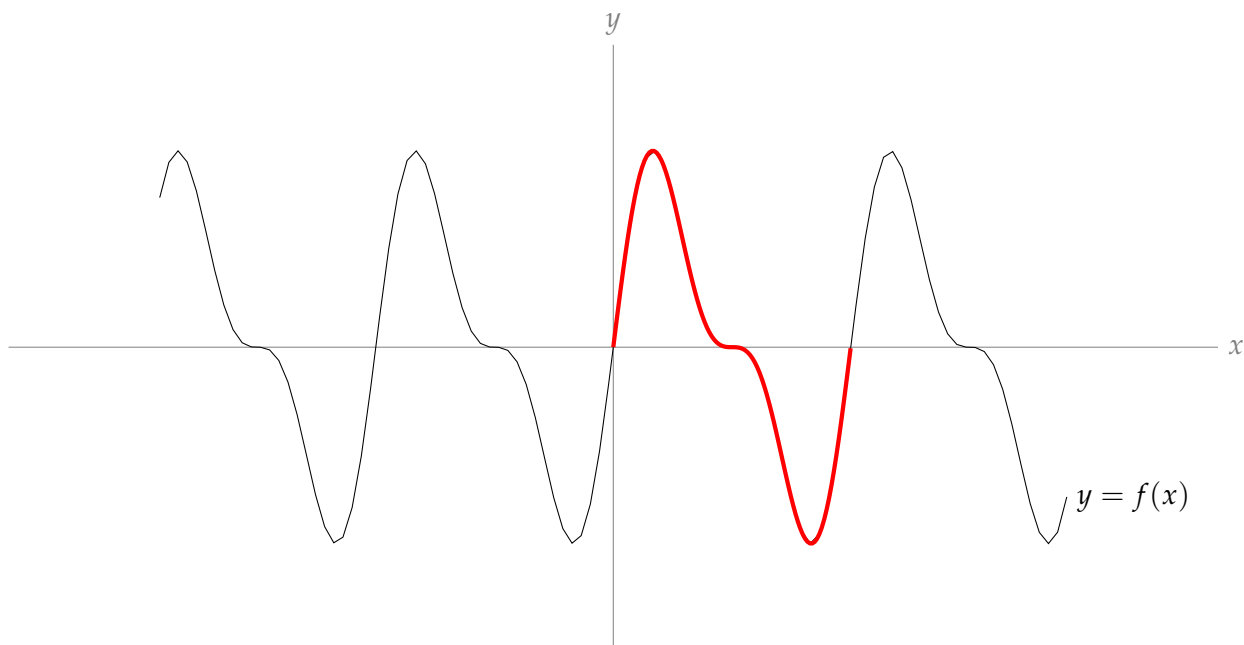
Remark: this is why, in general, we set $f''(x) = 0$ to find inflection points. (They can also occur where $f''(x)$ does not exist.)

Solutions to Exercises 3.6.4 — Jump to [TABLE OF CONTENTS](#)

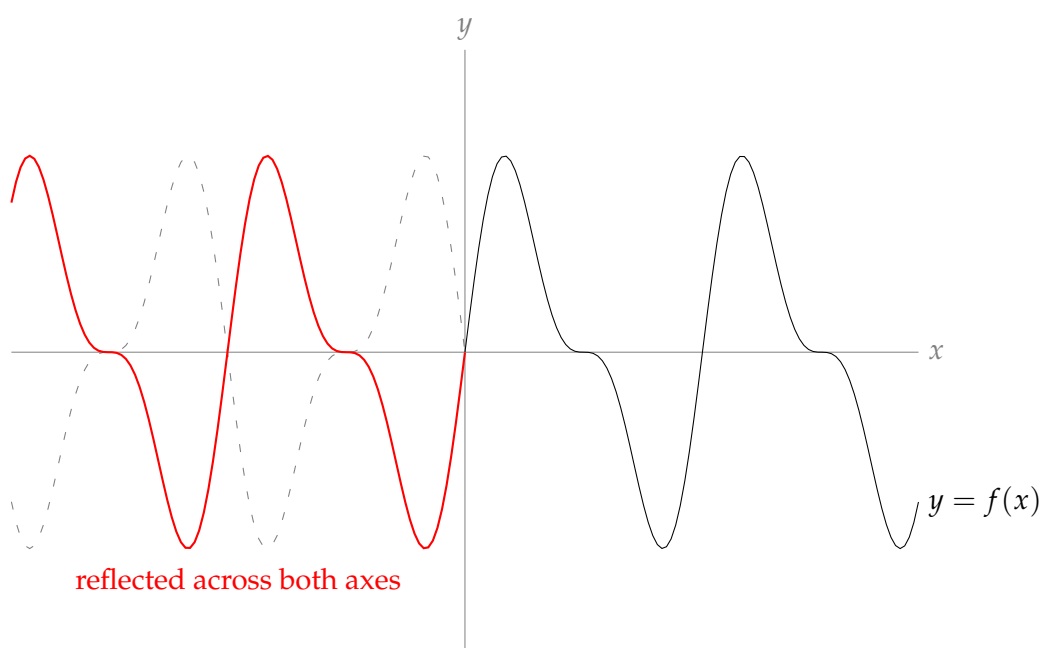
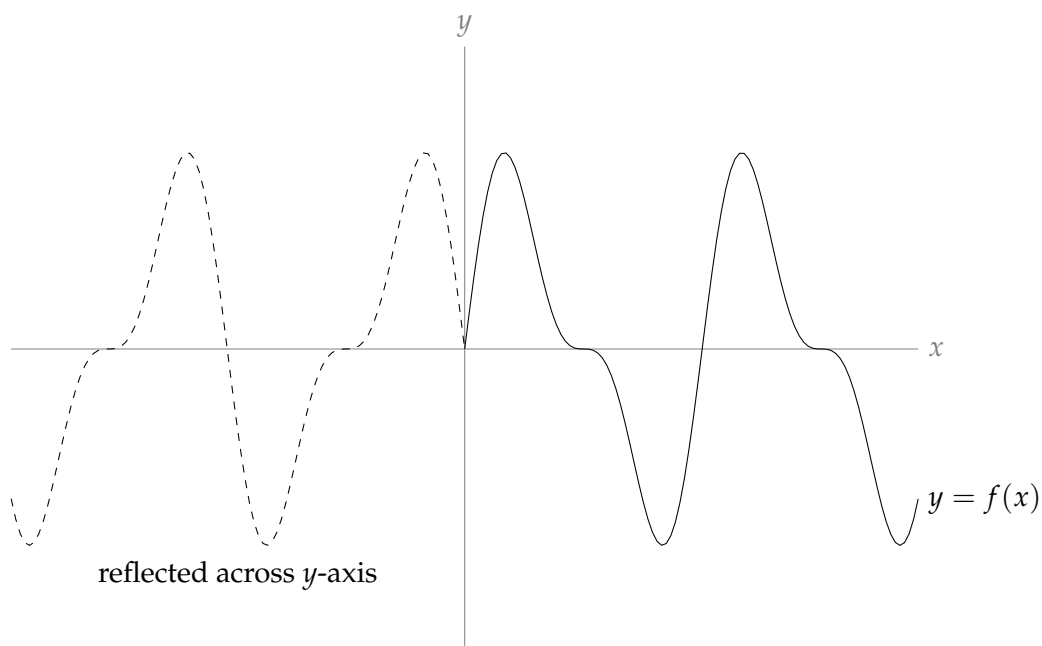
S-1: This function is symmetric across the y -axis, so it is even.

S-2: The function is not even, because it is not mirrored across the y -axis.

Assuming it continues as shown, the function is periodic, because the unit shown below is repeated:

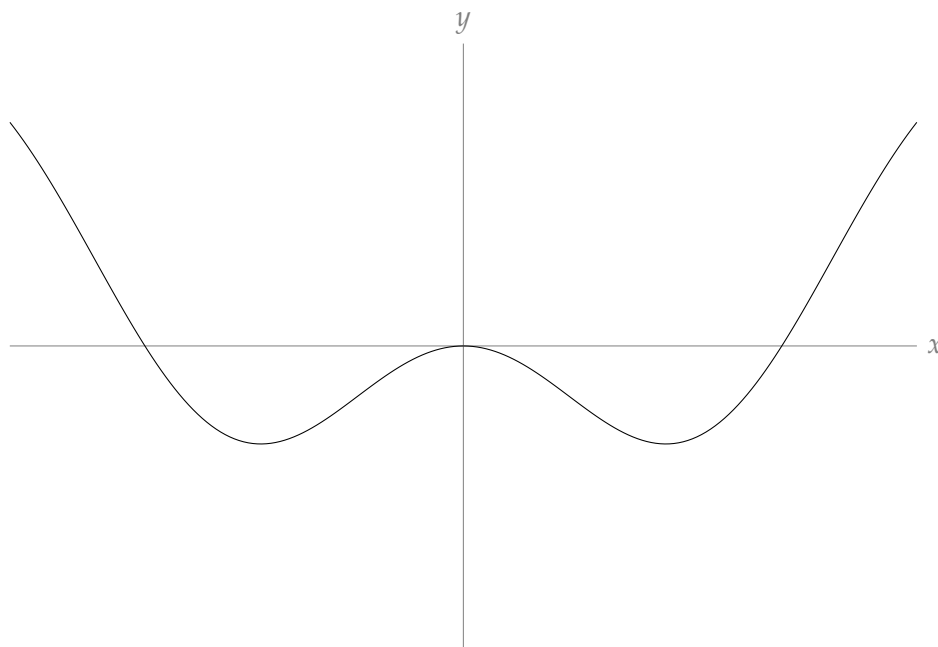


Additionally, $f(x)$ is odd. In a function with odd symmetry, if we mirror the right-hand portion of the curve (the portion to the right of the y -axis) across both the y -axis and the x -axis, it lines up with the left-hand portion of the curve.

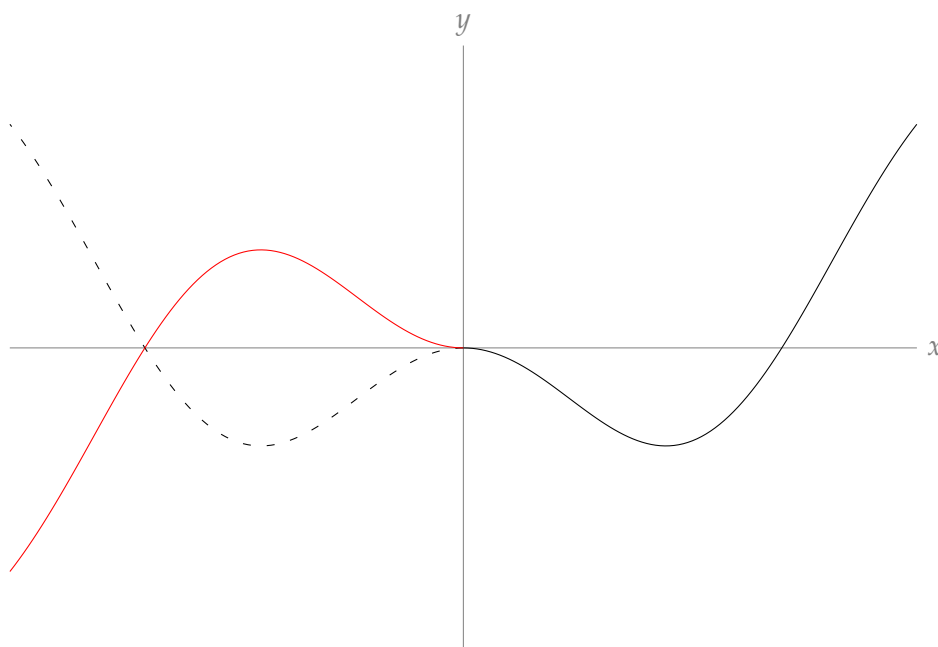


Since reflecting the right-hand portion of the graph across the y -axis, then the x -axis, gives us $f(x)$, we conclude $f(x)$ is odd.

S-3: Since the function is even, we simply reflect the portion shown across the y -axis to complete the sketch.



S-4: Since the function is odd, to complete the sketch, we reflect the portion shown across the y -axis (shown dashed), then the x -axis (shown in red).



S-5: A function is even if $f(-x) = f(x)$.

$$\begin{aligned} f(-x) &= \frac{(-x)^4 - (-x)^6}{e^{(-x)^2}} \\ &= \frac{x^4 - x^6}{e^{x^2}} \\ &= f(x) \end{aligned}$$

So, $f(x)$ is even.

S-6: For any real number x , we will show that $f(x) = f(x + 4\pi)$.

$$\begin{aligned} f(x + 4\pi) &= \sin(x + 4\pi) + \cos\left(\frac{x + 4\pi}{2}\right) \\ &= \sin(x + 4\pi) + \cos\left(\frac{x}{2} + 2\pi\right) \\ &= \sin(x) + \cos\left(\frac{x}{2}\right) \\ &= f(x) \end{aligned}$$

So, $f(x)$ is periodic.

S-7: $f(x)$ is not periodic. (You don't really have to justify this, but if you wanted to, you could say something like this. Notice $f(0) = 1$. Whenever $x > 10$, $f(x) > 1$. Then the value of $f(0)$ is *not* repeated indefinitely, so $f(x)$ is not periodic.)

To decide whether $f(x)$ is even, odd, or neither, simplify $f(-x)$:

$$\begin{aligned} f(-x) &= (-x)^4 + 5(-x)^2 + \cos((-x)^3) \\ &= x^4 + 5x^2 + \cos(-x^3) \\ &= x^4 + 5x^2 + \cos(x^3) \\ &= f(x) \end{aligned}$$

Since $f(-x) = f(x)$, our function is even.

S-8: It should be clear that $f(x)$ is not periodic. (If you wanted to justify this, you could note that $f(x) = 0$ has exactly two solutions, $x = 0, -5$. Since the value of $f(0)$ is repeated only twice, and not indefinitely, $f(x)$ is not periodic.)

To decide whether $f(x)$ is odd, even, or neither, we simplify $f(-x)$.

$$\begin{aligned} f(-x) &= (-x)^5 + 5(-x)^4 \\ &= -x^5 + 5x^4 \end{aligned}$$

We see that $f(-x)$ is not equal to $f(x)$ or to $-f(x)$. For instance, when $x = 1$:

- $f(-x) = f(-1) = 4$,
- $f(x) = f(1) = 6$, and

- $-f(x) = -f(1) = -6$.

Since $f(-x)$ is not equal to $f(x)$ or to $-f(x)$, $f(x)$ is neither even nor odd.

S-9: Recall the period of $g(X) = \tan X$ is π .

$$\tan(X + \pi) = \tan(X) \quad \text{for any } X \text{ in the domain of } \tan X$$

Replacing X with πx :

$$\begin{aligned} \tan(\pi x + \pi) &= \tan(\pi x) && \text{for any } x \text{ in the domain of } \tan(\pi x) \\ \tan(\pi(x + 1)) &= \tan(\pi x) && \text{for any } x \text{ in the domain of } \tan(\pi x) \\ f(x + 1) &= f(x) && \text{for any } x \text{ in the domain of } \tan(\pi x) \end{aligned}$$

The period of $f(x)$ is 1.

S-10: Let's consider $g(x) = \tan(3x)$ and $h(x) = \sin(4x)$ separately. Recall that π is the period of tangent.

$$\tan X = \tan(X + \pi) \quad \text{for every } X \text{ in the domain of } \tan X$$

Replacing X with $3x$:

$$\begin{aligned} \tan(3x) &= \tan(3x + \pi) && \text{for every } x \text{ in the domain of } \tan 3x \\ \tan(3x) &= \tan\left(3\left(x + \frac{\pi}{3}\right)\right) && \text{for every } x \text{ in the domain of } \tan 3x \\ g(x) &= g\left(x + \frac{\pi}{3}\right) && \text{for every } x \text{ in the domain of } \tan 3x \end{aligned}$$

So, the period of $g(x) = \tan(3x)$ is $\frac{\pi}{3}$.

Similarly, 2π is the period of sine.

$$\sin(X) = \sin(X + 2\pi) \quad \text{for every } X \text{ in the domain of } \sin(X)$$

Replacing X with $4x$:

$$\begin{aligned} \sin(4x) &= \sin(4x + 2\pi) && \text{for every } x \text{ in the domain of } \sin(4x) \\ \sin(4x) &= \sin\left(4\left(x + \frac{\pi}{2}\right)\right) && \text{for every } x \text{ in the domain of } \sin(4x) \\ h(x) &= h\left(x + \frac{\pi}{2}\right) && \text{for every } x \text{ in the domain of } \sin(4x) \end{aligned}$$

So, the period of $h(x) = \sin(4x)$ is $\frac{\pi}{2}$.

All together, $f(x) = g(x) + h(x)$ will repeat when both $g(x)$ and $h(x)$ repeat. The least common integer multiple of $\frac{\pi}{3}$ and $\frac{\pi}{2}$ is π . Since $g(x)$ repeats every $\frac{\pi}{3}$ units, and $h(x)$ repeats every $\frac{\pi}{2}$ units, they will not both repeat until we move π units. So, the period of $f(x)$ is π .

No exercises for Section 3.6.5. — Jump to [TABLE OF CONTENTS](#)

Solutions to Exercises 3.6.6 — Jump to [TABLE OF CONTENTS](#)

S-1: (a) Since we must have $3 - x \geq 0$, this tells us $x \leq 3$. So, the domain is $(-\infty, 3]$.

(b)

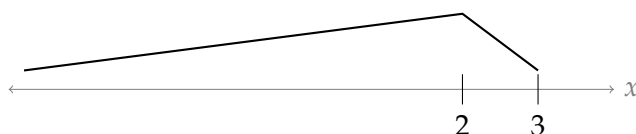
$$f'(x) = \sqrt{3-x} - \frac{x}{2\sqrt{3-x}} = 3 \frac{2-x}{2\sqrt{3-x}}$$

For every x in the domain of $f'(x)$, the denominator is positive, so the sign of $f'(x)$ depends only on the numerator.

x	$(-\infty, 2)$	2	$(2, 3)$	3
$f'(x)$	positive	0	negative	DNE
$f(x)$	increasing	maximum	decreasing	endpoint

So, f is increasing for $x < 2$, has a local (in fact global) maximum at $x = 2$, is decreasing for $2 < x < 3$, and has a local minimum at $x = 3$.

Remark: this shows us the basic skeleton of the graph. It consists of a single hump.



(c) When $x < 3$,

$$f''(x) = \frac{1}{4}(3x-12)(3-x)^{-3/2} < 0$$

The domain of $f''(x)$ is $(-\infty, -3)$, and over its domain it is always negative (the factor $(3x-12)$ is negative for all $x < 4$ and the factor $(3-x)^{-3/2}$ is positive for all $x < 3$). So, $f(x)$ has no inflection points and is concave down everywhere.

(d) We already found

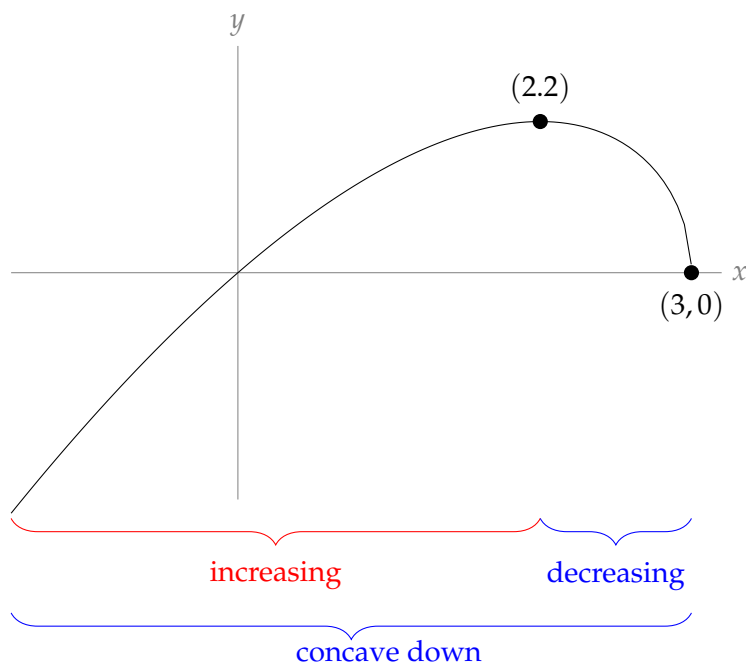
$$f'(x) = 3 \frac{2-x}{2\sqrt{3-x}}.$$

This is undefined at $x = 3$. Indeed,

$$\lim_{x \rightarrow 3^-} 3 \frac{2-x}{2\sqrt{3-x}} = -\infty,$$

so $f(x)$ has a vertical tangent line at $(3, 0)$.

(e) To sketch the curve $y = f(x)$, we already know its intervals of increase and decrease, and its concavity. We also note its intercepts are $(0, 0)$ and $(3, 0)$.



S-2:

- Asymptotes:

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} \frac{1}{x} - \frac{2}{x^4} = 0$$

So $y = 0$ is a horizontal asymptote both at $x = \infty$ and $x = -\infty$.

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{x^3 - 2}{x^4} = -\infty$$

So there is a vertical asymptote at $x = 0$, where the function plunges downwards from both the right and the left.

- Intervals of increase and decrease:

$$f'(x) = -\frac{1}{x^2} + \frac{8}{x^5} = \frac{8 - x^3}{x^5}$$

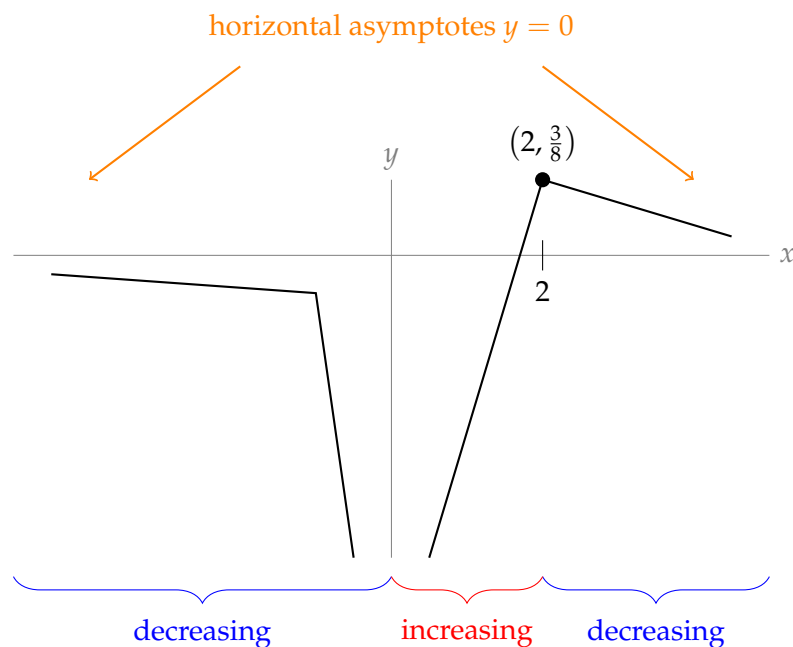
The only place where $f'(x)$ is zero only at $x = 2$. So $f(x)$ has a horizontal tangent at $x = 2$, $y = \frac{3}{8}$. This is a critical point.

The derivative is undefined at $x = 0$, as is the function.

x	$(-\infty, 0)$	0	$(0, 2)$	2	$(2, \infty)$
$f'(x)$	negative	DNE	positive	0	negative
$f(x)$	decreasing	vertical asymptote	increasing	local max	decreasing

Since the function changes from increasing to decreasing at $x = 2$, the only local maximum is at $x = 2$.

At this point, we get a rough sketch of $f(x)$.



- Concavity:

$$f''(x) = \frac{2}{x^3} - \frac{40}{x^6} = \frac{2x^3 - 40}{x^6}$$

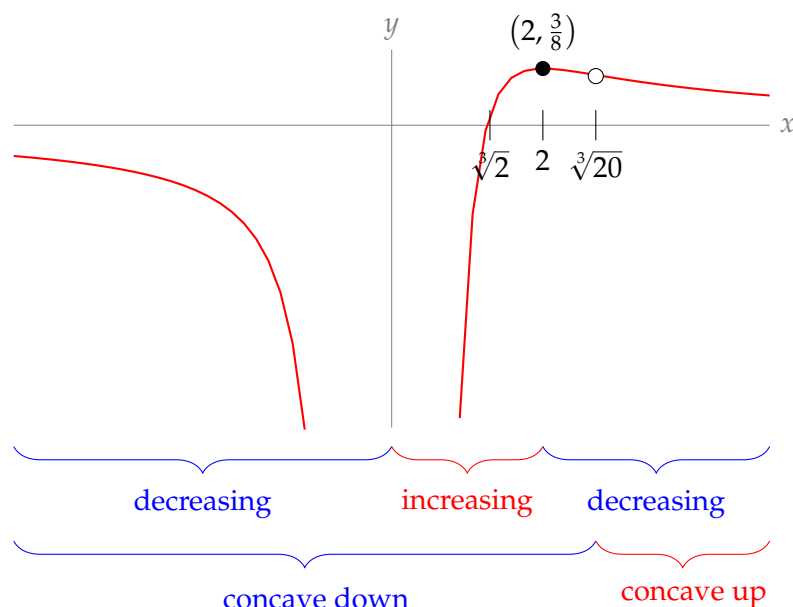
The second derivative of $f(x)$ is positive for $x > \sqrt[3]{20}$ and negative for $x < \sqrt[3]{20}$. So the curve is concave up for $x > \sqrt[3]{20}$ and concave down for $x < \sqrt[3]{20}$. There is an inflection point at $x = \sqrt[3]{20} \approx 2.7$, $y = \frac{18}{20^{4/3}} \approx 0.3$.

- Intercepts:

Since $f(x)$ is not defined at $x = 0$, there is no y -intercept. The only x -intercept is $x = \sqrt[3]{2} \approx 1.3$.

- Sketch:

We can add concavity to our skeleton sketched above, and label our intercept and inflection point (the open dot).



S-3:

- Asymptotes:

When $x = -1$, the denominator $1 + x^3$ of $f(x)$ is zero while the numerator is 1, so $x = -1$ is a vertical asymptote. More precisely,

$$\lim_{x \rightarrow -1^-} f(x) = -\infty \quad \lim_{x \rightarrow -1^+} f(x) = \infty$$

There are no horizontal asymptotes, because

$$\lim_{x \rightarrow \infty} \frac{x^4}{1 + x^3} = \infty \quad \lim_{x \rightarrow -\infty} \frac{x^4}{1 + x^3} = -\infty$$

- Intervals of increase and decrease:

We note that $f'(x)$ is defined for all $x \neq -1$ and is not defined for $x = -1$. Therefore, the only singular point for $f(x)$ is $x = -1$.

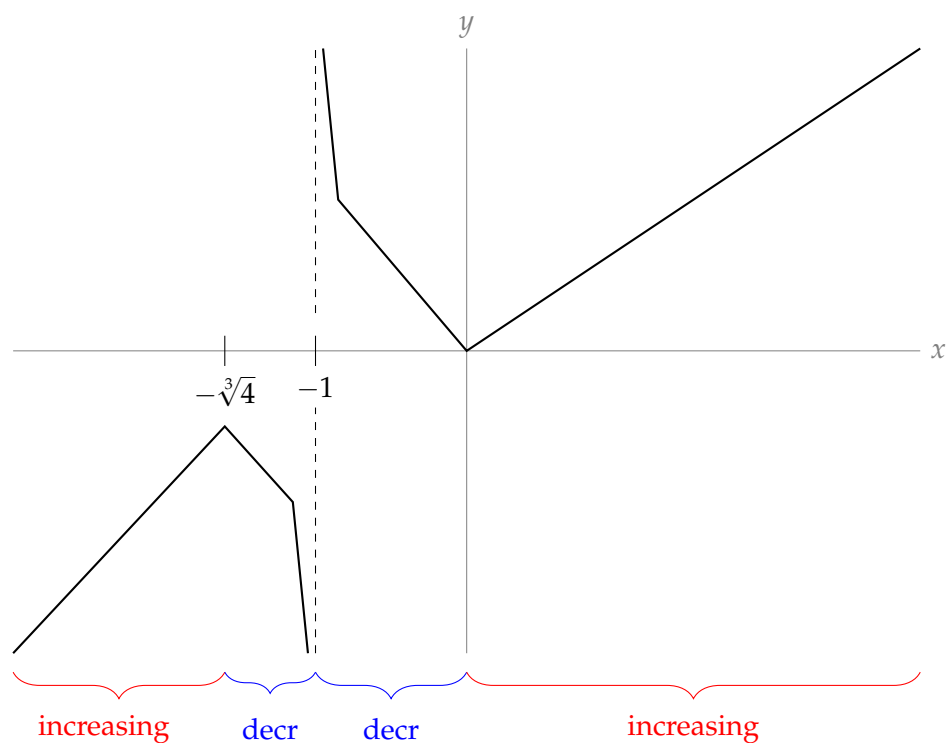
To find critical points, we set

$$\begin{aligned} f'(x) &= 0 \\ 4x^3 + x^6 &= 0 \\ x^3(4 + x^3) &= 0 \\ x^3 &= 0 \quad \text{or} \quad 4 + x^3 = 0 \\ x &= 0 \quad \text{or} \quad x = -\sqrt[3]{4} \approx -1.6 \end{aligned}$$

At these critical points, $f(0) = 0$ and $f(-\sqrt[3]{4}) = \frac{4\sqrt[3]{4}}{-3} < 0$. The denominator of $f'(x)$ is never negative, so the sign of $f'(x)$ is the same as the sign of its numerator, $x^3(4 + x^3)$.

x	$(-\infty, -\sqrt[3]{4})$	$-\sqrt[3]{4}$	$(-\sqrt[3]{4}, -1)$	-1	$(-1, 0)$	0	$(0, \infty)$
$f'(x)$	positive	0	negative	DNE	negative	0	positive
$f(x)$	increasing	l. max	decreasing	VA	decreasing	l. min	increasing

Now, we have enough information to make a skeleton of our graph.

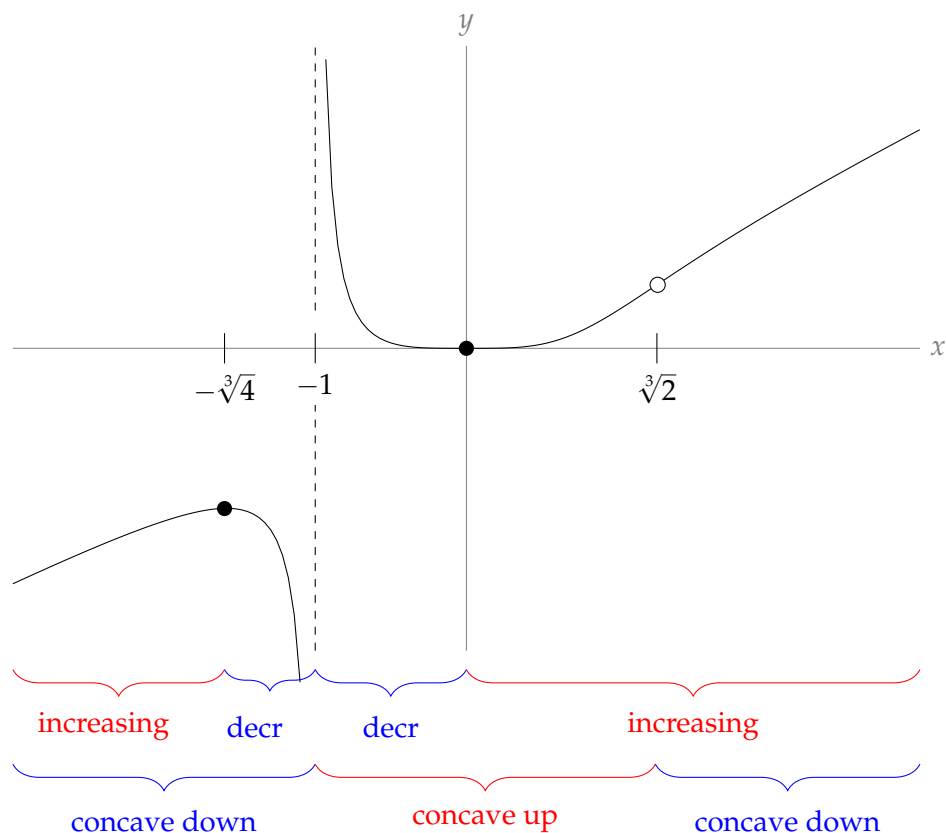


- Concavity:

The second derivative is undefined when $x = -1$. It is zero when $12x^2 - 6x^5 = 6x^2(2 - x^3) = 0$. That is, at $x = \sqrt[3]{2} \approx 1.3$ and $x = 0$. Notice that the sign of $f''(x)$ does not change at $x = 0$, so $x = 0$ is not an inflection point.

x	$(-\infty, -1)$	-1	$(-1, 0)$	0	$(0, \sqrt[3]{2})$	$\sqrt[3]{2}$	$(\sqrt[3]{2}, \infty)$
$f''(x)$	negative	DNE	positive	0	positive	0	negative
$f(x)$	concave down	VA	concave up		concave up	IP	concave down

Now we can refine our skeleton by adding concavity. In the sketch below, closed dots are critical points, and the open dot is an inflection point.



S-4:

- Asymptotes:

$$\lim_{x \rightarrow -\infty} \frac{x^3}{1-x^2} = \infty \quad \lim_{x \rightarrow \infty} \frac{x^3}{1-x^2} = -\infty$$

So, $f(x)$ has no horizontal asymptotes.

On the other hand $f(x)$ blows up at both $x = 1$ and $x = -1$, so there are vertical asymptotes at $x = 1$ and $x = -1$. More precisely,

$$\begin{aligned} \lim_{x \rightarrow -1^-} \frac{x^3}{1-x^2} &= \infty & \lim_{x \rightarrow -1^+} \frac{x^3}{1-x^2} &= -\infty \\ \lim_{x \rightarrow 1^-} \frac{x^3}{1-x^2} &= \infty & \lim_{x \rightarrow 1^+} \frac{x^3}{1-x^2} &= -\infty \end{aligned}$$

- Symmetry:

$f(x)$ is an odd function, because

$$f(-x) = \frac{(-x)^3}{1-(-x)^2} = \frac{-x^3}{1-x^2} = -f(x)$$

- Intercepts:

The only intercept of $f(x)$ is the origin. In particular, that means that out of the three intervals where it is continuous, namely $(-\infty, -1)$, $(-1, 1)$ and $(1, \infty)$, in two of them $f(x)$ is always positive or always negative.

-
- When $x < -1$: $1 - x^2 < 0$ and $x^3 < 0$, so $f(x) > 0$.
 - When $x > 1$: $1 - x^2 < 0$ and $x^3 > 0$, so $f(x) < 0$.
 - When $-1 < x < 0$, $1 - x^2 > 0$ and $x^3 < 0$, so $f(x) < 0$.
 - When $0 < x < 1$, $1 - x^2 > 0$ and $x^3 > 0$, so $f(x) > 0$.

- Intervals of increase and decrease:

$$f'(x) = \frac{3x^2 - x^4}{(1 - x^2)^2} = \frac{x^2(3 - x^2)}{(1 - x^2)^2}$$

The only singular points are $x = \pm 1$, where $f(x)$, and hence $f'(x)$, is not defined. The critical points are:

$$\begin{aligned} f'(x) &= 0 \\ x^2 &= 0 \quad \text{or} \quad 3 - x^2 = 0 \\ x &= 0 \quad \text{or} \quad x = \pm\sqrt{3} \approx \pm 1.7 \end{aligned}$$

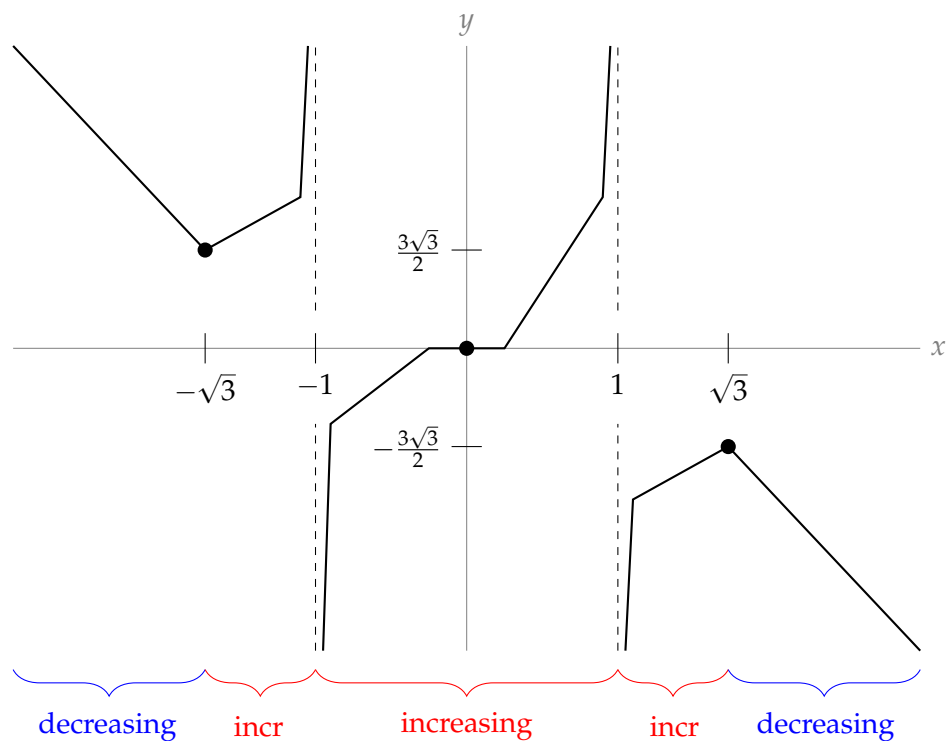
The values of f at its critical points are $f(0) = 0$, $f(\sqrt{3}) = -\frac{3\sqrt{3}}{2} \approx -2.6$ and $f(-\sqrt{3}) = \frac{3\sqrt{3}}{2} \approx 2.6$.

Notice the sign of $f'(x)$ is the same as the sign of $3 - x^2$.

x	$(-\infty, -\sqrt{3})$	$-\sqrt{3}$	$(-\sqrt{3}, -1)$	-1	$(-1, 0)$
$f'(x)$	negative	0	positive	DNE	positive
$f(x)$	decreasing	local min	increasing	VA	increasing

x	0	$(0, 1)$	1	$(1, \sqrt{3})$	$\sqrt{3}$	$(\sqrt{3}, \infty)$
$f'(x)$	0	positive	DNE	positive	0	negative
$f(x)$		increasing	VA	increasing	local max	decreasing

Now we have enough information to sketch a skeleton of $f(x)$.



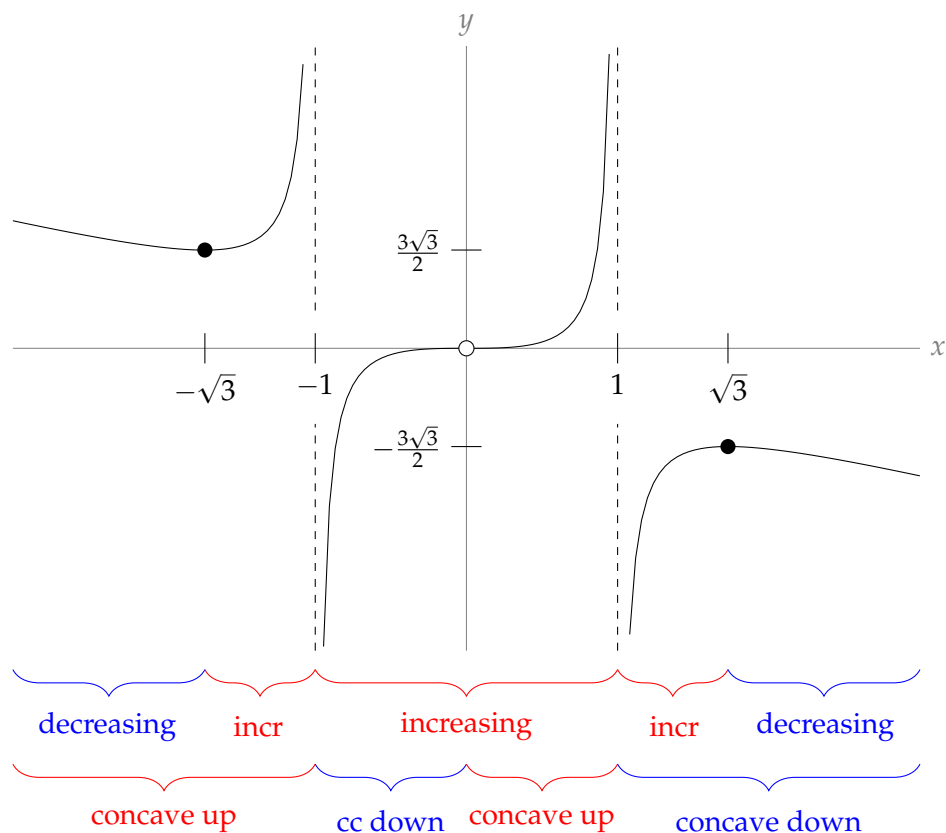
- Concavity:

$$f''(x) = \frac{2x(3+x^2)}{(1-x^2)^3}$$

The second derivative is zero when $x = 0$, and is undefined when $x = \pm 1$.

x	$(-\infty, -1)$	$(-1, 0)$	0	$(0, 1)$	$(1, \infty)$
$f''(x)$	positive	negative	0	positive	negative
$f(x)$	concave up	concave down	inflection point	concave up	concave down

Now, we can refine our skeleton. In the sketch below, closed dots are critical points, and the open dot is an inflection point.



S-5: (a) One branch of the function, the exponential function e^x , is continuous everywhere. So $f(x)$ is continuous for $x < 0$. When $x \geq 0$, $f(x) = \frac{x^2 + 3}{3(x + 1)}$, which is continuous whenever $x \neq -1$ (so it's continuous for all $x > 0$). So, $f(x)$ is continuous for $x > 0$. To see that $f(x)$ is continuous at $x = 0$, we see:

$$\begin{aligned}\lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} e^x = 1 \\ \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} \frac{x^2 + 3}{3(x + 1)} = 1 \\ \text{So, } \lim_{x \rightarrow 0} f(x) &= 1 = f(0)\end{aligned}$$

Hence $f(x)$ is continuous at $x = 0$, so $f(x)$ is continuous everywhere.

(b) We differentiate the function twice. Notice

$$\begin{aligned}\frac{d}{dx} \left\{ \frac{x^2 + 3}{3(x+1)} \right\} &= \frac{3(x+1)(2x) - (x^2 + 3)(3)}{9(x+1)^2} \\ &= \frac{x^2 + 2x - 3}{3(x+1)^2} \\ &= \frac{(x-1)(x+3)}{3(x+1)^2} \quad \text{where } x \neq -1\end{aligned}$$

$$\text{Then } \lim_{x \rightarrow 0^+} f'(x) = \frac{(0-1)(0+3)}{3(0+1)^2} = -1 \neq 1 = e^0 = \lim_{x \rightarrow 0^-} f'(x)$$

$$\text{so } f'(x) = \begin{cases} e^x & x < 0 \\ DNE & x = 0 \\ \frac{(x-1)(x+3)}{3(x+1)^2} & x > 0 \end{cases}$$

Differentiating again,

$$\begin{aligned}\frac{d^2}{dx^2} \left\{ \frac{x^2 + 3}{3(x+1)} \right\} &= \frac{d}{dx} \left\{ \frac{x^2 + 2x - 3}{3(x+1)^2} \right\} \\ &= \frac{3(x+1)^2(2x+2) - (x^2 + 2x - 3)(6)(x+1)}{9(x+1)^4} \left(\frac{\div 3(x+1)}{\div 3(x+1)} \right) \\ &= \frac{(x+1)(2x+2) - 2(x^2 + 2x - 3)}{3(x+1)^3} \\ &= \frac{8}{3(x+1)^3} \quad \text{where } x \neq -1\end{aligned}$$

$$\text{so } f''(x) = \begin{cases} e^x & x < 0 \\ DNE & x = 0 \\ \frac{8}{3(x+1)^3} & x > 0 \end{cases}$$

i. The only singular point is $x = 0$, and the only critical point is $x = 1$. (When you're reading off the expression for $f'(x)$, remember that the bottom line only applies when $x > 0$.)

x	$(-\infty, 0)$	0	$(0, 1)$	1	$(1, \infty)$
$f'(x)$	positive	DNE	negative	0	positive
$f(x)$	increasing	local max	decreasing	local min	increasing

The coordinates of the local maximum are $(0, 1)$ and the coordinates of the local minimum are $(1, \frac{2}{3})$.

ii.

When $x \neq 0$, $f''(x)$ is always positive, so $f(x)$ is concave up.

iii.

$$\begin{aligned}\lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{x^2 + 3}{3x + 3} \\ &= \lim_{x \rightarrow \infty} \frac{x + \frac{3}{x}}{1 + \frac{3}{x}} = \infty\end{aligned}$$

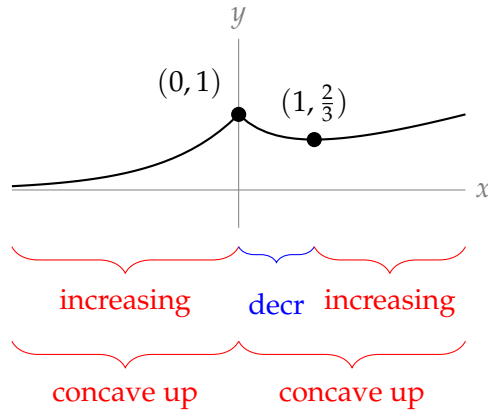
So, there is no horizontal asymptote to the right.

$$\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow -\infty} e^x = 0$$

So, $y = 0$ is a horizontal asymptote to the left.

Since $f(x)$ is continuous everywhere, there are no vertical asymptotes.

(c)



S-6:

- Asymptotes: In the problem statement, we are told:

$$\lim_{x \rightarrow \pm\infty} \frac{1 + 2x}{e^{x^2}} = 0$$

So, $y = 0$ is a horizontal asymptote both at $x = \infty$ and at $x = -\infty$.

Since $f(x)$ is continuous, it has no vertical asymptotes.

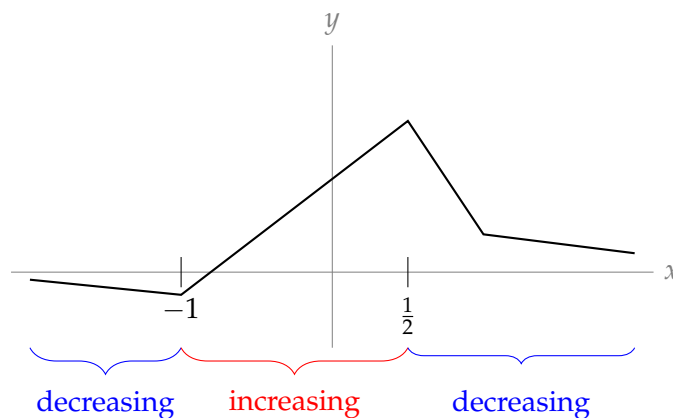
- Intervals of increase and decrease:

The critical points are the zeroes of $1 - x - 2x^2 = (1 - 2x)(1 + x)$. That is, $x = \frac{1}{2}, -1$.

x	$(-\infty, -1)$	-1	$(-1, \frac{1}{2})$	$\frac{1}{2}$	$(\frac{1}{2}, \infty)$
$f'(x)$	negative	0	positive	0	negative
$f(x)$	decreasing	local min	increasing	local max	decreasing

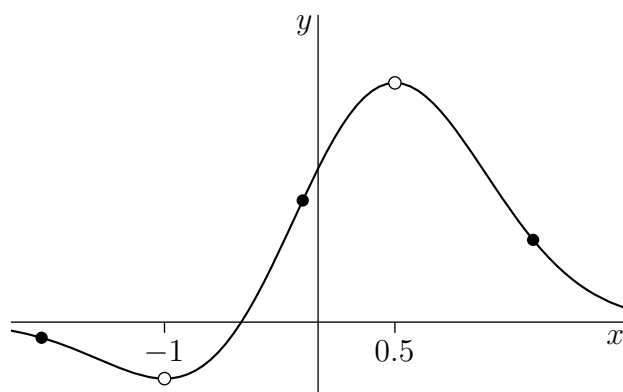
At these critical points, $f(\frac{1}{2}) = 2e^{-1/4} > 0$ and $f(-1) = -e^{-1} < 0$.

From here, we can sketch a skeleton of the graph.



- Concavity:

We are told that we don't have to actually solve for the inflection points. We just need to know enough to get a basic idea. So, we'll turn the skeleton of the graph into smooth curve.



Inflection points are points where the convexity changes from up to down or vice versa. It looks like our graph is convex down for x from $-\infty$ to about -1.8 , convex up from about $x = -1.8$ to about $x = -0.1$, convex down from about $x = -0.1$ to about $x = 1.4$ and convex up from about $x = 1.4$ to infinity. So there are three inflection points at roughly $x = -1.8, -0.1, 1.4$.

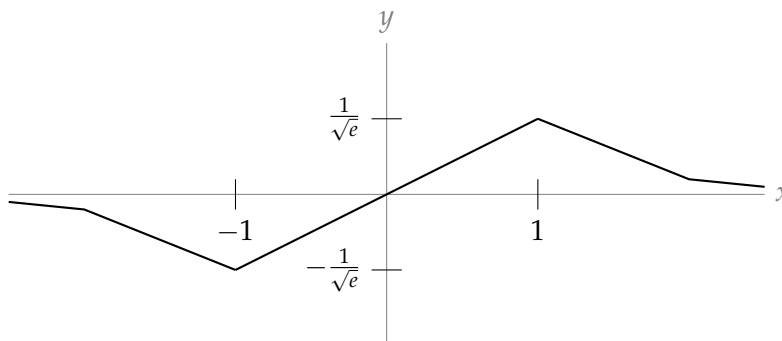
S-7:

(a) We need to know the first and second derivative of $f(x)$. Using the product and chain rules, $f'(x) = e^{-x^2/2}(1 - x^2)$. Given to us is $f''(x) = (x^3 - 3x)e^{-x^2/2}$. (These derivatives are also easy to find using the formula developed in Question 19, Section 2.14.)

Since $e^{-x^2/2}$ is always positive, the sign of $f'(x)$ is the same as the sign of $1 - x^2$. $f(x)$ has no singular points and its only critical points are $x = \pm 1$. At these critical points, $f(-1) = -\frac{1}{\sqrt{e}}$ and $f(1) = \frac{1}{\sqrt{e}}$.

x	$(-\infty, -1)$	-1	$(-1, 1)$	1	$(1, \infty)$
$f'(x)$	negative	0	positive	0	negative
$f(x)$	decreasing	local min	increasing	local max	decreasing

This, together with the observations that $f(x) < 0$ for $x < 0$, $f(0) = 0$ and $f(x) > 0$ for $x > 0$ (in fact f is an odd function), is enough to sketch a skeleton of our graph.



We can factor $f''(x) = (x^3 - 3x)e^{-x^2/2} = x(x + \sqrt{3})(x - \sqrt{3})e^{-x^2/2}$. Since $e^{-x^2/2}$ is always positive, the sign of $f''(x)$ is the same as the sign of $x(x + \sqrt{3})(x - \sqrt{3})$.

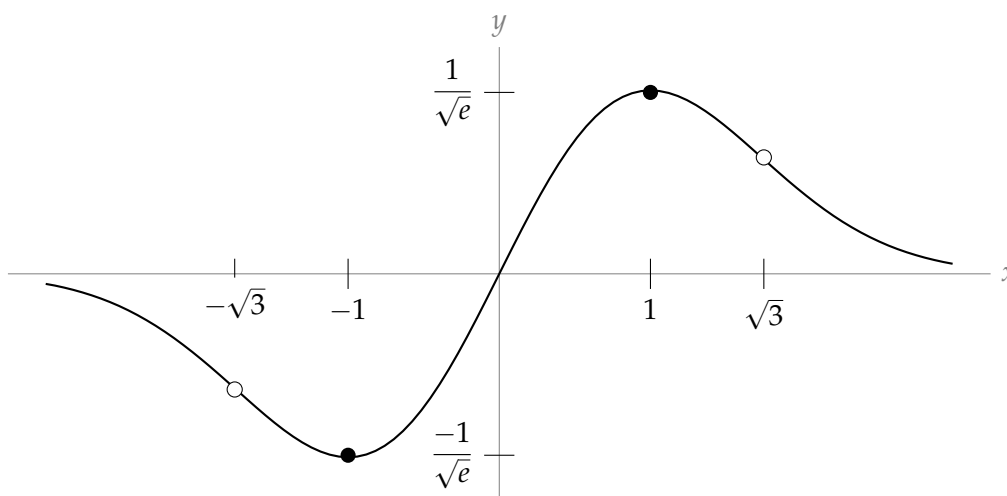
x	$(-\infty, -\sqrt{3})$	$-\sqrt{3}$	$(-\sqrt{3}, 0)$	0	$(0, \sqrt{3})$	$\sqrt{3}$	$(\sqrt{3}, \infty)$
$f''(x)$	negative	0	positive	0	negative	0	positive
$f(x)$	concave down	IP	concave up	IP	concave down	IP	concave up

(b) We've already seen that $f(x)$ has a local min at $x = -1$ and a local max at $x = 1$.

As x tends to negative infinity, $f(x)$ tends to 0, and $f(x)$ is decreasing on $(-\infty, -1)$. Then $f(x)$ is between 0 and $f(-1) = \frac{-1}{\sqrt{e}}$ on $(-\infty, -1)$. Then $f(x)$ is increasing on $(-1, 1)$ from $f(-1) = \frac{-1}{\sqrt{e}}$ to $f(1) = \frac{1}{\sqrt{e}}$. Finally, for $x > 1$, $f(x)$ is decreasing from $f(1) = \frac{1}{\sqrt{e}}$ and tending to 0. So when $x > 1$, $f(x)$ is between $\frac{1}{\sqrt{e}}$ and 0.

So, over its entire domain, $f(x)$ is between $\frac{-1}{\sqrt{e}}$ and $\frac{1}{\sqrt{e}}$, and it only achieves those values at $x = -1$ and $x = 1$, respectively. Therefore, the local and global min of $f(x)$ is at $(-1, \frac{-1}{\sqrt{e}})$, and the local and global max of $f(x)$ is at $(1, \frac{1}{\sqrt{e}})$.

(c) In the graph below, open dots are inflection points, and solid dots are extrema.



S-8:

- Symmetry:

$$f(-x) = -x + 2 \sin(-x) = -x - 2 \sin x = -f(x)$$

So, $f(x)$ is an odd function. If we can sketch $y = f(x)$ for nonnegative x , we can use symmetry to complete the curve for all x .

- Asymptotes:

Since $f(x)$ is continuous, it has no vertical asymptotes. It also has no horizontal asymptotes, since

$$\lim_{x \rightarrow -\infty} f(x) = -\infty \quad \lim_{x \rightarrow \infty} f(x) = \infty$$

- Intervals of increase and decrease:

Since $f(x)$ is differentiable everywhere, there are no singular points.

$$f'(x) = 1 + 2 \cos x$$

So, the critical points of $f(x)$ occur when

$$\cos x = -\frac{1}{2}$$

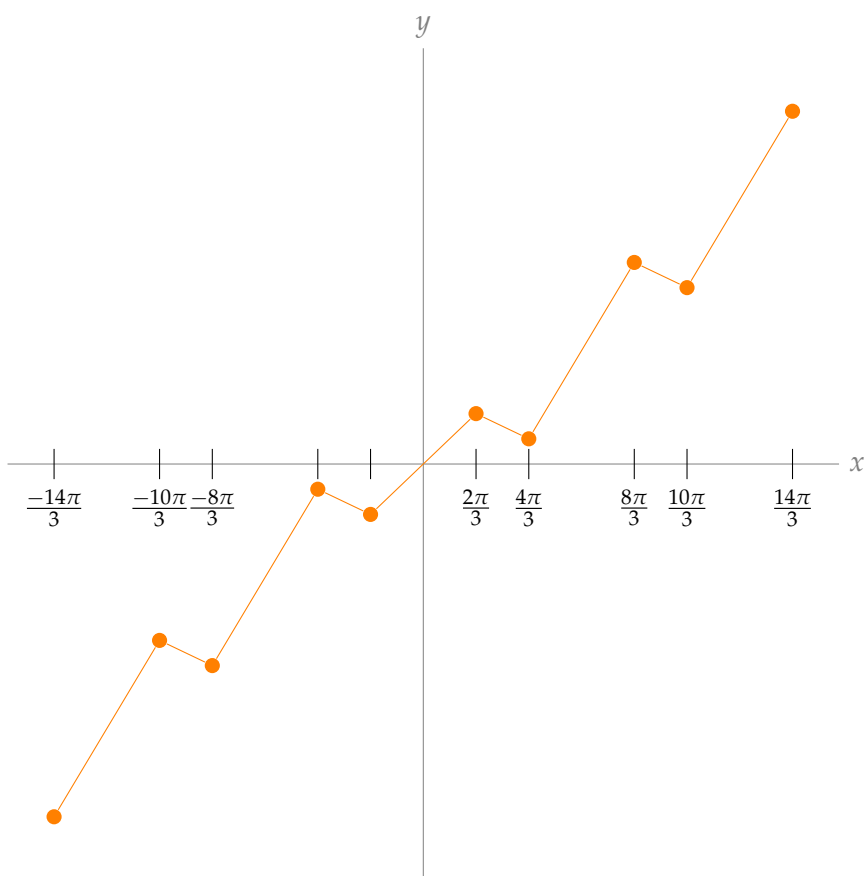
$$x = 2\pi n \pm \frac{2\pi}{3} \text{ for any integer } n$$

For instance, $f(x)$ has critical points at $x = \frac{2\pi}{3}, x = \frac{4\pi}{3}, x = \frac{8\pi}{3}$, and $x = \frac{10\pi}{3}$.

From the unit circle, or the graph of $y = 1 + 2 \cos x$, we see:

x	$(-\frac{2\pi}{3}, \frac{2\pi}{3})$	$\frac{2\pi}{3}$	$(\frac{2\pi}{3}, \frac{4\pi}{3})$	$\frac{4\pi}{3}$	$(\frac{4\pi}{3}, \frac{8\pi}{3})$	$\frac{8\pi}{3}$	$(\frac{8\pi}{3}, \frac{10\pi}{3})$
$f'(x)$	positive	0	negative	0	positive	0	negative
$f(x)$	increasing	l. max	decreasing	l. min	increasing	l. max	decreasing

We have enough information to sketch a skeleton of the curve $y = f(x)$. We use the pattern above for the graph to the right of the y -axis, and use odd symmetry for the graph to the left of the y -axis.



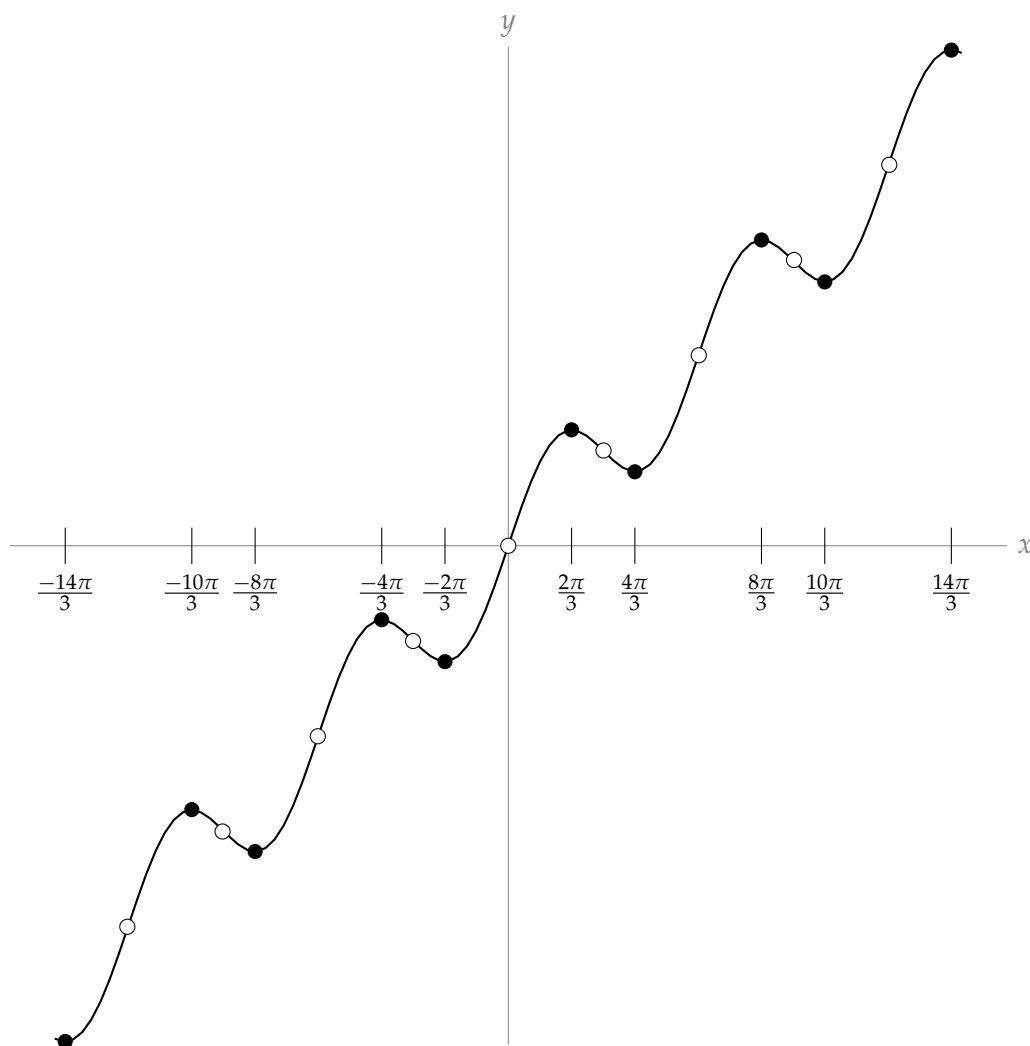
- Concavity:

$$f''(x) = -2 \sin x$$

So, $f''(x)$ exists everywhere, and is zero for $x = \pi + \pi n$ for every integer n .

x	$(0, \pi)$	π	$(\pi, 2\pi)$	2π	$(2\pi, 3\pi)$	3π	$(3\pi, 4\pi)$
$f''(x)$	negative	0	positive	0	negative	0	positive
$f(x)$	concave down	IP	concave up	IP	concave down	IP	concave up

Using these values, and the odd symmetry of $f(x)$, we can refine our skeleton. The closed dots are local extrema, and the open dots are inflection points occurring at every integer multiple of π .



S-9: We first compute the derivatives $f'(x)$ and $f''(x)$.

$$f'(x) = 4 \cos x + 4 \sin 2x = 4 \cos x + 8 \sin x \cos x = 4 \cos x(1 + 2 \sin x)$$

$$f''(x) = -4 \sin x + 8 \cos 2x = -4 \sin x + 8 - 16 \sin^2 x = -4(4 \sin^2 x + \sin x - 2)$$

The graph has the following features.

- Symmetry: $f(x)$ is periodic of period 2π . We'll consider only $-\pi \leq x \leq \pi$. (Any interval of length 2π will do.)
- y -intercept: $f(0) = -2$

- Intervals of increase and decrease: $f'(x) = 0$ when $\cos x = 0$, i.e. $x = \pm \frac{\pi}{2}$, and when $\sin x = -\frac{1}{2}$, i.e. $x = -\frac{\pi}{6}, -\frac{5\pi}{6}$.

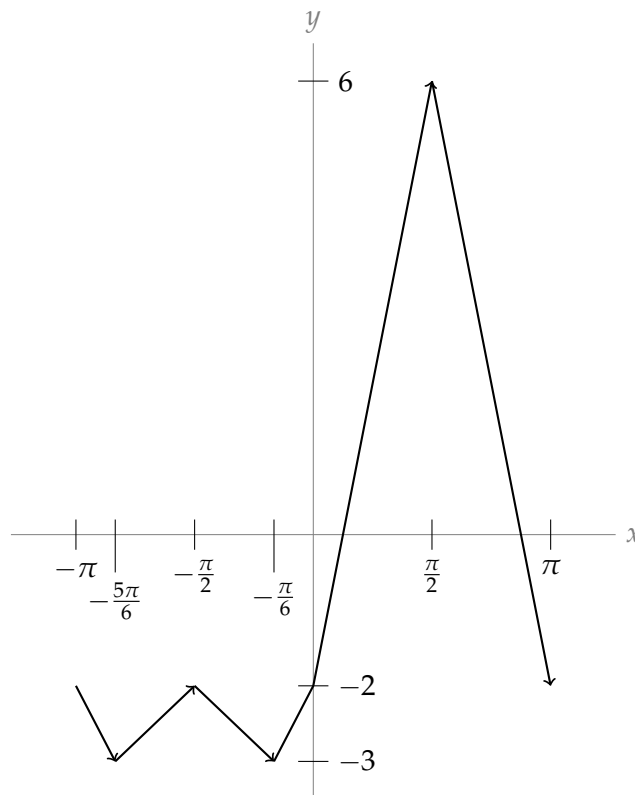
x	$(-\pi, -\frac{5\pi}{6})$	$(-\frac{5\pi}{6}, -\frac{\pi}{2})$	$(-\frac{\pi}{2}, -\frac{\pi}{6})$	$(-\frac{\pi}{6}, \frac{\pi}{2})$	$(\frac{\pi}{2}, \pi)$
$f'(x)$	negative	positive	negative	positive	negative
$f(x)$	decreasing	increasing	decreasing	increasing	decreasing

This tells us local maxima occur at $x = \pm \frac{\pi}{2}$ and local minima occur at $x = -\frac{5\pi}{6}$ and $x = -\frac{\pi}{6}$.

Here is a table giving the value of f at each of its critical points.

x	$-\frac{5\pi}{6}$	$-\frac{\pi}{2}$	$-\frac{\pi}{6}$	$\frac{\pi}{2}$
$\sin x$	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	1
$\cos 2x$	$\frac{1}{2}$	-1	$\frac{1}{2}$	-1
$f(x)$	-3	-2	-3	6

From here, we can graph a skeleton of $f(x)$:



- Concavity: To find the points where $f''(x) = 0$, set $y = \sin x$, so $f''(x) = -4(4y^2 + y - 2)$. Then we really need to solve

$$4y^2 + y - 2 = 0$$

which gives us

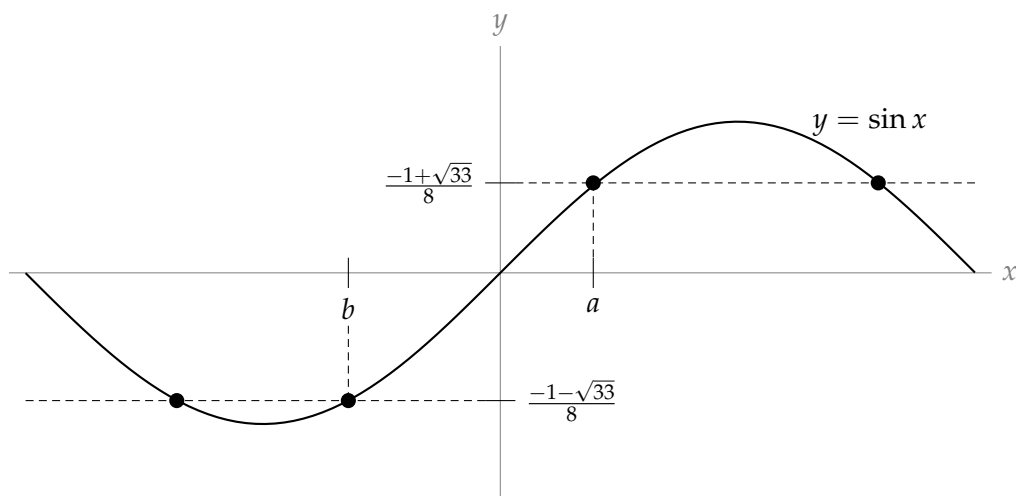
$$y = \frac{-1 \pm \sqrt{33}}{8}$$

These two y -values map to the following two x -values, which we'll name a and b for convenience:

$$a = \arcsin\left(\frac{-1 + \sqrt{33}}{8}\right) \approx 0.635$$

$$b = \arcsin\left(\frac{-1 - \sqrt{33}}{8}\right) \approx -1.003$$

However, these are not the only values of x in $[-\pi, \pi]$ with $\sin x = \frac{-1 + \sqrt{33}}{8}$. The analysis above misses the others because the arcsine function only returns numbers in the range $[-\frac{\pi}{2}, \frac{\pi}{2}]$. The graph below shows that there should be other values of x with $\sin x = \frac{-1 + \sqrt{33}}{8}$, and hence $f''(x) = 0$.



We can recover the other solutions in $[-\pi, \pi]$ by recalling that

$$\sin(x) = \sin(\pi - x)$$

(see CLP appendix A7). So, if we choose $x = \arcsin\left(\frac{-1 + \sqrt{33}}{8}\right) \approx 0.635$ to make $\sin(x) = \frac{-1 + \sqrt{33}}{8}$ so that $f''(x) = 0$, then setting

$$x = \pi - a = \pi - \arcsin\left(\frac{-1 + \sqrt{33}}{8}\right) \approx 2.507$$

will also give us $\sin(x) = \frac{-1 + \sqrt{33}}{8}$ and $f''(x) = 0$. Similarly, setting

$$x = \pi - b = \pi - \arcsin\left(\frac{-1 - \sqrt{33}}{8}\right) \approx 4.145$$

would give us $f''(x) = 0$. However, this value is outside $[-\pi, \pi]$. To find another solution inside $[-\pi, \pi]$ we use the identity

$$\sin(x) = \sin(-\pi - x)$$

(which we can obtain from the identity we used above and the fact that $\sin(\theta) = \sin(\theta \pm 2\pi)$ for any angle θ). Using this, we can show that

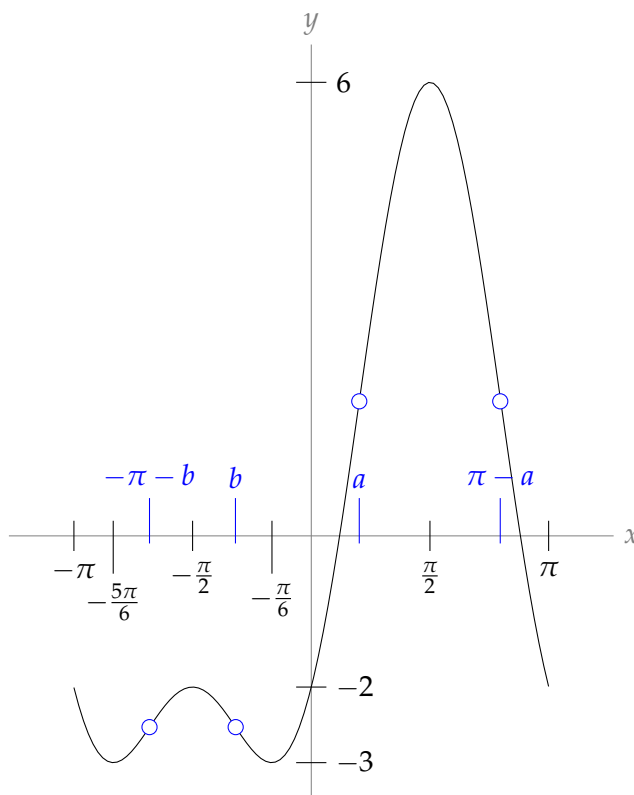
$$x = -\pi - b = -\pi - \arcsin\left(\frac{-1 - \sqrt{33}}{8}\right) \approx -2.139$$

also gives $f''(x) = 0$.

So, all together, $f''(x) = 0$ when $x = -\pi - b$, $x = b$, $x = a$, and $x = \pi - a$.

Now, we should compute the sign of $f''(x)$ while x is between $-\pi$ and π . Recall that, if $y = \sin x$, then $f''(x) = -4(4y^2 + y - 2)$. So, in terms of y , f'' is a parabola pointing down, with intercepts $y = \frac{-1 \pm \sqrt{33}}{8}$. Then f'' is positive when y is in the interval $(\frac{-1 - \sqrt{33}}{8}, \frac{-1 + \sqrt{33}}{8})$, and f'' is negative otherwise. From the graph of sine, we see that y is between $\frac{-1 - \sqrt{33}}{8}$ and $\frac{-1 + \sqrt{33}}{8}$ precisely on the intervals $(-\pi, -\pi - b)$, (b, a) , and $(\pi - a, \pi)$.

Therefore, $f(x)$ is concave up on the intervals $(-\pi, -\pi - b)$, (b, a) , and $(\pi - a, \pi)$, and $f(x)$ is concave down on the intervals $(-\pi - b, b)$ and $(a, \pi - a)$. So, the inflection points of f occur at $x = -\pi - b$, $x = b$, $x = a$, and $x = \pi - a$.



To find the maximum and minimum values of $f(x)$ on $[0, \pi]$, we compare the values of $f(x)$ at its critical points in this interval (only $x = \frac{\pi}{2}$) with the values of $f(x)$ at its endpoints $x = 0$, $x = \pi$.

Since $f(0) = f(\pi) = -2$, the minimum value of f on $[0, \pi]$ is -2 , achieved at $x = 0, \pi$ and the maximum value of f on $[0, \pi]$ is 6 , achieved at $x = \frac{\pi}{2}$.

S-10: Let $f(x) = \sqrt[3]{\frac{x+1}{x^2}}$.

- Asymptotes: Since $\lim_{x \rightarrow 0} f(x) = \infty$, $f(x)$ has a vertical asymptote at $x = 0$ where the curve reaches steeply upward from both the left and the right.

$\lim_{x \rightarrow \pm\infty} f(x) = 0$, so $y = 0$ is a horizontal asymptote for $x \rightarrow \pm\infty$.

- Intercepts: $f(-1) = 0$.

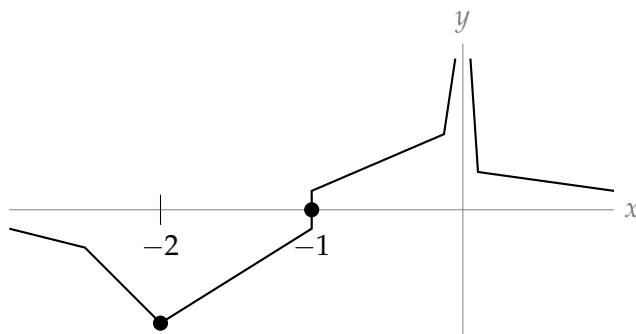
- Intervals of increase and decrease:

$$f'(x) = \frac{-(x+2)}{3x^{5/3}(x+1)^{2/3}}$$

There is a singular point at $x = -1$ and a critical point at $x = -2$, in addition to a discontinuity at $x = 0$. Note that $(x+1)^{2/3} = (\sqrt[3]{x+1})^2$, which is never negative. Note also that $\lim_{x \rightarrow -1} f'(x) = \infty$, so $f(x)$ has a vertical tangent line at $x = -1$.

x	$(-\infty, -2)$	-2	$(-2, -1)$	-1	$(-1, 0)$	0	$(0, \infty)$
$f'(x)$	negative	0	positive	DNE	positive	DNE	negative
$f(x)$	decreasing	l. min	increasing	vertical	increasing	VA	decreasing

This gives us enough information to sketch a skeleton of the curve.



- Concavity:

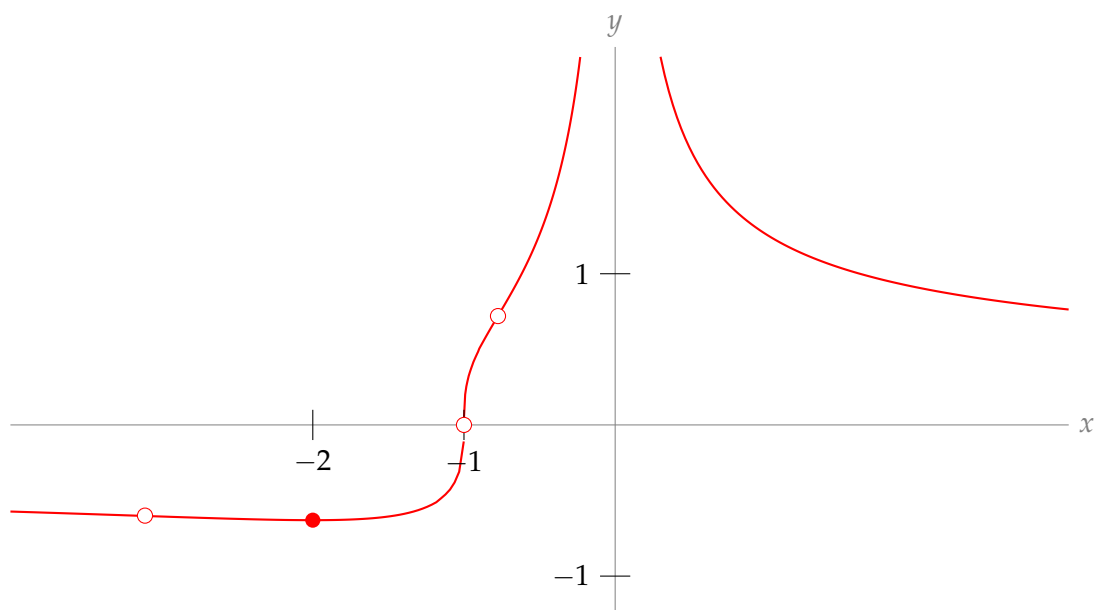
$$f''(x) = \frac{4x^2 + 16x + 10}{9x^{8/3}(x+1)^{5/3}}$$

We still have a discontinuity at $x = 0$, and $f''(x)$ does not exist at $x = -1$. The second derivative is zero when $4x^2 + 16x + 10 = 0$. Using the quadratic formula, we find this occurs when $x = -2 \pm \sqrt{1.5} \approx -0.8, -3.2$. Note $x^{8/3} = (\sqrt[3]{x})^8$ is never negative.

x	$(-\infty, -2 - \sqrt{1.5})$	$-2 - \sqrt{1.5}$	$(-2 - \sqrt{1.5}, -1)$	-1
$f''(x)$	negative	0	positive	DNE
$f(x)$	concave down	IP	concave up	IP

x	$(-1, -2 + \sqrt{1.5})$	$-2 + \sqrt{1.5}$	$(-2 + \sqrt{1.5}, 0)$	$(0, \infty)$
$f''(x)$	negative	0	positive	positive
$f(x)$	concave down	IP	concave up	concave up

Now, we can refine our skeleton. The closed dot is the local minimum, and the open dots are inflection points.



S-11: The parts of the question are just scaffolding to lead you through sketching the curve. Their answers are given explicitly, in an organized manner, in the “answers” section. In this solution, they are scattered throughout.

- Asymptotes:

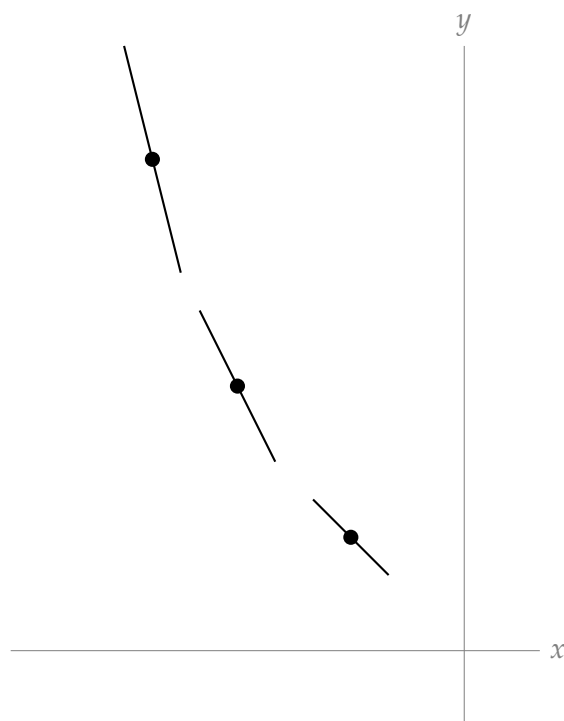
Since the function has a derivative at every real number, the function is continuous for every real number, so it has no vertical asymptotes. In the problem statement, you are told $\lim_{x \rightarrow \infty} f(x) = 0$, so $y = 0$ is a horizontal asymptote as x goes to infinity. It remains to evaluate $\lim_{x \rightarrow -\infty} f(x)$. Let's consider the limit of $f'(x)$ instead. Recall K is a positive constant.

$$\begin{aligned}\lim_{x \rightarrow -\infty} e^{-x} &= \lim_{x \rightarrow \infty} e^x = \infty \\ \lim_{x \rightarrow -\infty} K(2x - x^2) &= -\infty\end{aligned}$$

So,

$$\lim_{x \rightarrow -\infty} K(2x - x^2)e^{-x} = -\infty$$

That is, as x becomes a hugely negative number, $f'(x)$ also becomes a hugely negative number. As we move left along the x -axis, $f(x)$ is decreasing with a steeper and steeper slope, as in the sketch below. That means $\lim_{x \rightarrow -\infty} f(x) = \infty$.



Sketch: various tangent lines to $f(x)$,
their slopes getting more strongly negative
as x gets more strongly negative.

- Intervals of increase and decrease:

We are given $f'(x)$ (although we don't know $f(x)$):

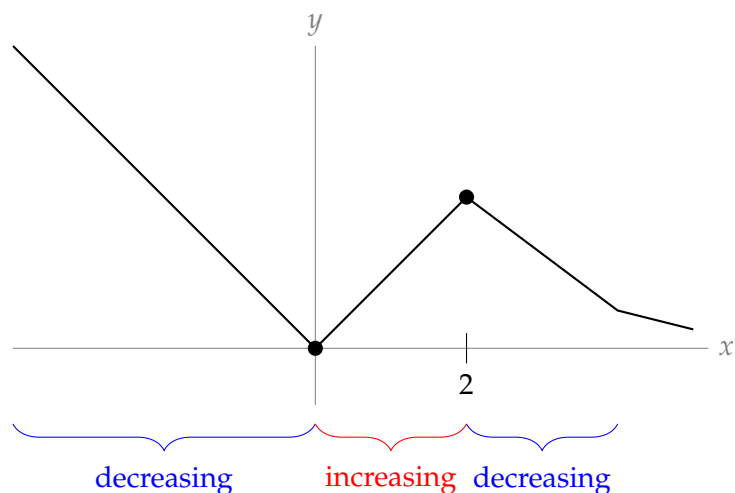
$$f'(x) = Kx(2-x)e^{-x}$$

The critical points of $f(x)$ are $x = 0$ and $x = 2$, and there are no singular points. Recall e^{-x} is always positive, and K is a positive constant.

x	$(-\infty, 0)$	0	$(0, 2)$	2	$(2, \infty)$
$f'(x)$	negative	0	positive	0	negative
$f(x)$	decreasing	local min	increasing	local max	decreasing

So, $f(0) = 0$ is a local minimum, and $f(2) = 2$ is a local maximum.

Looking ahead to part (d), we have a skeleton of the curve.



- Concavity:

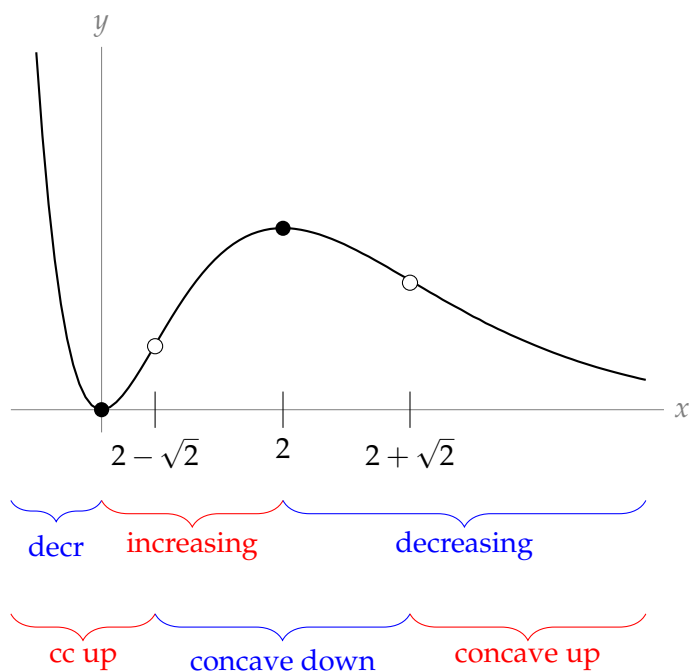
Since we're given $f'(x)$, we can find $f''(x)$.

$$\begin{aligned} f''(x) &= K(2 - 2x - 2x + x^2)e^{-x} \\ &= K(2 - 4x + x^2)e^{-x} \\ &= K(x - 2 - \sqrt{2})(x - 2 + \sqrt{2})e^{-x} \end{aligned}$$

where the last line can be found using the quadratic equation. So, $f''(x) = 0$ for $x = 2 \pm \sqrt{2}$, and $f''(x)$ exists everywhere.

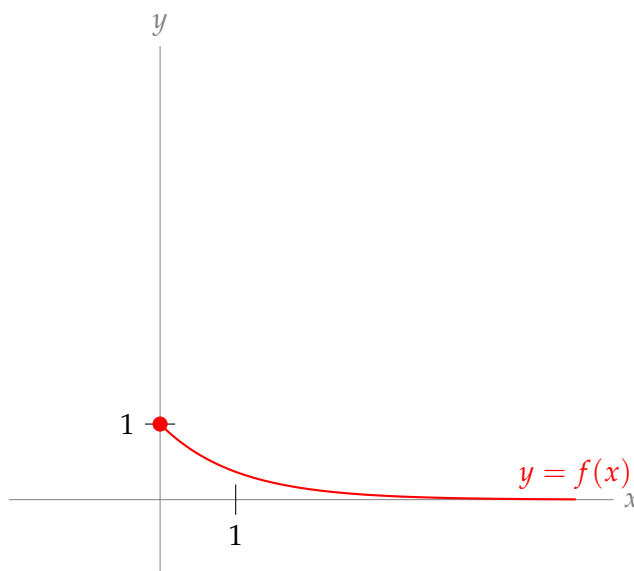
x	$(-\infty, 2 - \sqrt{2})$	$2 - \sqrt{2}$	$(2 - \sqrt{2}, 2 + \sqrt{2})$	$2 + \sqrt{2}$	$(2 + \sqrt{2}, \infty)$
$f'(x)$	positive	0	negative	0	positive
$f(x)$	concave up	IP	concave down	IP	concave up

Now, we can add concavity to our sketch.



S-12: (a) You should be familiar with the graph of $y = e^x$. You can construct the graph of $y = e^{-x}$ just by reflecting the graph of $y = e^x$ across the y -axis. To see why this is the case, imagine swapping each value of x with its negative: for example, swapping the point at $x = -1$ with the point at $x = 1$, etc. Alternatively, you can graph $y = f(x) = e^{-x}$, $x \geq 0$, using the methods of this section: at $x = 0$, $y = f(0) = 1$; as x increases, $y = f(x) = e^{-x}$ decreases, with no local extrema; and as $x \rightarrow +\infty$, $y = f(x) \rightarrow 0$.

There are no inflection points or extrema, except the endpoint $(0, 1)$.



(b)

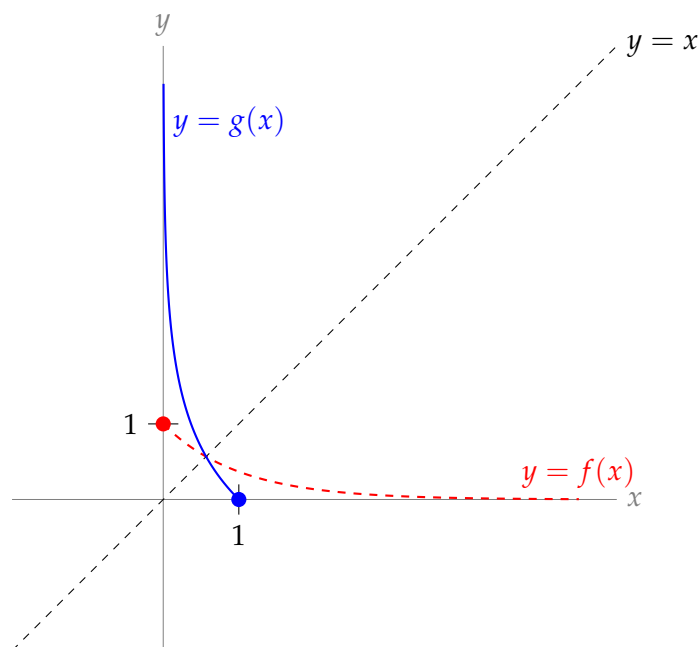
Recall that, to graph the inverse of a function, we reflect the original function across the line $y = x$. To see why this is true, consider the following. By definition, the inverse function g of f is obtained by solving $y = f(x)$ for x as a function of y . So, for any pair of numbers x and y , we have

$$f(x) = y \text{ if and only if } g(y) = x$$

That is, g is the function that swaps the input and output of f . Now the point (x, y) lies on the graph of f if and only if $y = f(x)$. Similarly, the point (X, Y) lies on the graph of g if and only if $Y = g(X)$. Choosing $Y = x$ and $X = y$, we see that the point $(X, Y) = (y, x)$ lies on the graph of g if and only if $x = g(y)$, which in turn is the case if and only if $y = f(x)$. So

(y, x) is on the graph of g if and only if (x, y) is on the graph of f .

To get from the point (x, y) to the point (y, x) we have to exchange $x \leftrightarrow y$, which we can do by reflecting over the line $y = x$. Thus we can construct the graph of g by reflecting the curve $y = f(x)$ over the line $y = x$.



(c) The domain of g is the range of f , which is $(0, 1]$. The range of g is the domain of f , which is $[0, \infty)$.

(d) Since g and f are inverses,

$$g(f(x)) = x$$

Using the chain rule,

$$g'(f(x)) \cdot f'(x) = 1$$

Since $f'(x) = -e^{-x} = -f(x)$:

$$g'(f(x)) \cdot f(x) = -1$$

We plug in $f(x) = \frac{1}{2}$.

$$\begin{aligned} g'\left(\frac{1}{2}\right) \cdot \frac{1}{2} &= -1 \\ g'\left(\frac{1}{2}\right) &= -2 \end{aligned}$$

S-13: (a) First, we differentiate.

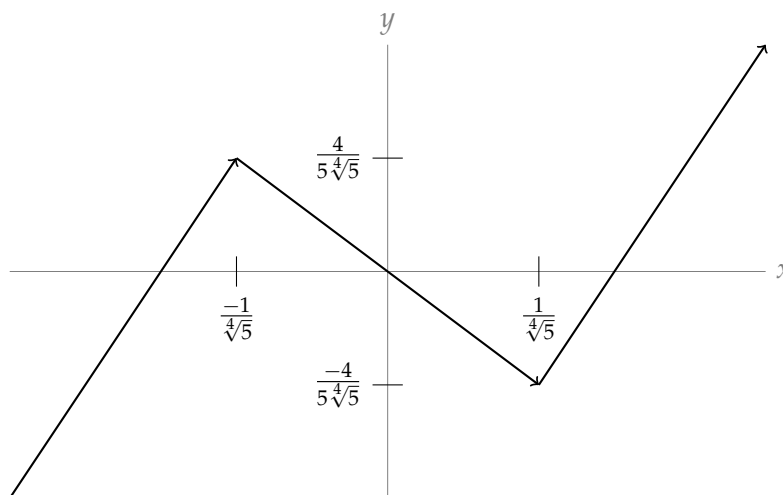
$$f(x) = x^5 - x \quad f'(x) = 5x^4 - 1 \quad f''(x) = 20x^3$$

The function and its first derivative tells us the following:

- $\lim_{x \rightarrow \infty} f(x) = \infty$, $\lim_{x \rightarrow -\infty} f(x) = -\infty$
- $f'(x) > 0$ (i.e. f is increasing) for $|x| > \frac{1}{\sqrt[4]{5}}$

- $f'(x) = 0$ (i.e. f has critical points) for $x = \pm \frac{1}{\sqrt[4]{5}} \approx \pm 0.67$
- $f'(x) < 0$ (i.e. f is decreasing) for $|x| < \frac{1}{\sqrt[4]{5}}$
- $f\left(\pm \frac{1}{\sqrt[4]{5}}\right) = \mp \frac{4}{5\sqrt[4]{5}} \approx \mp 0.53$

This gives us a first idea of the shape of the graph.

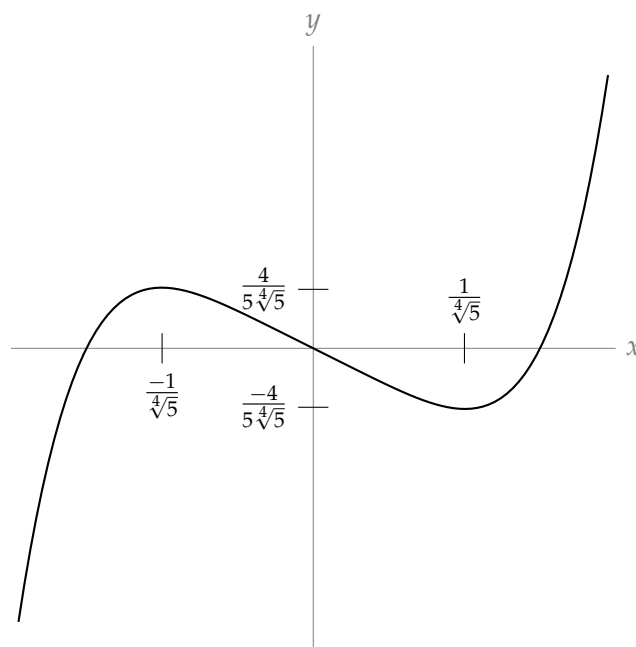


We refine this skeleton using information from the second derivative.

- $f''(x) > 0$ (i.e. f is concave up) for $x > 0$,
- $f''(x) = 0$ (i.e. f has an inflection point) for $x = 0$, and
- $f''(x) < 0$ (i.e. f is concave down) for $x < 0$

Thus

- f has no asymptotes
- f has a local maximum at $x = -\frac{1}{\sqrt[4]{5}}$ and a local minimum at $x = \frac{1}{\sqrt[4]{5}}$
- f has an inflection point at $x = 0$
- f is concave down for $x < 0$ and concave up for $x > 0$



(b) The function $x^5 - x + k$ has a root at $x = x_0$ if and only if $x^5 - x = -k$ at $x = x_0$. So the number of distinct real roots of $x^5 - x + k$ is the number of times the curve $y = x^5 - x$ crosses the horizontal line $y = -k$. The local maximum of $x^5 - x$ (when $x = -\frac{1}{\sqrt[4]{5}}$) is $\frac{4}{5\sqrt[4]{5}}$, and the local minimum of $x^5 - x$ (when $x = \frac{1}{\sqrt[4]{5}}$) is $-\frac{4}{5\sqrt[4]{5}}$. So, looking at the graph of $x^5 - x$ above, we see that the number of distinct real roots of $x^5 - x + k$ is

- 1 when $|k| > \frac{4}{5\sqrt[4]{5}}$
- 2 when $|k| = \frac{4}{5\sqrt[4]{5}}$
- 3 when $|k| < \frac{4}{5\sqrt[4]{5}}$

S-14:

(a) You might not be familiar with hyperbolic sine and cosine, but you don't need to be. We can graph them using the same methods as the other curves in this section. The derivatives are given to us:

$$\begin{aligned} \frac{d}{dx} \{\sinh x\} &= \cosh x = \frac{e^x + e^{-x}}{2} & \frac{d}{dx} \{\cosh x\} &= \sinh x = \frac{e^x - e^{-x}}{2} \\ \left(\frac{d}{dx}\right)^2 \{\sinh x\} &= \sinh x = \frac{e^x - e^{-x}}{2} & \left(\frac{d}{dx}\right)^2 \{\cosh x\} &= \cosh x = \frac{e^x + e^{-x}}{2} \end{aligned}$$

Observe that:

- $\sinh(x)$ has a derivative that is always positive, so $\sinh(x)$ is always increasing. The second derivative of $\sinh(x)$ is negative to the left of $x = 0$ and positive to the right of $x = 0$, so $\sinh(x)$ is concave down to the left of the y -axis and concave up to its right, with an inflection point at $x = 0$.

- $\cosh(x)$ has a derivative that is positive when $x > 0$ and negative when $x < 0$. The second derivative of $\cosh(x)$ is always positive, so it is always concave up.

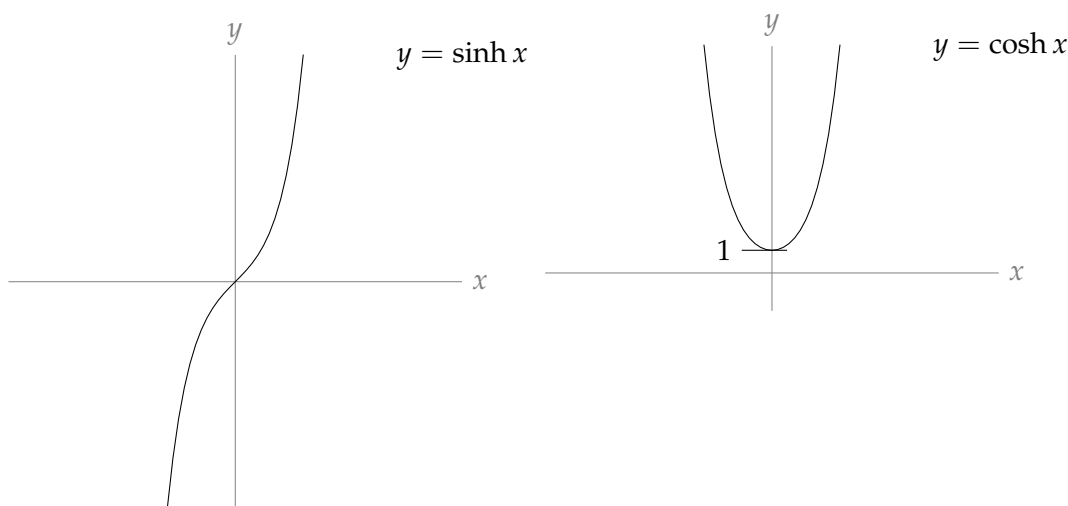
- $\cosh(0) = 1$ and $\sinh(0) = 0$.

- $\lim_{x \rightarrow \infty} \sinh x = \lim_{x \rightarrow \infty} \cosh x = \lim_{x \rightarrow \infty} \frac{e^x}{2} = \infty$, since $\lim_{x \rightarrow \infty} e^{-x} = 0$

- $\lim_{x \rightarrow -\infty} \sinh x = \lim_{x \rightarrow -\infty} \left(\frac{e^x}{2} - \frac{e^{-x}}{2} \right) = \lim_{x \rightarrow -\infty} \left(\frac{e^{-x}}{2} - \frac{e^x}{2} \right) = -\infty$ and

$$\lim_{x \rightarrow -\infty} \cosh x = \lim_{x \rightarrow -\infty} \left(\frac{e^x}{2} + \frac{e^{-x}}{2} \right) = \lim_{x \rightarrow -\infty} \left(\frac{e^{-x}}{2} + \frac{e^x}{2} \right) = \infty$$

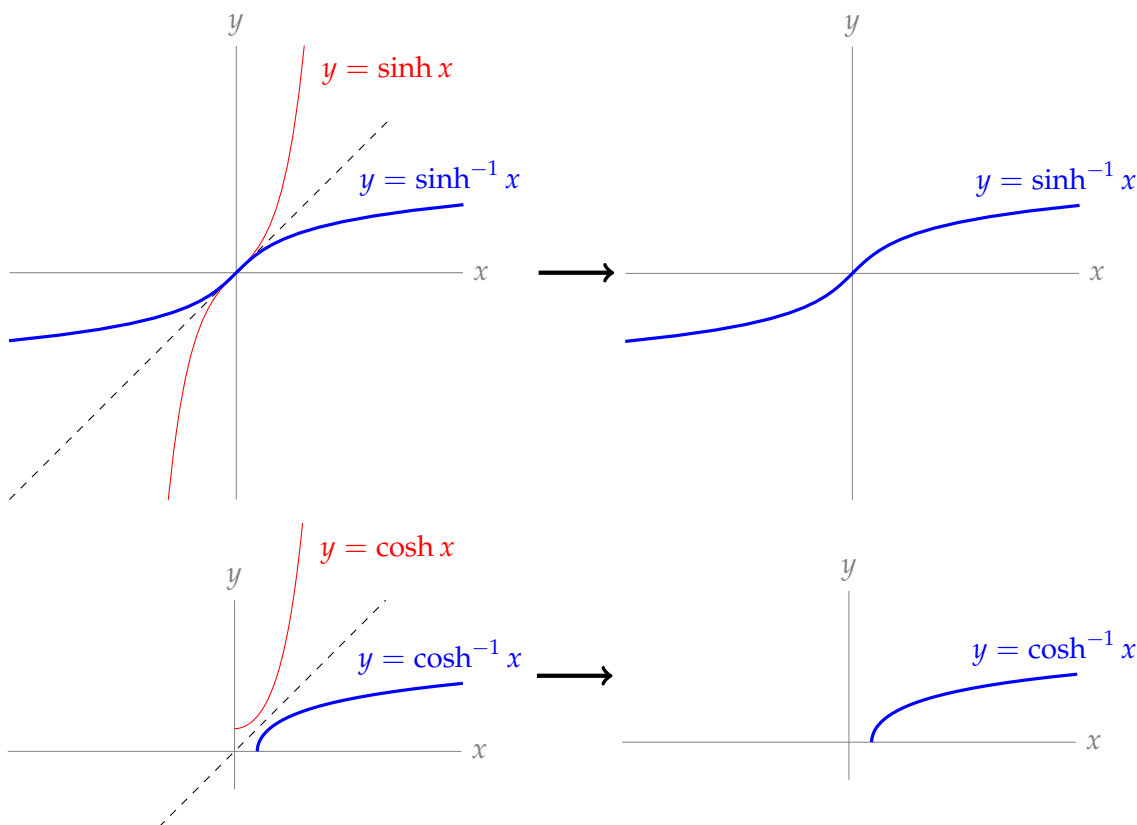
- $\cosh(x)$ is even, since $\cosh(-x) = \frac{e^{-x} + e^{-(-x)}}{2} = \frac{e^{-x} + e^x}{2} = \cosh(x)$, and $\sinh(x)$ is odd, since $\sinh(-x) = \frac{e^{-x} - e^{-(-x)}}{2} = \frac{e^{-x} - e^x}{2} = \frac{-(e^x - e^{-x})}{2} = -\sinh(x)$



(b)

- As y runs over $(-\infty, \infty)$ the function $\sinh(y)$ takes every real value exactly once. So, for each $x \in (-\infty, \infty)$, define $\sinh^{-1}(x)$ to be the unique solution of $\sinh(y) = x$.
- As y runs over $[0, \infty)$ the function $\cosh(y)$ takes every real value in $[1, \infty)$ exactly once. In particular, the smallest value of $\cosh(y)$ is $\cosh(0) = 1$. So, for each $x \in [1, \infty)$, define $\cosh^{-1}(x)$ to be the unique $y \in [0, \infty)$ that obeys $\cosh(y) = x$.

To graph the inverse of a (one-to-one) function, we reflect the original function over the line $y = x$. Using this method to graph $y = \sinh^{-1}(x)$ is straightforward. To graph $y = \cosh^{-1}(x)$, we need to be careful of the domains: we are restricting $\cosh(x)$ to values of x in $[0, \infty)$. The graphs are



(c) Let $y(x) = \cosh^{-1}(x)$. Then, using the definition of $\cosh^{-1}$,

$$\cosh y(x) = x$$

We differentiate with respect to x using the chain rule.

$$\begin{aligned} \frac{d}{dx} \{\cosh y(x)\} &= \frac{d}{dx} \{x\} \\ y'(x) \sinh y(x) &= 1 \end{aligned}$$

We solve for $y'(x)$.

$$y'(x) = \frac{1}{\sinh y(x)}$$

We want to have our answer in terms of x , not y . We know that $\cosh y = x$, so if we can convert hyperbolic sine into hyperbolic cosine, we can get rid of y . Our tool for this is the identity, given in the question statement, $\cosh^2(x) - \sinh^2(x) = 1$. This tells us $\sinh^2(y) = 1 - \cosh^2(y)$. Now we have to decide whether $\sinh(y)$ is the positive or negative square root of $1 - \cosh^2(y)$ in our context. Looking at the graph of $y(x) = \cosh^{-1}(x)$, we see $y'(x) > 0$. So we use the positive square root:

$$y'(x) = \frac{1}{\sqrt{\cosh^2 y(x) - 1}} = \frac{1}{\sqrt{x^2 - 1}}$$

Remark: $\frac{d}{dx} \{\arccos(x)\} = \frac{-1}{\sqrt{1-x^2}}$, so again the hyperbolic trigonometric function has properties similar to (but not exactly the same as) its trigonometric counterpart.